Squares in Squares

SVG, high-precision, and other updates by David Ellsworth
based on original compiled by Erich Friedman

The following pictures show n unit squares packed inside the smallest known square (of side length s). If a pictured packing has multiple numbers in its label above, the picture represents the largest; each smaller is represented by removing any square. For the $n \leq 324$ not pictured, the trivial packing (with no tilted squares) is the best known packing. Where a polynomial root is known for s of degree 3 or higher (and has no concise closed-form expression), a 🔒 icon is shown; click this to see the polynomial root form of s.

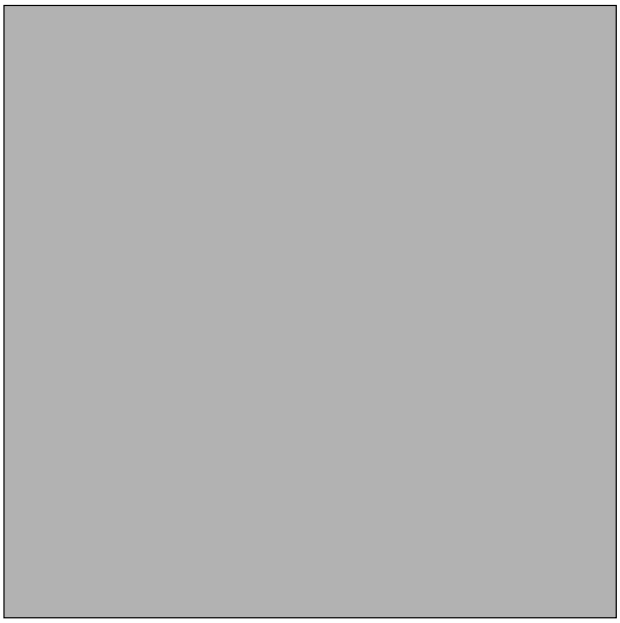
See also triangular table view (recommended) and older records and/or alternative packings. For more information on each packing, view its SVG's source code. In browsers that don't provide an easy way to do this, you can prepend the URL with "`view-source:`" (without the quotes).

In SVG Edit Mode, which can be turned on either with the button or by pressing [E] while viewing an SVG, the squares can be dragged using the mouse or other pointer device. (Note, recursive pushing isn't yet implemented.) Holding Shift constrains motion to an axis parallel to the square's edges. Holding Ctrl enables rotation-only movement. Pressing Delete will delete a square while the left mouse button is held on it. Pressing [S] will download/save the current edited packing.

Many of these SVGs do not encode their squares as distinct objects (for efficiency, readability, and avoiding visual double-stroke artifacts). To convert them into a format that does individually encode the coordinates and angle of each square, either use the SVG Edit Mode as described above (which is limited to double precision), or the parse_svg_packing.py tool, which preserves full precision.

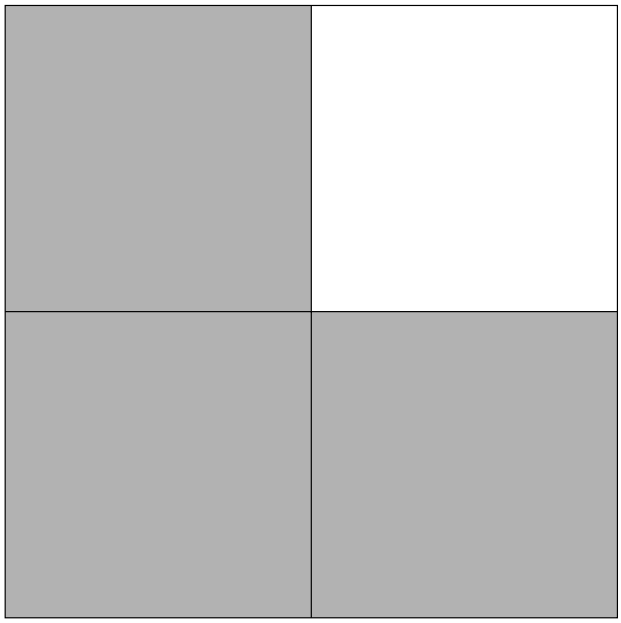
Zoom: 0.25× 1/3× 0.5× 2/3× 3/4× 1× 1.5× 2× SVG Edit Mode: OFF

1



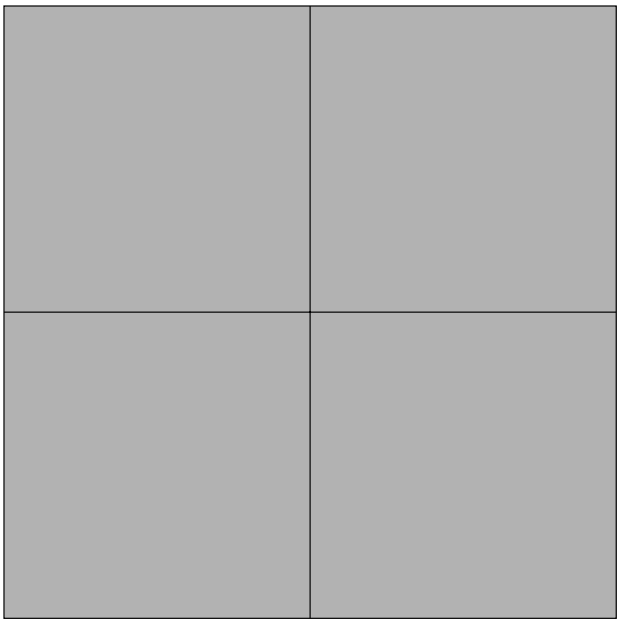
$s = 1$
Trivial.

2, 3



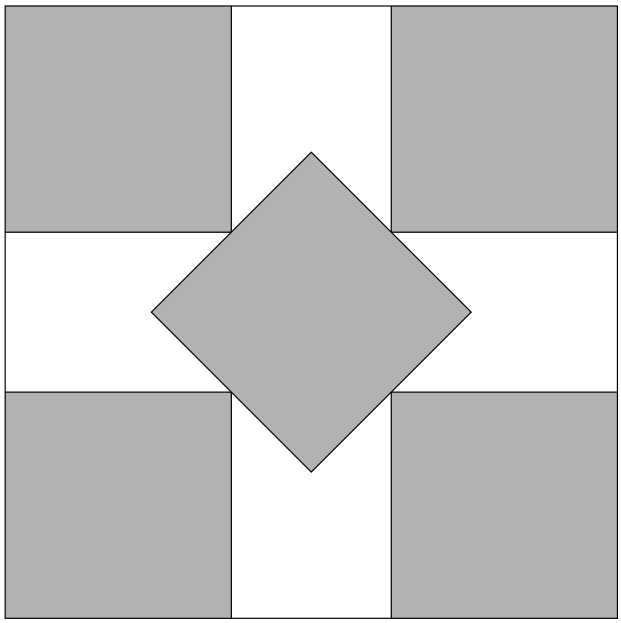
$s = 2$
Proved by Frits Göbel
in early 1979.

4



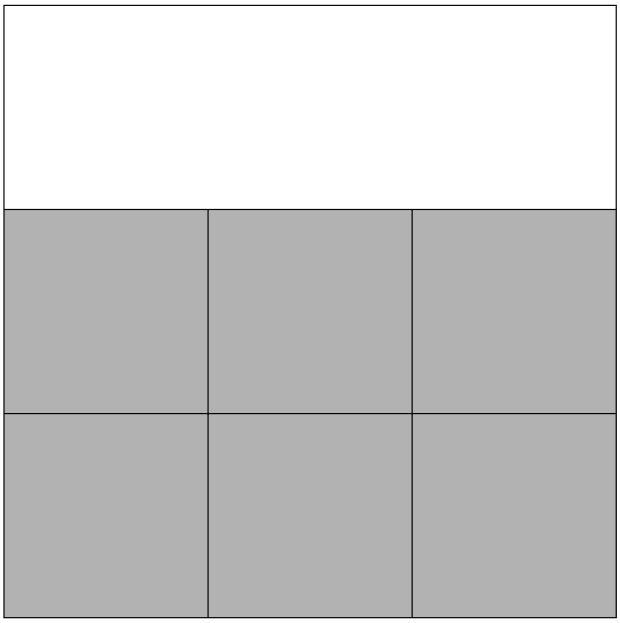
$s = 2$
Trivial.

5



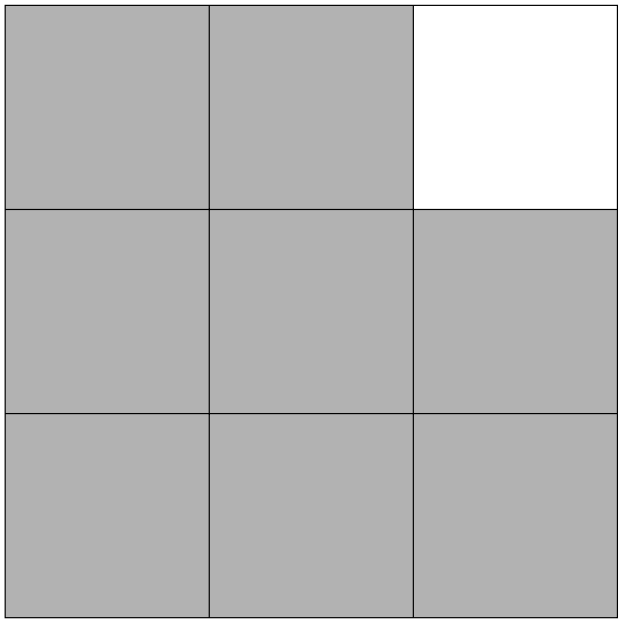
$s = 2 + \frac{1}{2}\sqrt{2} = 2.70710678118654\ldots$
Rigid.
Proved by Frits Göbel
in early 1979.

6



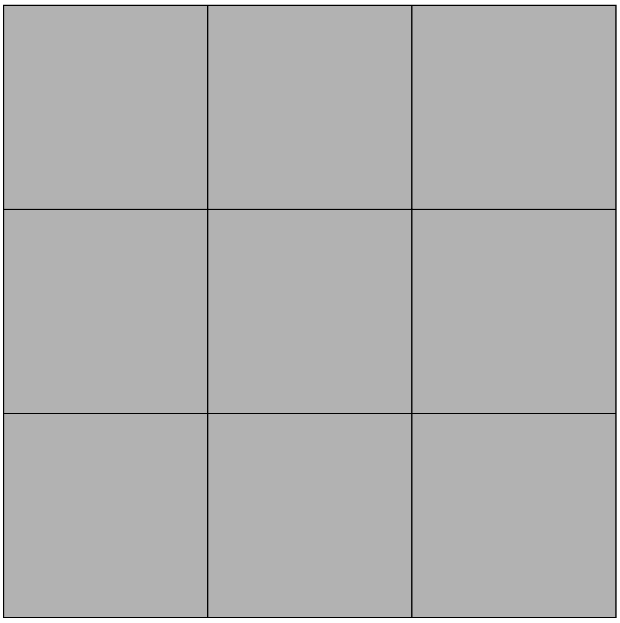
$s = 3$
Proved by Michael Kearney
and Peter Shiu in June 2001.

7, 8



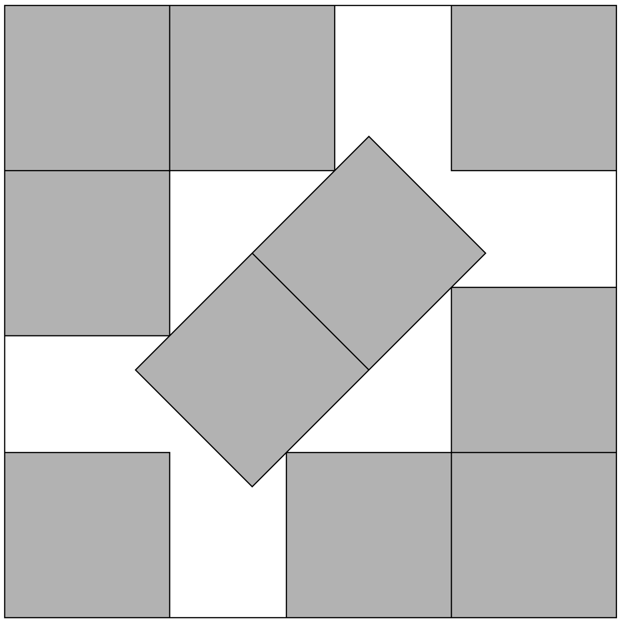
$s = 3$
Proved by Erich Friedman
in 1999.

9



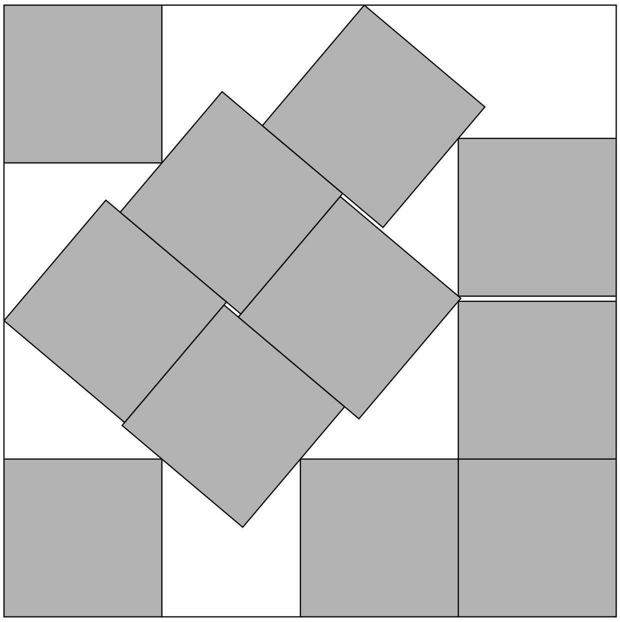
$s = 3$
Trivial.

10



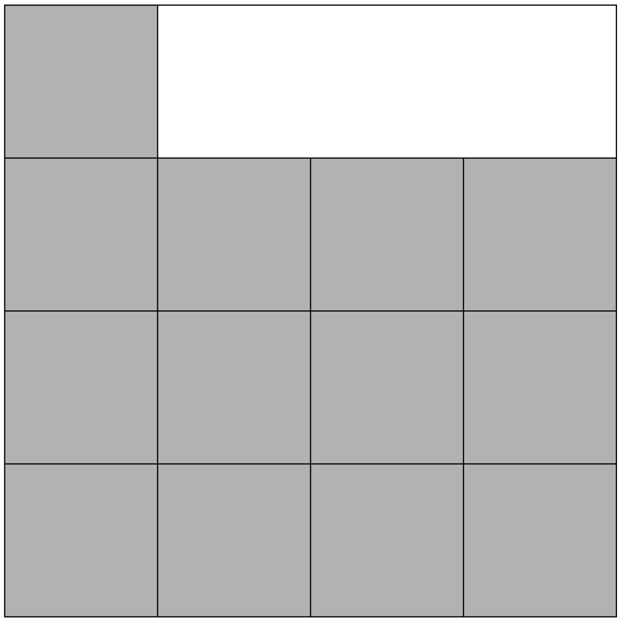
$s = 3 + \frac{1}{2}\sqrt{2} = 3.70710678118654\ldots$
Found by Frits Göbel in early 1979.
Proved by Walter Stromquist in 2003.
Explore group

11



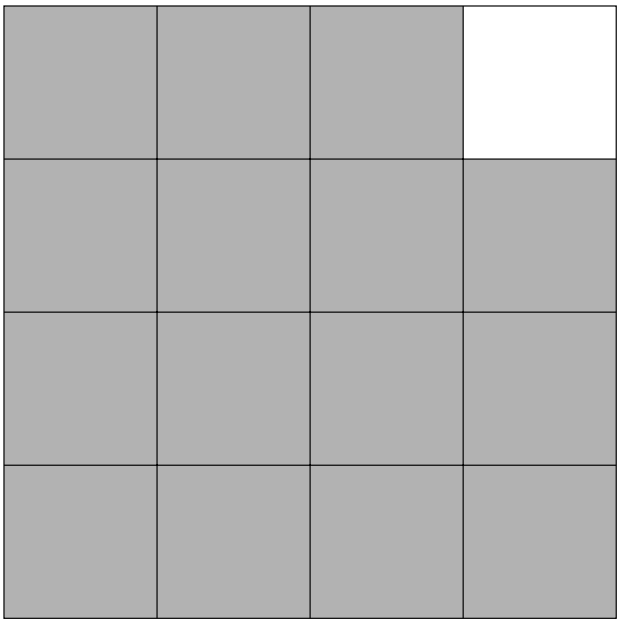
$s = {}^{8}$🔒 $= 3.87708359002281\ldots$
Rigid.
Found by Walter Trump
in 1979.

13



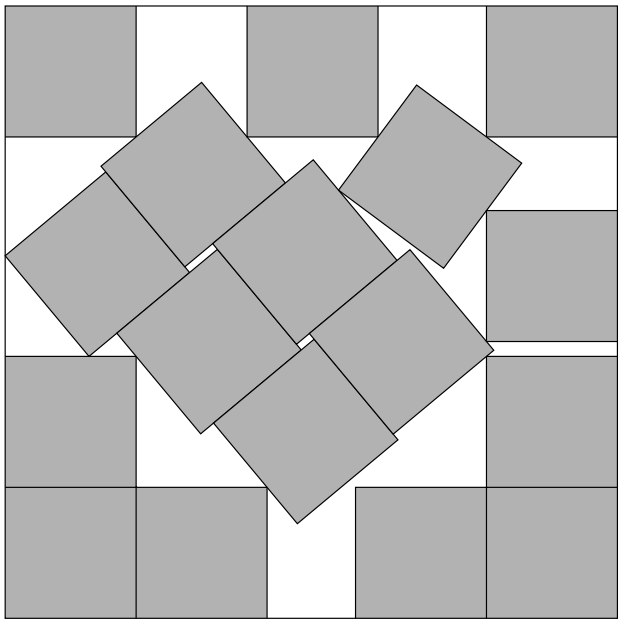
$s = 4$
Proved by Wolfram Bentz
in August 2009.

14, 15



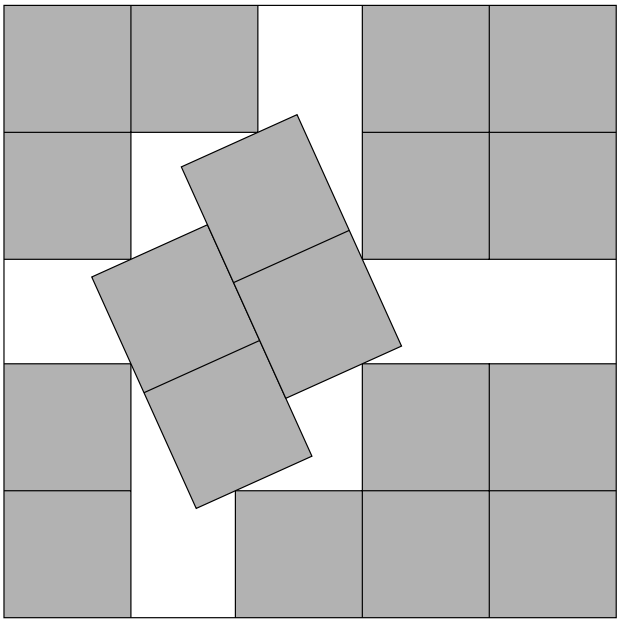
$s = 4$
Proved by Erich Friedman
in 1999.

17



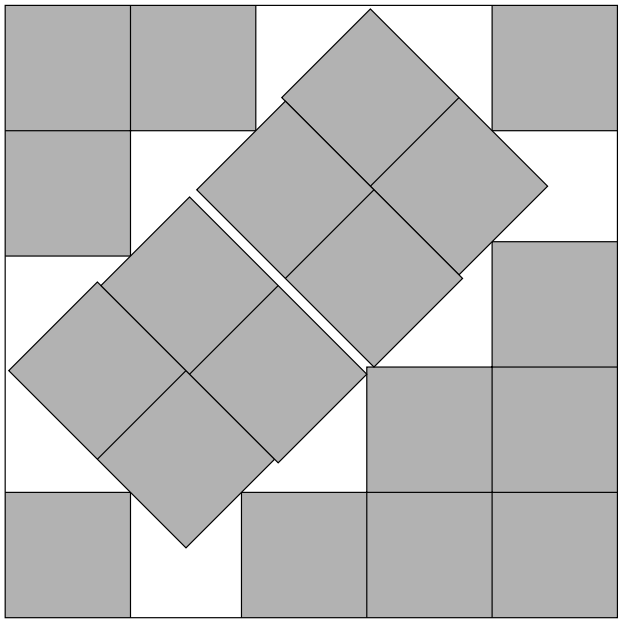
$s = {}^{18}\!\!\blacksquare = 4.67553009360455\ldots$
Found by John Bidwell
in 1998.
Based on packing found by Pertti Hämäläinen
in 1980.

18



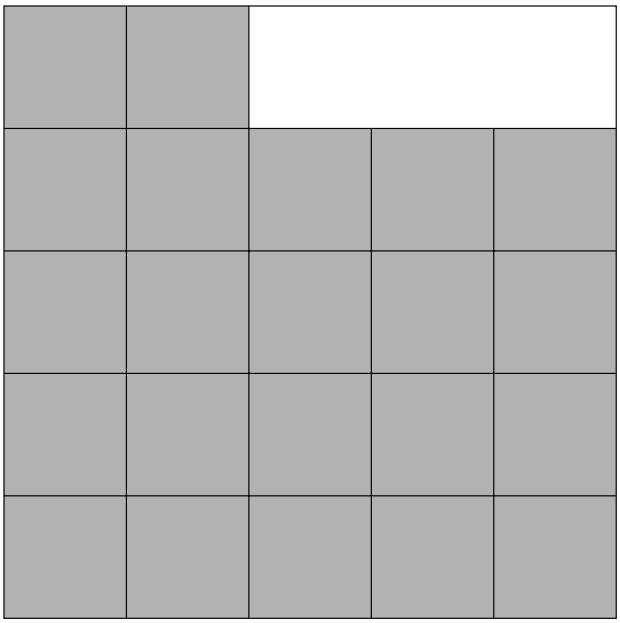
$s = \frac{7}{2} + \frac{1}{2}\sqrt{7} = 4.82287565553229\ldots$
Found by Pertti Hämäläinen
in 1980.
Pictured alternative with minimal rotated
squares found by Mats Gustafsson
in 1981.

19



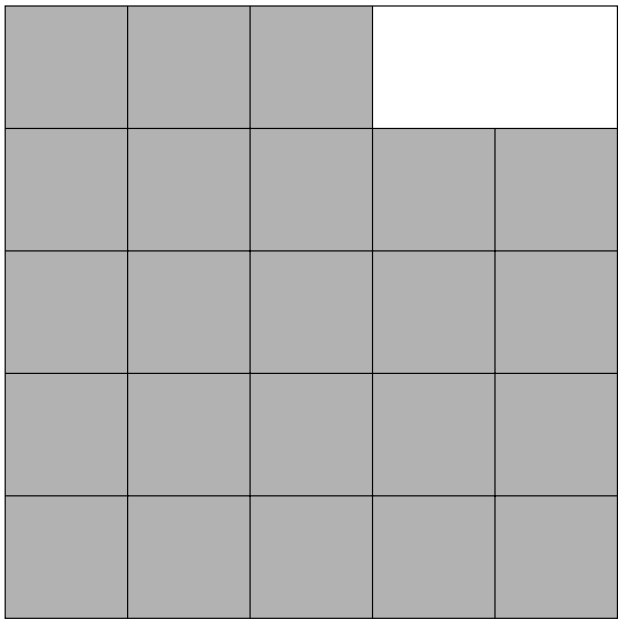
$s = 3 + \frac{4}{3}\sqrt{2} = 4.88561808316412\ldots$
Found first by Robert Wainwright
in late 1979.
Based on packing found by
Charles F. Cottingham in early 1979.

22



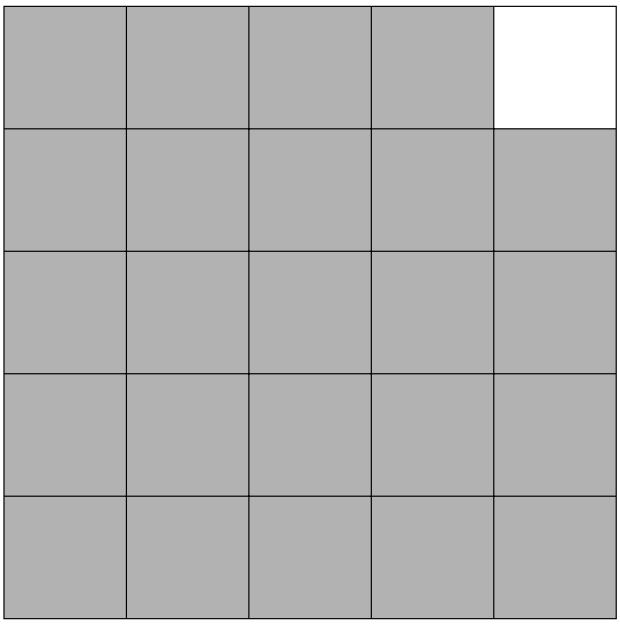
$s = 5$
Proved by Wolfram Bentz
in October 2018.

23



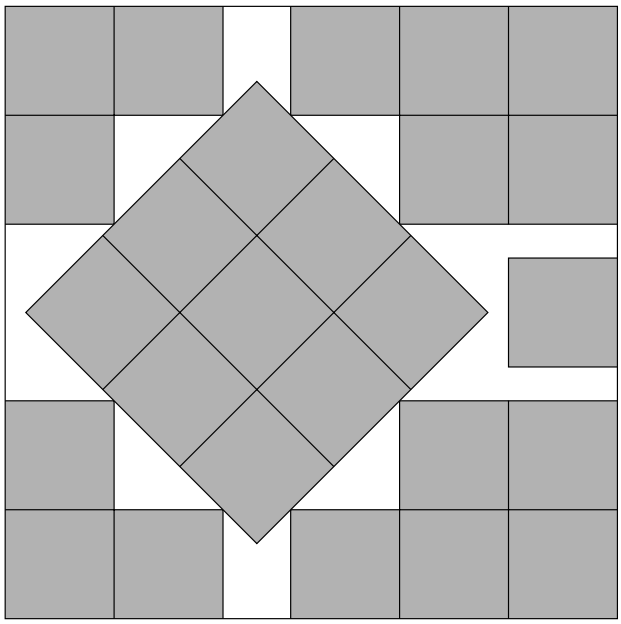
$s = 5$
Proved by Hiroshi Nagamochi
in 2005.

24



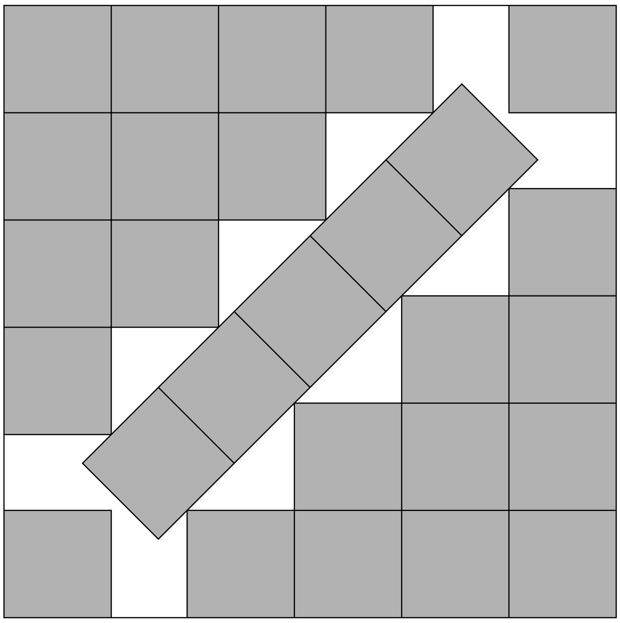
$s = 5$
Proved by Erich Friedman
in 1999.

26



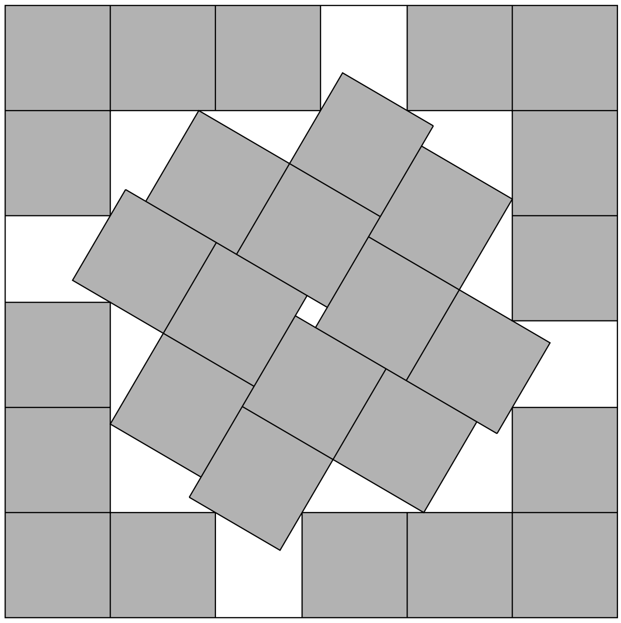
$s = \frac{7}{2} + \frac{3}{2}\sqrt{2} = 5.62132034355964\ldots$
Found by Erich Friedman
in 1997.
Unextends the $s(37)$ found by Evert Stenlund
in early 1980.

27



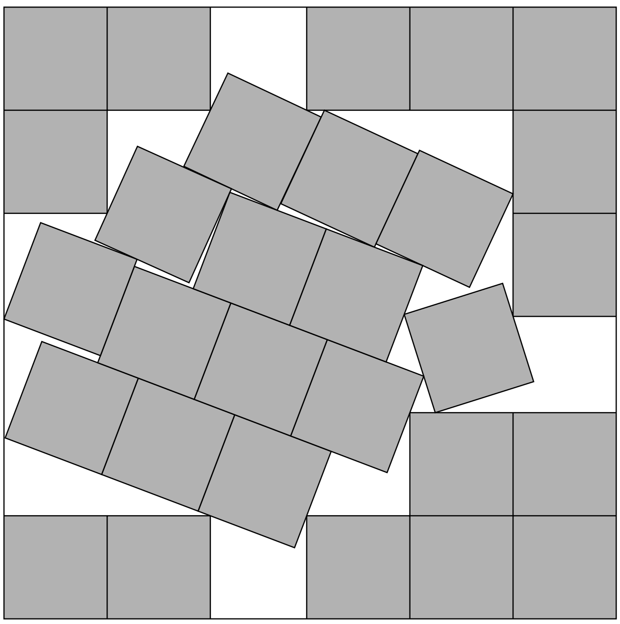
$s = 5 + \frac{1}{2}\sqrt{2} = 5.70710678118654...$
Found by Frits Göbel
in early 1979.
Explore group

28



$s = {}^{6}$ 🔒 $= 5.82444461667405...$
Rigid.
Found by David Ellsworth
in December 2025, using his modified version of Thomas Schadt's simulated annealing program, starting from randomness.

29

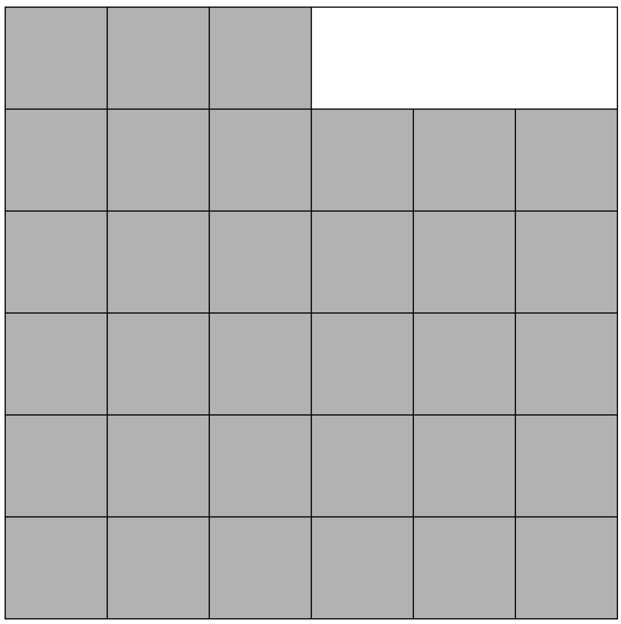


$s = 5.93383346267692...$
Found by Thomas Schadt
in December 2025, using a simulated annealing program he wrote, starting from randomness.

Similar to the packing found by Thierry Gensane and Philippe Ryckelynck in April 2004, using a computer program they wrote.

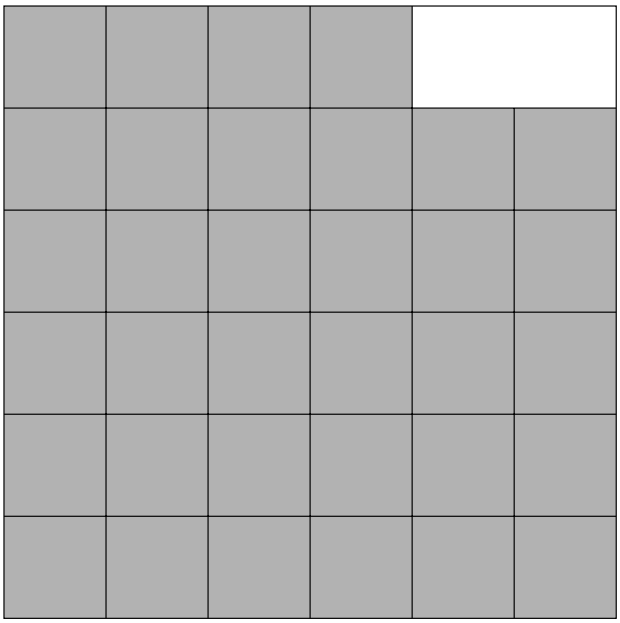
Optimized by David Ellsworth
in December 2025.

33



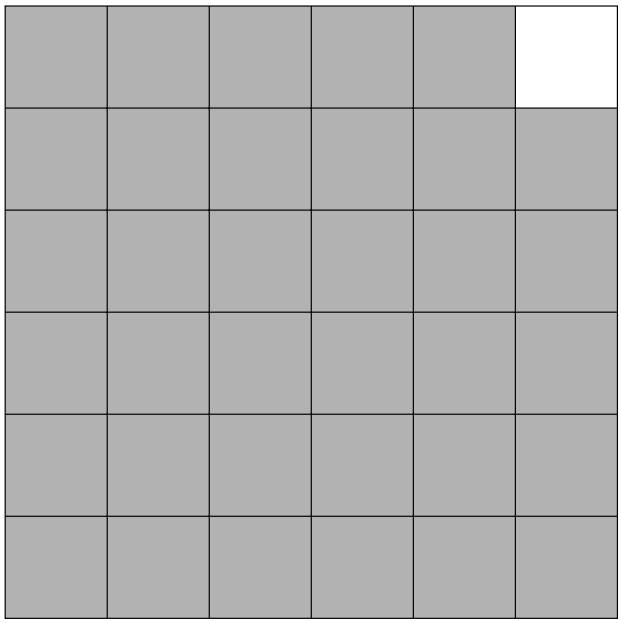
$s = 6$
Proved by Wolfram Bentz
in October 2018.

34



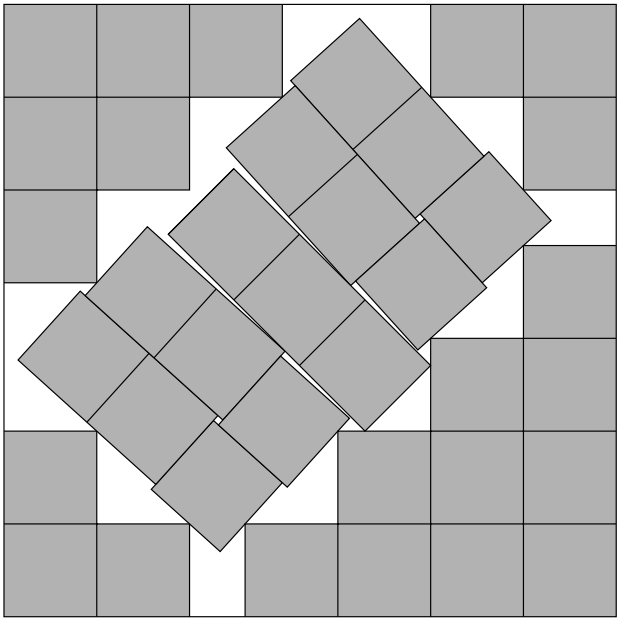
$s = 6$
Proved by Hiroshi Nagamochi
in 2005.

35



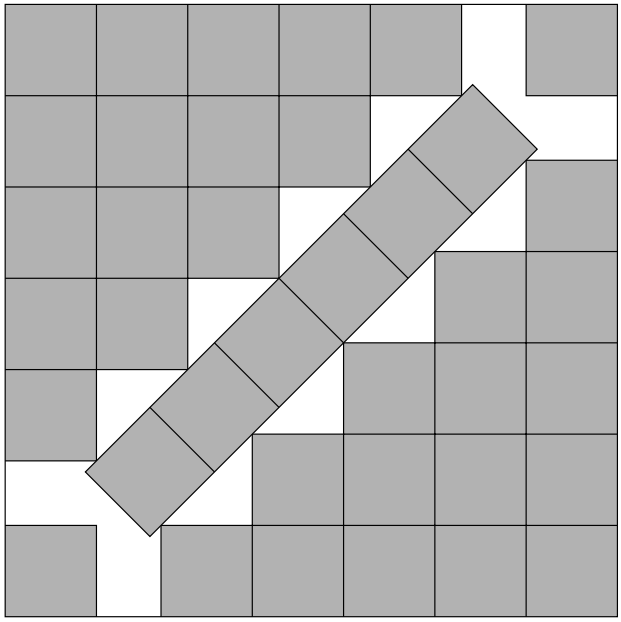
$s = 6$
Proved by Erich Friedman
in 1999.

37



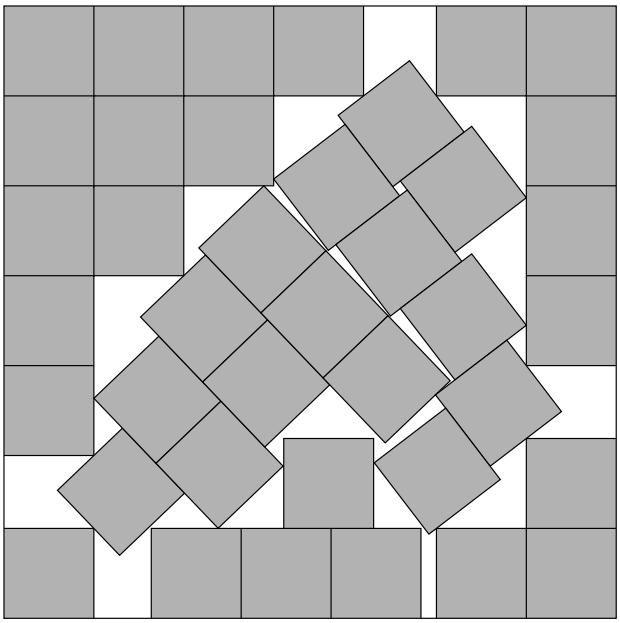
$s = {}^{8}\!\!\text{🔒} = 6.59861960924436\ldots$
Found by David W. Cantrell
in September 2002.
Improves upon the $s(37)$ found by
Evert Stenlund in early 1980.

38



$s = 6 + \frac{1}{2}\sqrt{2} = 6.70710678118654\ldots$
Found by Frits Göbel
in early 1979.
Explore group

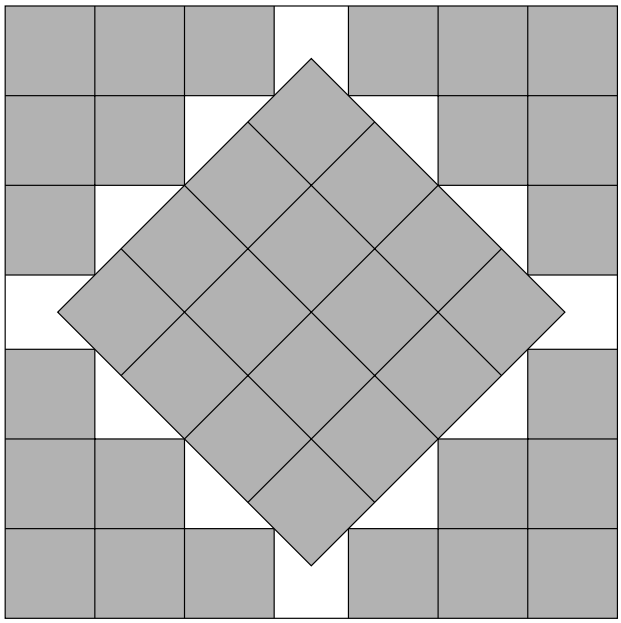
39



$s = {}^{5}\!\!\color{gray}\blacksquare = 6.81072208306864\ldots$

Found by Thomas Schadt in January 2026, using a simulated annealing program he wrote, starting from randomness.
Refound and refined by David Ellsworth in January 2026, starting from randomness.
Optimized by David Ellsworth in January 2026.

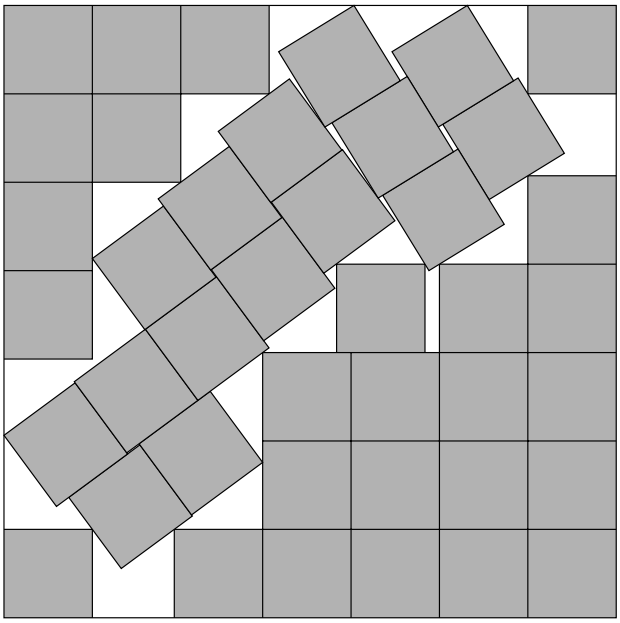
40



$s = 4 + 2\sqrt{2} = 6.82842712474619\ldots$

Rigid.
Found by Frits Göbel in early 1979.
Explore group

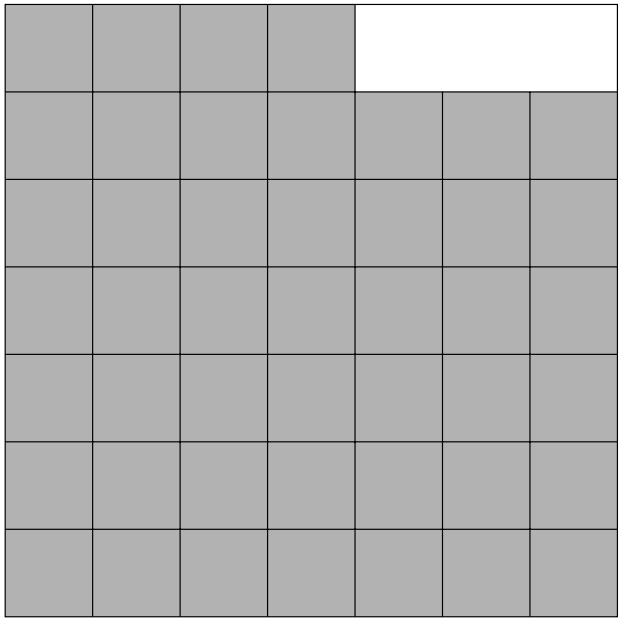
41



$s = {}^{42}\!\!\color{gray}\blacksquare = 6.92669309446880\ldots$

Found by Thomas Schadt in December 2025, using a simulated annealing program he wrote, starting from randomness.
Refound and refined by David Ellsworth in December-January 2025-2026, using his modified version of Thomas Schadt's

46

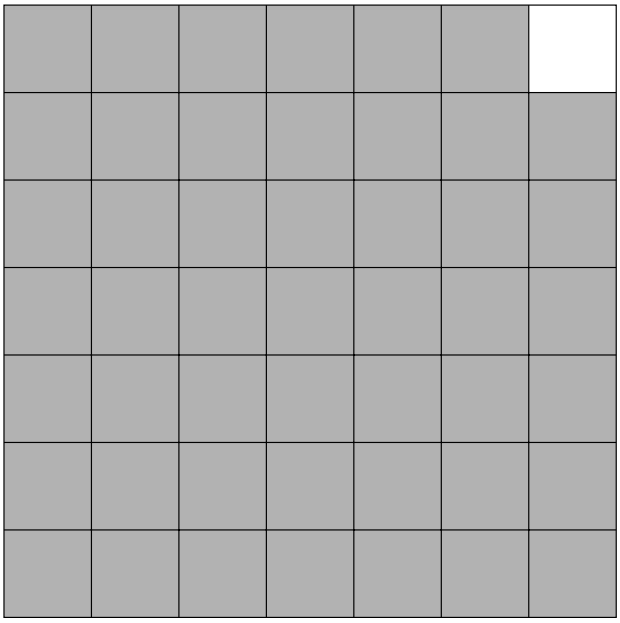


$s = 7$

Proved by Wolfram Bentz in August 2009.

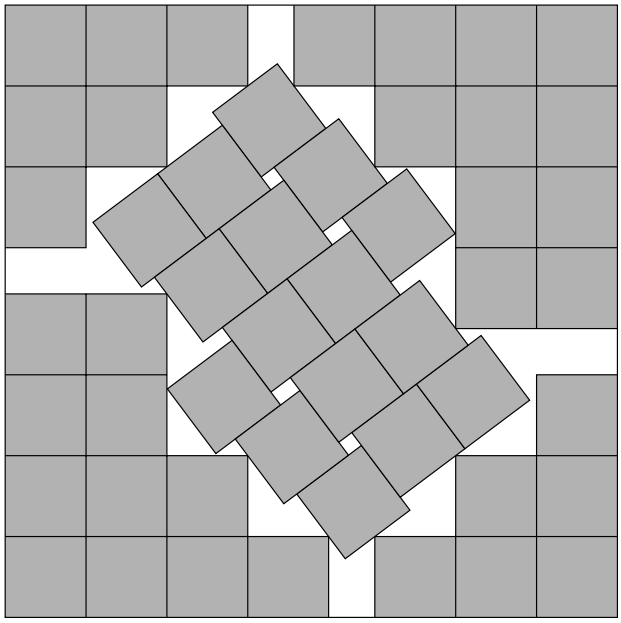
simulated annealing program, starting from randomness.
Optimized by David Ellsworth in January 2026.

47, 48



$s = 7$
Proved by Hiroshi Nagamochi in 2005.

50



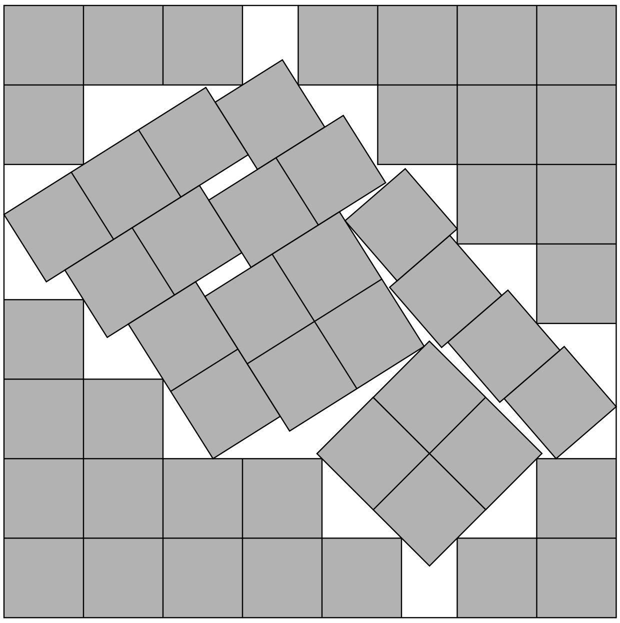
$s = 7 + \frac{4}{7} = 7.57142857142857\ldots$
Found by Thomas Schadt in December 2025, using a simulated annealing program he wrote, starting from randomness. Optimized by David Ellsworth.

This is the third record-setting packing found with a rational side length, thanks to the Pythagorean triple $\{3, 4, 5\}$ determining its tilt angle. It is the first record-setting packing to use the same Pythagorean triple as the $s(104)$ found by David Ellsworth in December 2024 (the first competitively-intended packing found with a rational side length, though not record-setting).

It's the second record-setting packing found that is doubly semi-primitive, i.e. has only non-rotated squares on its leftmost and rightmost sides, but has rotated square(s) poking into its top and bottom sides.

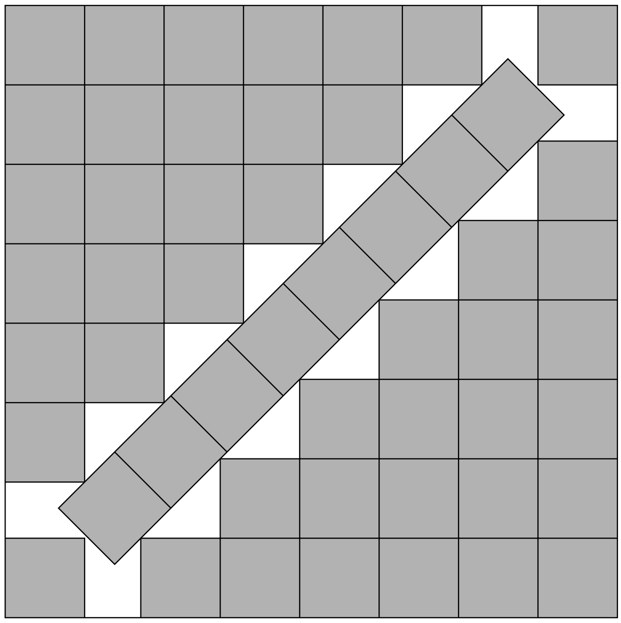
It's the first to have both of these properties.

51



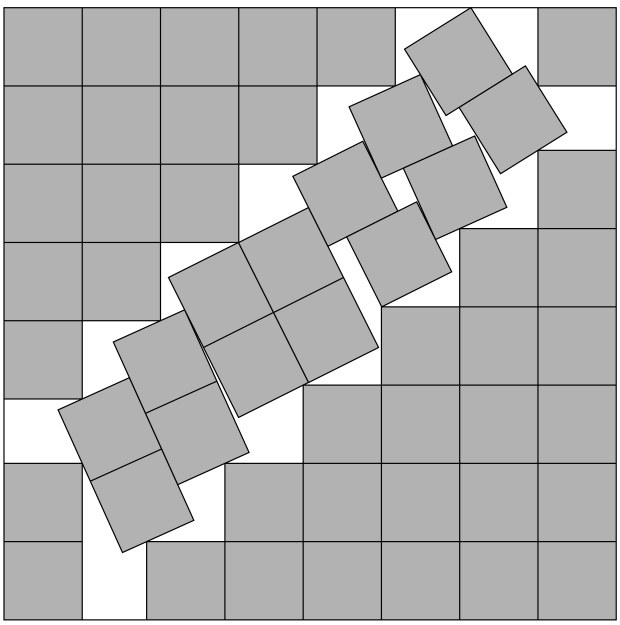
$s =$ 12 🔒 $= 7.70079923541701\ldots$
Found by Thomas Schadt in January 2026, using a simulated annealing program he wrote, starting from randomness.
Refound by David Ellsworth in January 2026, using his modified version #2 of Thomas Schadt's simulated annealing program, starting from randomness (statistics gathered).
Optimized by David Ellsworth in February 2026.

52



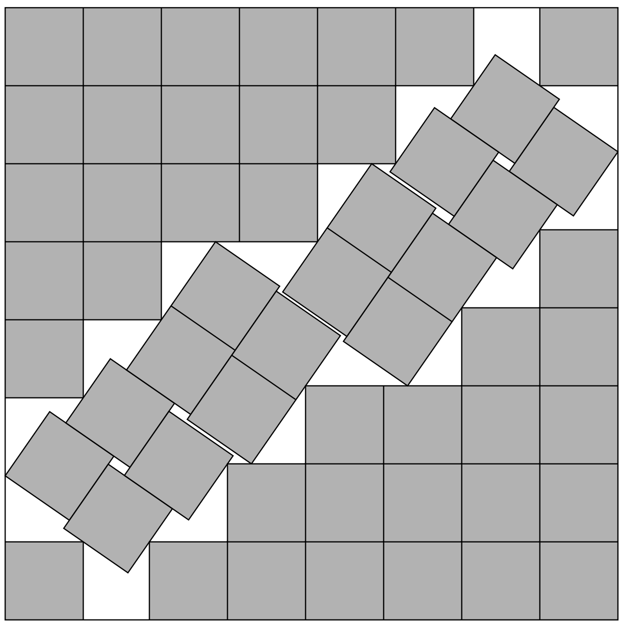
$s = 7 + \frac{1}{2}\sqrt{2} = 7.70710678118654\ldots$
Found by Frits Göbel in early 1979.
Explore group

53



$s = \frac{13}{2} + \frac{1}{2}\sqrt{7} = 7.82287565553229\ldots$
Found by David W. Cantrell in September 2002.
Improved by David W. Cantrell in December 2024.

54



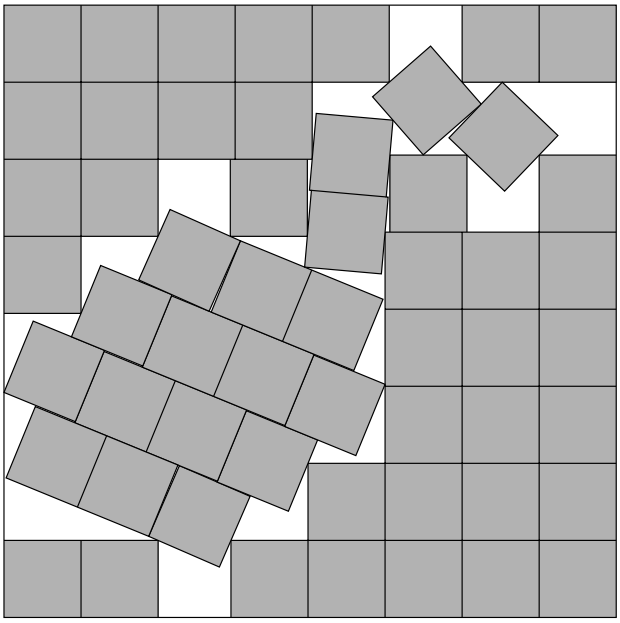
$$s = 7 - \frac{1}{2}\sqrt{2} + \sqrt{1 + \sqrt{2}}$$
$$= 7.84666719284348\ldots$$

Found by David W. Cantrell in October 2005.

Improved by David Ellsworth
in February 2026, using his modified
version #3 of Thomas Schadt's simulated
annealing program, starting from
randomness.
Optimized by David Ellsworth
in February 2026.

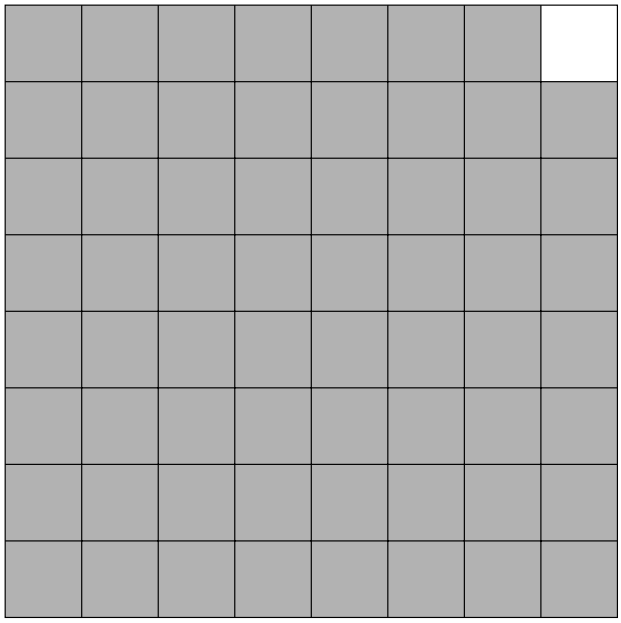
Improved by Joe DeVincentis
in April 2014.

55



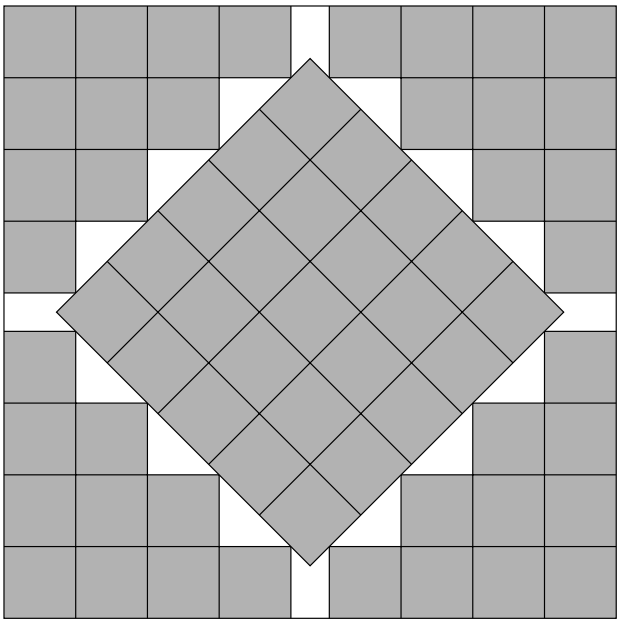
$s = 7.94577100750391\ldots$
Found by Thomas Schadt in January 2026,
using a simulated annealing program he
wrote, starting from a cherry-picked state
found by his earlier program (started from
randomness) which reimplements the
Gensane & Ryckelynck algorithm.
Refound by David Ellsworth
in February 2026, using his modified
version #3 of Thomas Schadt's simulated
annealing program, starting from
randomness (statistics gathered).
Improved by David Ellsworth
in February 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Optimized by David Ellsworth
in February 2026.

62, 63



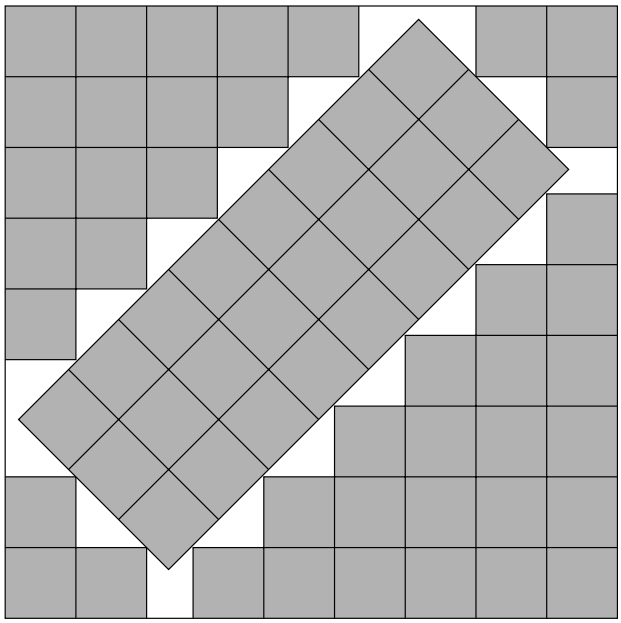
$s = 8$
Proved by Hiroshi Nagamochi
in 2005.

65



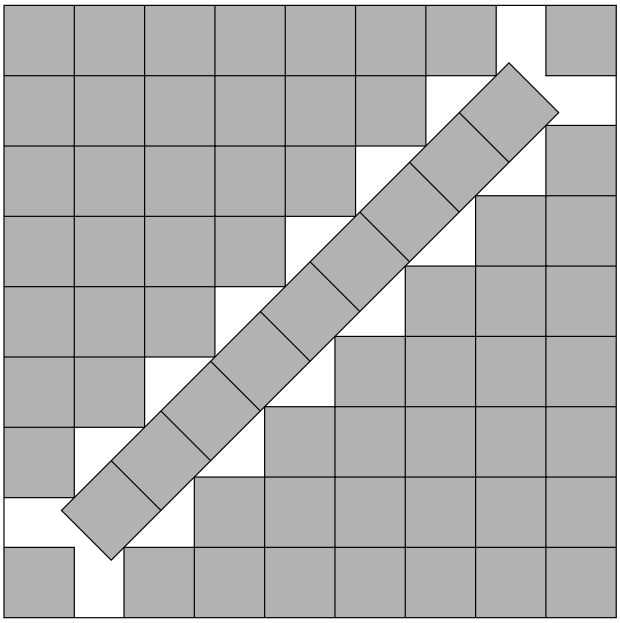
$s = 5 + \frac{5}{2}\sqrt{2} = 8.53553390593273\ldots$
Found by Frits Göbel
in early 1979.
Explore group

66



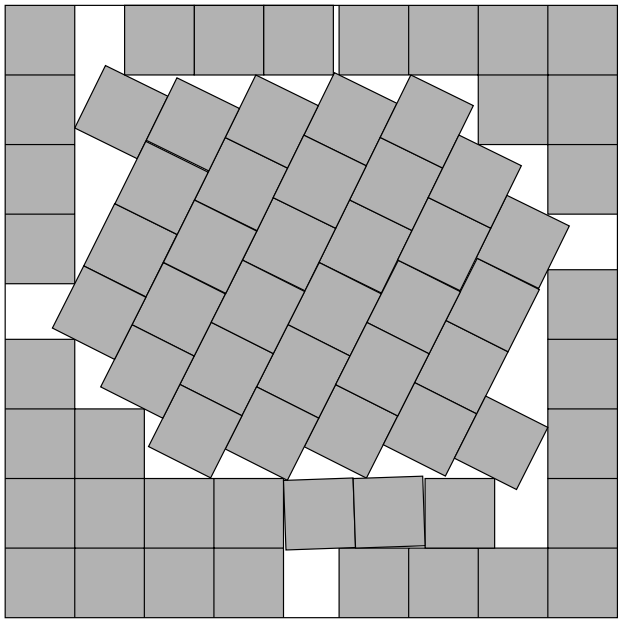
$s = 3 + 4\sqrt{2} = 8.65685424949238\ldots$
Found by Evert Stenlund in early 1980.

67



$s = 8 + \frac{1}{2}\sqrt{2} = 8.70710678118654\ldots$
Found by Evert Stenlund
in early 1980, extending the $s(52)$
found by Frits Göbel in early 1979.
Explore group

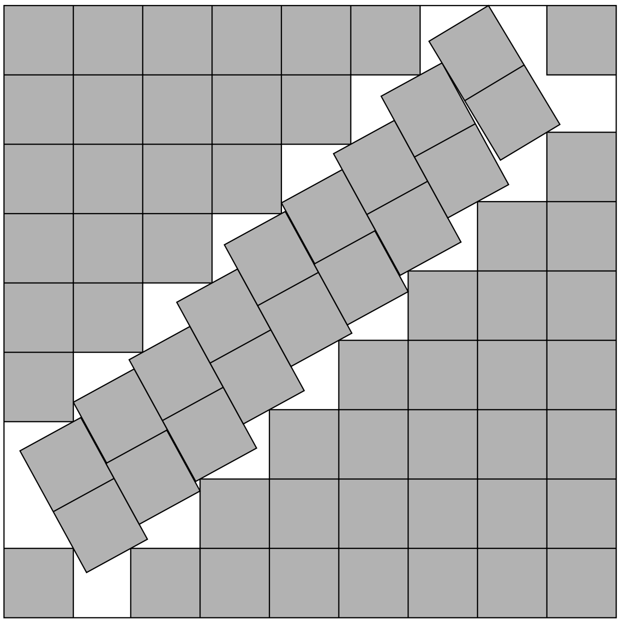
68



$s = 8.7987961402601$
Found by Sigvart Brendberg in June 2023,
using a computer program he wrote
followed by manual optimization.
Improved by Thomas Schadt in December 2025, using a simulated annealing program he wrote, starting from randomness.
Optimized by David Ellsworth in December 2025.
Improved by Joost de Winter in August 2026, working with unspecified AI.
Improved and optimized by Allen Chang
in September 2026, working with GPT-6 Astra, with help from "TheMagicAnimals".

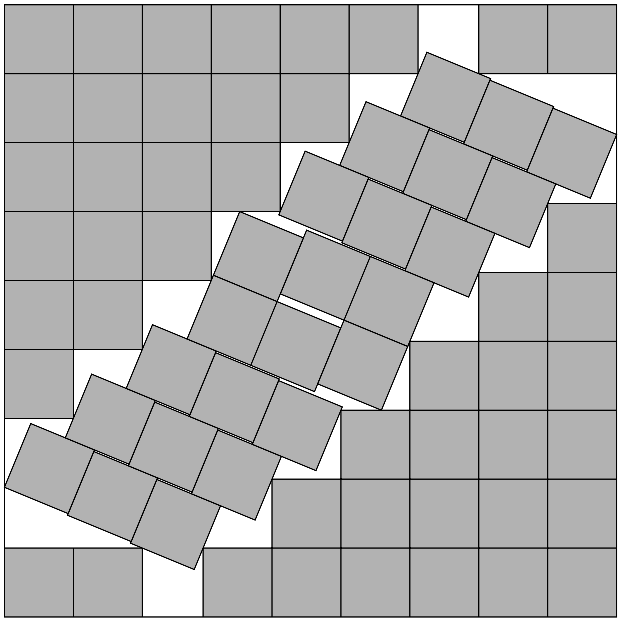
Improved by Ruilin Wang in September 2026, using ChatGPT-6 Astra.
Optimized by David Ellsworth in September 2026.
Improved by Jake Loyd in September 2026, using a L-BFGS program he wrote, starting from randomness.
Not yet analytically optimized.

69



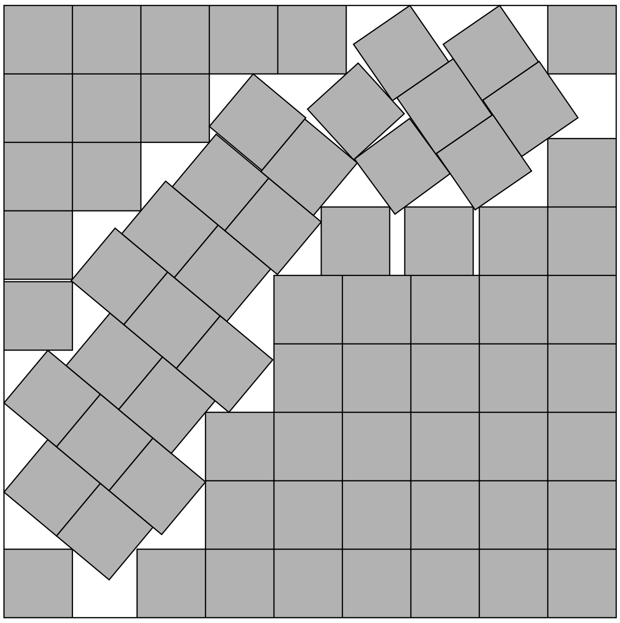
$s = {}^{38}$ 🔒 $= 8.82719465572973\ldots$
Found by Maurizio Morandi in June 2010.
Improved by David W. Cantrell in August 2023.
Improved by hmbelvedere in July 2026, using an undisclosed iterative method (probably with AI).
Refound by David Ellsworth in September 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from $s(69)$ found by Maurizio Morandi in June 2010.
Optimized by David Ellsworth in September 2026.

70



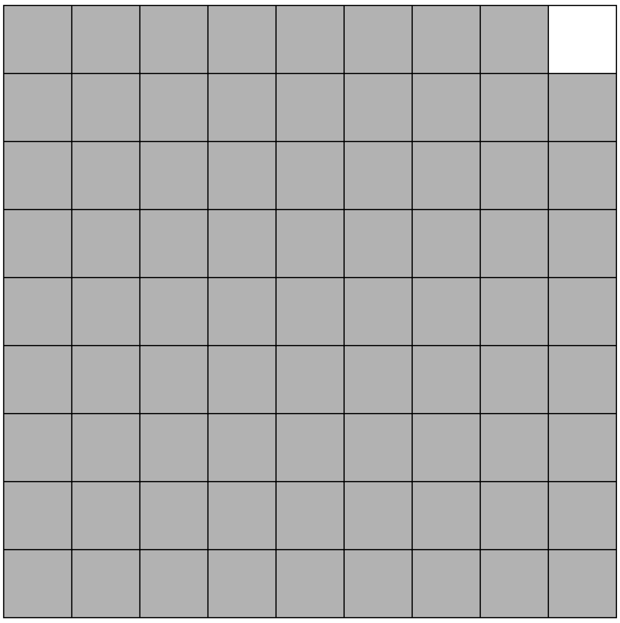
$s = {}^{4}$ 🔒 $= 8.88166675700900\ldots$
Found by Joe DeVincentis in April 2014.

71



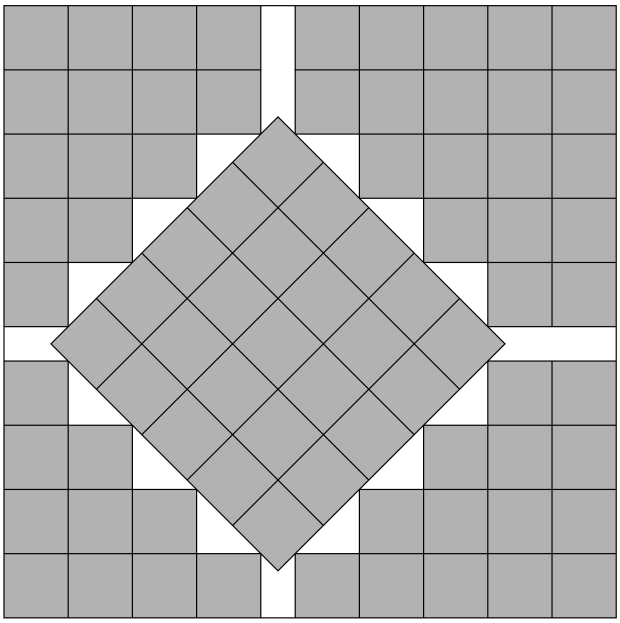
$s = 8.94407155757031\ldots$
Found by Thomas Schadt in December 2025, using a simulated annealing program he wrote, improving upon the $s(71)$ found by David W. Cantrell in October 2005.
Optimized by David Ellsworth in January 2026.

79, 80



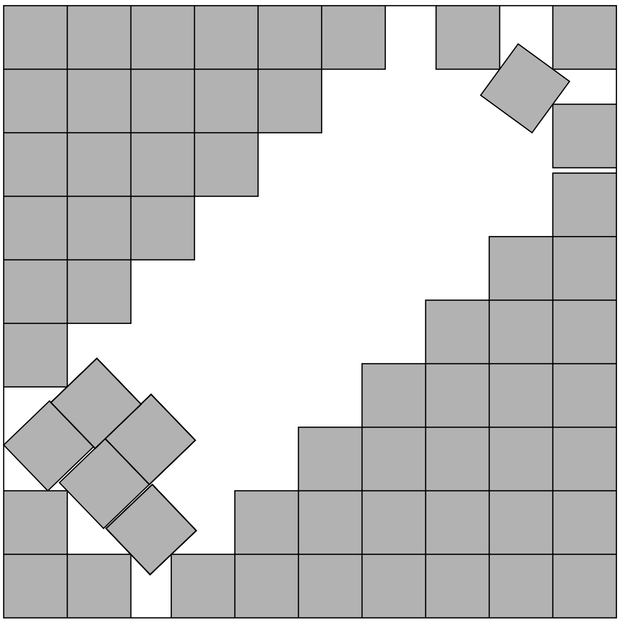
$s = 9$
Proved by Hiroshi Nagamochi in 2005.

82



$s = 6 + \frac{5}{2}\sqrt{2} = 9.53553390593273\ldots$
Found by Frits Göbel in early 1979.
Adds two "L"s to $s(65)$.

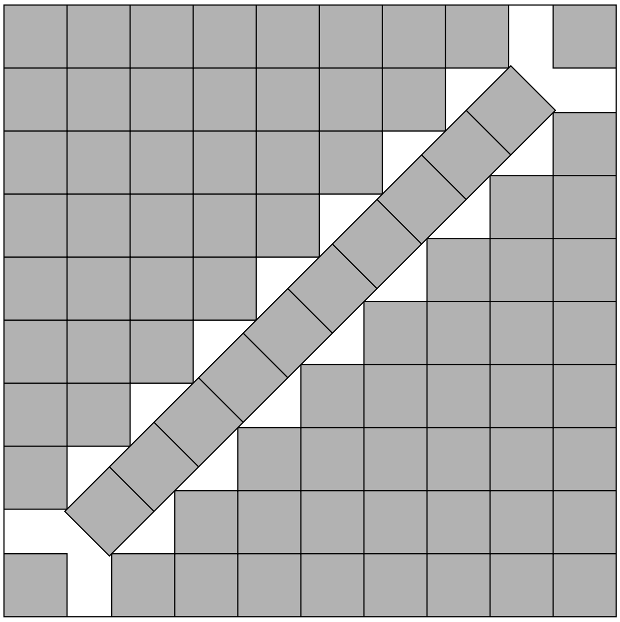
83



$s = {}^{672}$ 🔒 $= 9.63475764863108\ldots$
Found by Károly Hajba in September 2024.
Improved upon the $s(83)$ found by Evert Stenlund in early 1980.
Improved by David W. Cantrell in November 2024. Extends the $s(17)$ found by John Bidwell in 1998.
Improved by Allen Chang

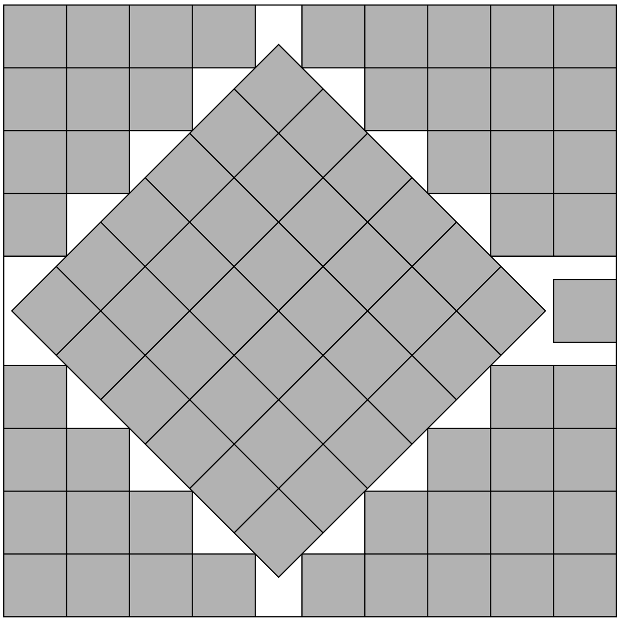
in September 2026, with GPT-5.6 Sol and GPT-6 Astra.
Optimized by David Ellsworth
in September 2026.
Polynomial root solution found by Allen Chang in September 2026 (see SVG source).

84



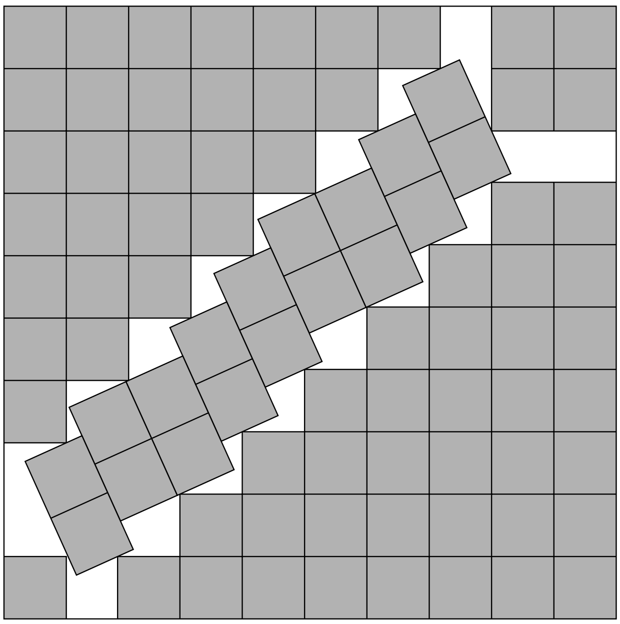
$s = 9 + \frac{1}{2}\sqrt{2} = 9.70710678118654\ldots$
Found by Evert Stenlund
in early 1980, extending the $s(52)$
found by Frits Göbel in early 1979.
Explore group

85



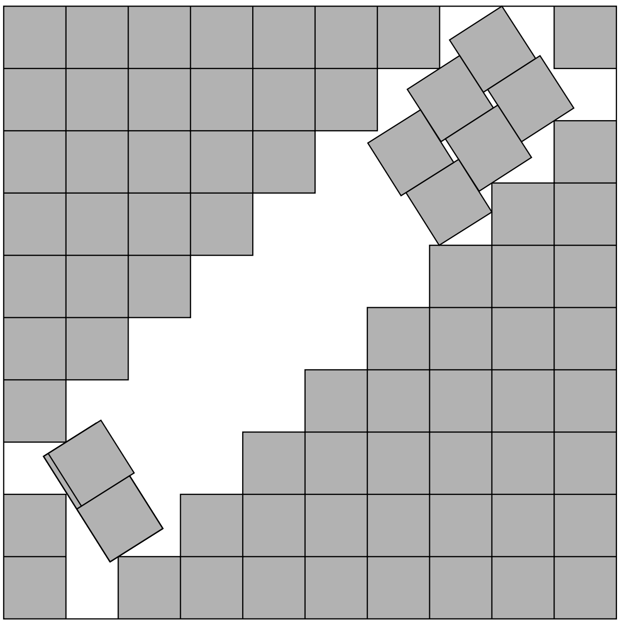
$s = \frac{11}{2} + 3\sqrt{2} = 9.74264068711928\ldots$
Found by Erich Friedman
in 1997.

86



$s = \frac{17}{2} + \frac{1}{2}\sqrt{7} = 9.82287565553229\ldots$
Found by Erich Friedman
in 1997.

87

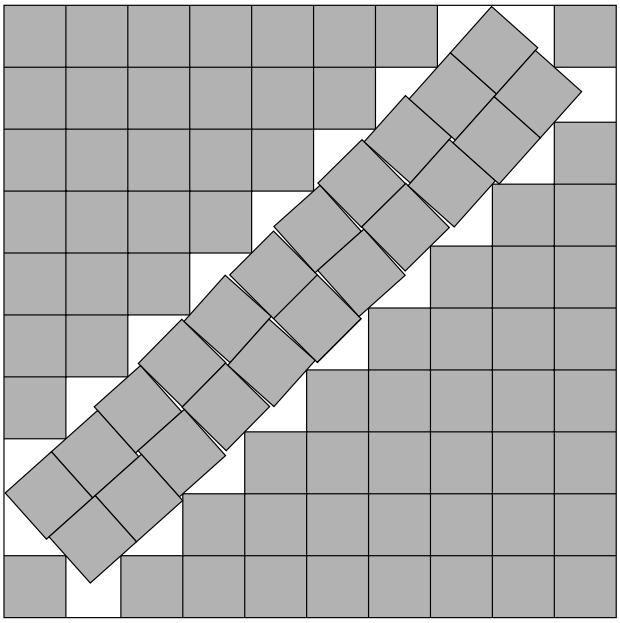


$s =$ 41 🔒 $= 9.83881526994826\ldots$
Found by David Ellsworth in
December 2024, based on the $s(107)$

Extends the alternative packing of the $s(18)$ found by Pertti Hämäläinen in 1980 found by Mats Gustafsson in 1981.

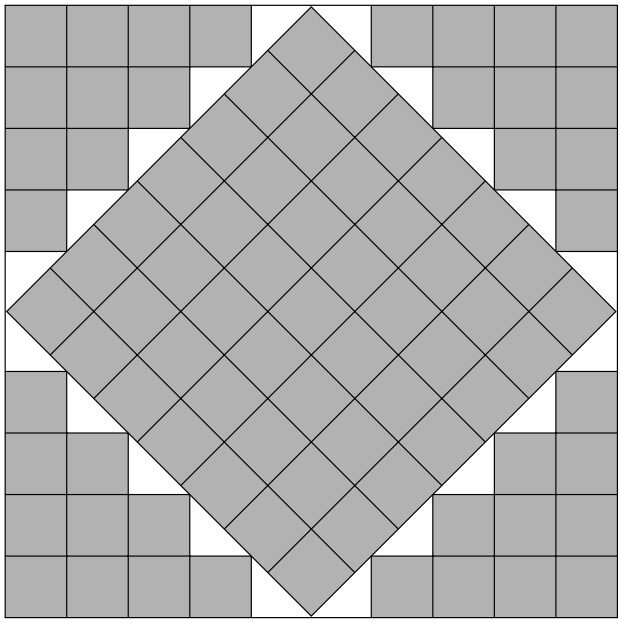
found by Károly Hajba in November 2024 and the $s(54)$ found by Joe DeVincentis in April 2014.
Improved by David W. Cantrell in January 2025.
Improved by David Ellsworth in February 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(87)$ found by David Ellsworth in December 2024.
Optimized by David Ellsworth in February 2026.
Improved and optimized by Allen Chang in September 2026, working with GPT-6 Astra, with help from "TheMagicAnimals".

88



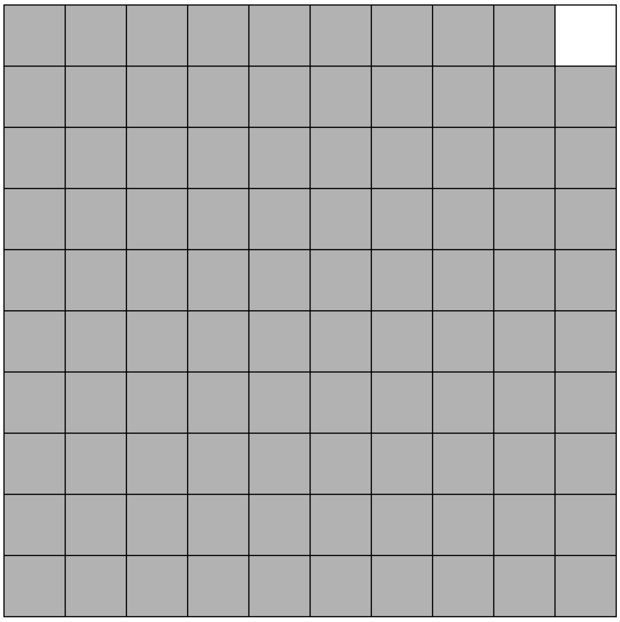
$s = {}^{20}\!\!🔒 = 9.88815305375857\ldots$
Found by David Ellsworth in November 2024, by adapting and extending the $s(37)$ improvement found by David W. Cantrell in September 2002.
Improves upon the $s(88)$ found by Erich Friedman in 1997, which extended the $s(41)$ found by Charles F. Cottingham in 1979.
Improved independently by both David W. Cantrell and David Ellsworth in January 2025.
This new technique, which will need to be applied to about 13 additional packings previously thought to be finished, independently found by both David W. Cantrell and David Ellsworth.
Improvement by Thomas Schadt pending.

89



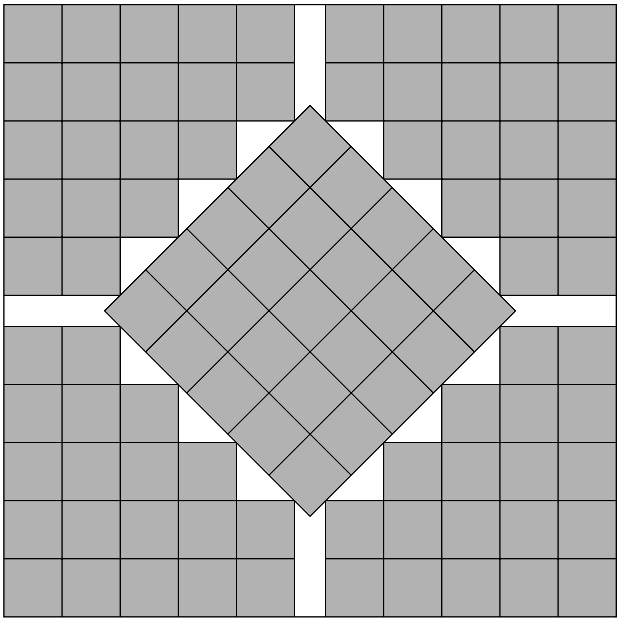
$s = 5 + \frac{7}{2}\sqrt{2} = 9.94974746830583\ldots$
Found by Evert Stenlund in early 1980, by continuing a pattern found by Frits Göbel in early 1979.
Explore group

98, 99



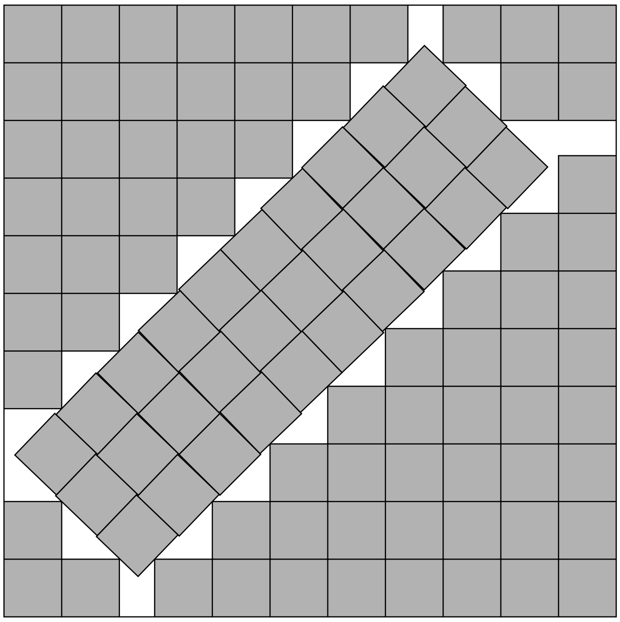
$s = 10$
Proved by Hiroshi Nagamochi
in 2005.

101



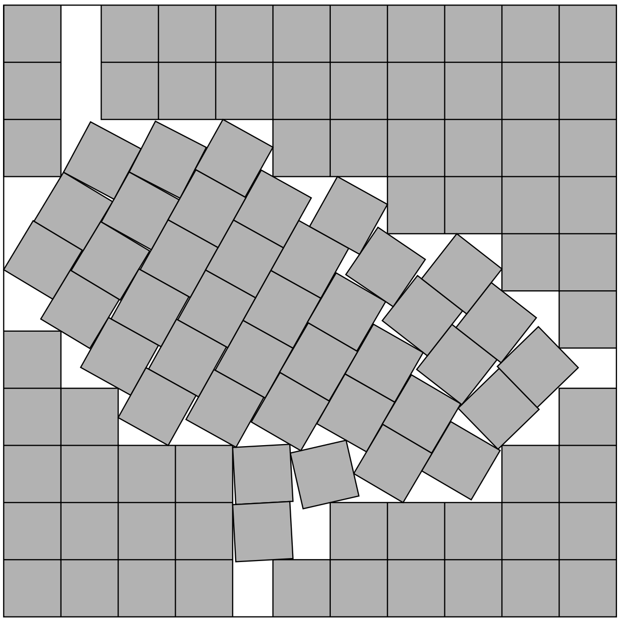
$s = 7 + \frac{5}{2}\sqrt{2} = 10.53553390593273\ldots$
Adds two "L"s to the $s(65)$ found
by Frits Göbel in early 1979.

102



$s =$ ^{8}🔒 $= 10.61138794077522\ldots$
Found by Károly Hajba
in September 2024.
Extended the $s(37)$ found by
Evert Stenlund in early 1980.
Improved by David W. Cantrell and
David Ellsworth in November 2024,
by extending the $s(37)$ found by
David W. Cantrell in September 2002.
Improved by David Ellsworth in

103

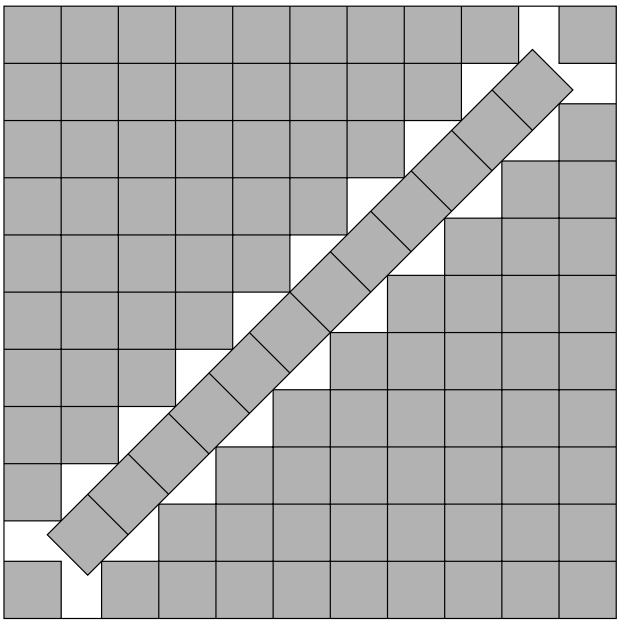


$s = 10.70378195534367\ldots$
Found by Thomas Schadt
in December 2025, using a simulated
annealing program he wrote, starting from
randomness.
Refound and refined by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program, starting from
randomness.
The third record-setting packing found

December 2024.
Improved by David W. Cantrell
in January 2025.
This new technique, which will need to be
applied to about 13 additional packings
previously thought to be finished,
independently found by both
David W. Cantrell and David Ellsworth in
January 2025.

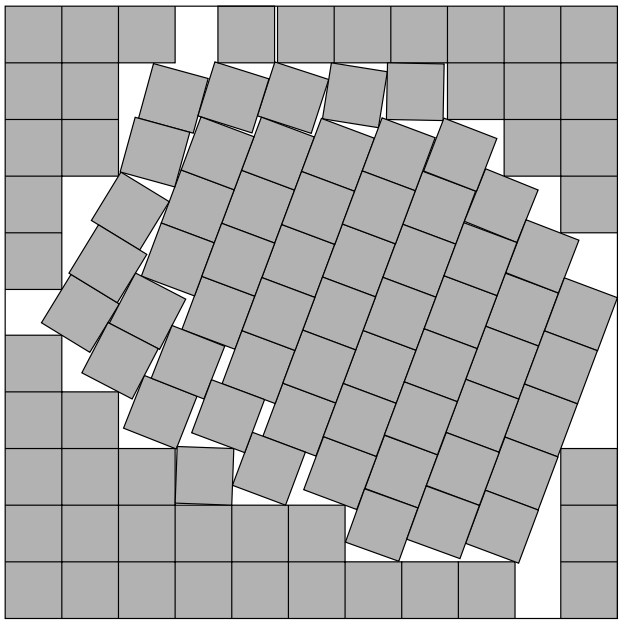
that is doubly semi-primitive.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

104



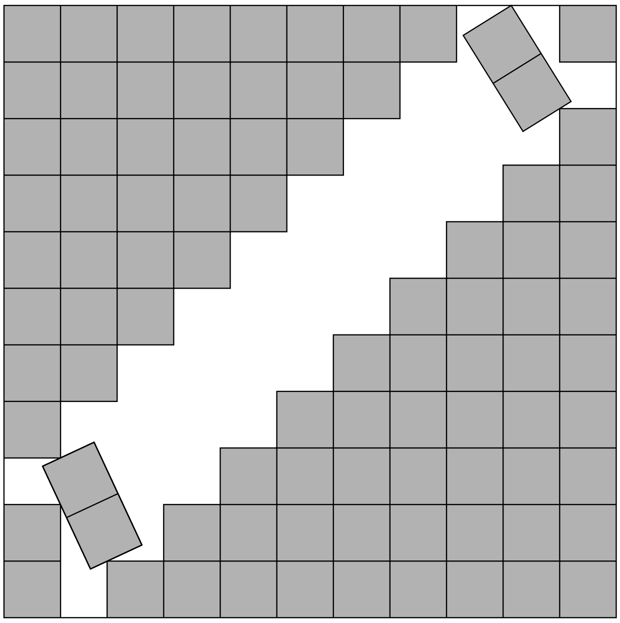
$s = 10 + \frac{1}{2}\sqrt{2} = 10.70710678118654\ldots$
Extends the $s(52)$ found by
Frits Göbel in early 1979.
Explore group

105



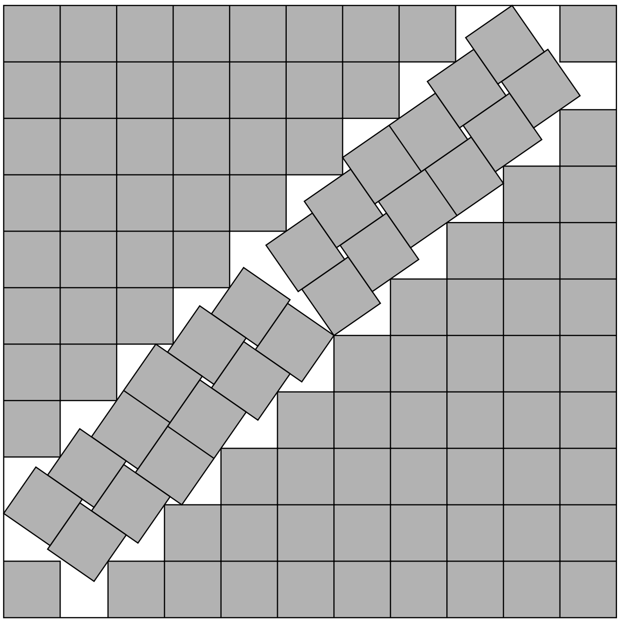
$s = 10.80761933330707\ldots$
Found by Thomas Schadt
in December 2025, using a simulated
annealing program he wrote, starting from
randomness.
Refound and refined by David Ellsworth
in March 2026, using his modified version
#3 of Thomas Schadt's simulated
annealing program, starting from
randomness.
Not yet analytically optimized.

106



$s = {}^{32}$🔒 $= 10.82297973416944\ldots$
Found by David Ellsworth
in November 2024, based on the
$s(53)$ found by David W. Cantrell in
September 2002.
Improved by David W. Cantrell
in December 2024.

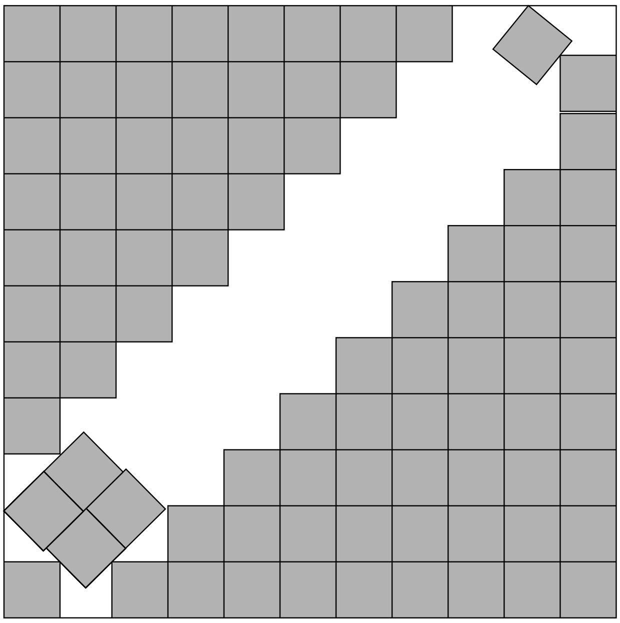
107



$$s = 10 - \frac{1}{2}\sqrt{2} + \sqrt{1 + \sqrt{2}}$$
$$= 10.8466671928434\ldots$$

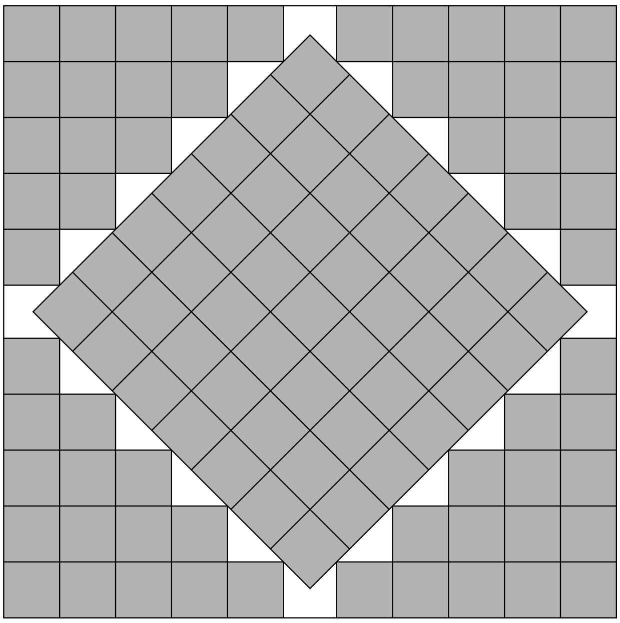
Found by Károly Hajba
in November 2024.
Is an alternative packing for an extension of the
$s(54)$ found by Joe DeVincentis in April 2014.

108



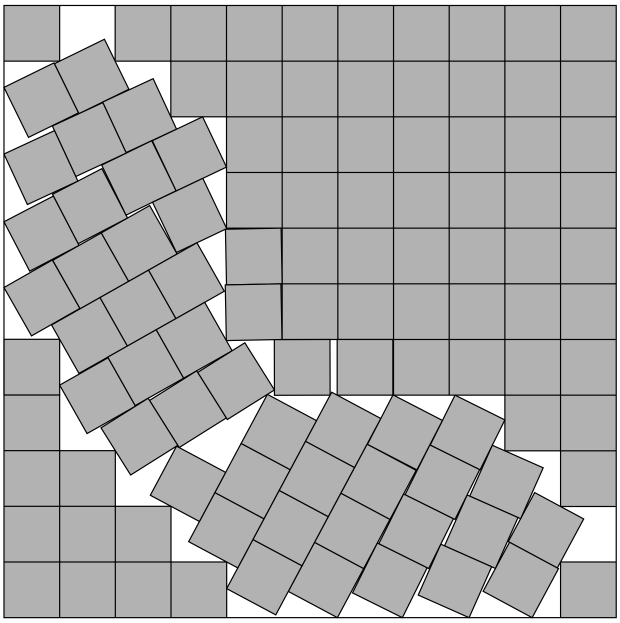
$s = {}^{144}$🔒 $= 10.92591939016138\ldots$
Found by Károly Hajba
in October 2024.
Improved by David Ellsworth
in November 2024.
Extends the $s(11)$ found by Walter Trump
in 1979.

109



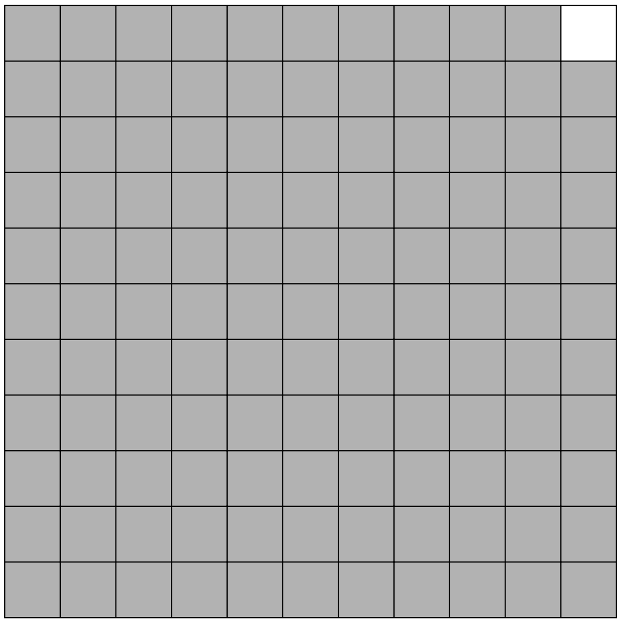
$s = 6 + \frac{7}{2}\sqrt{2} = 10.94974746830583\ldots$
Continues a pattern found
by Frits Göbel in early 1979.
Explore group

110



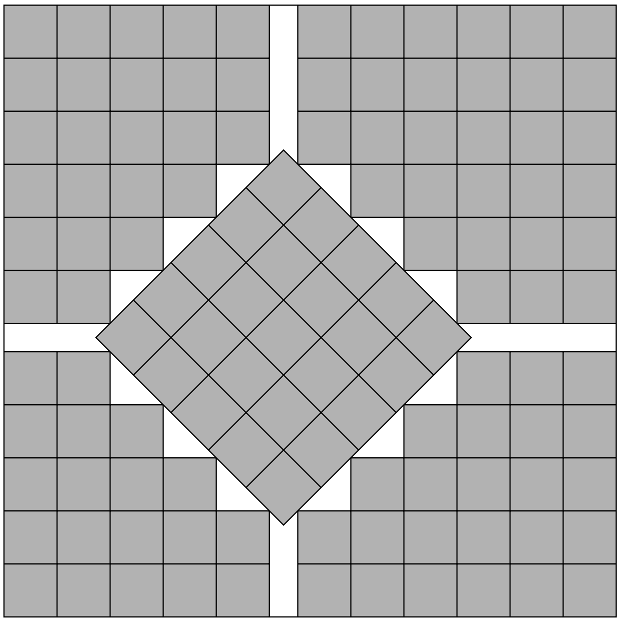
$s = 10.99679327401957\ldots$
Found by David W. Cantrell
in February 2025.
Bounds the $s(n^2 - n) = n$ conjecture to
$n < 11$.
Improved by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

119, 120



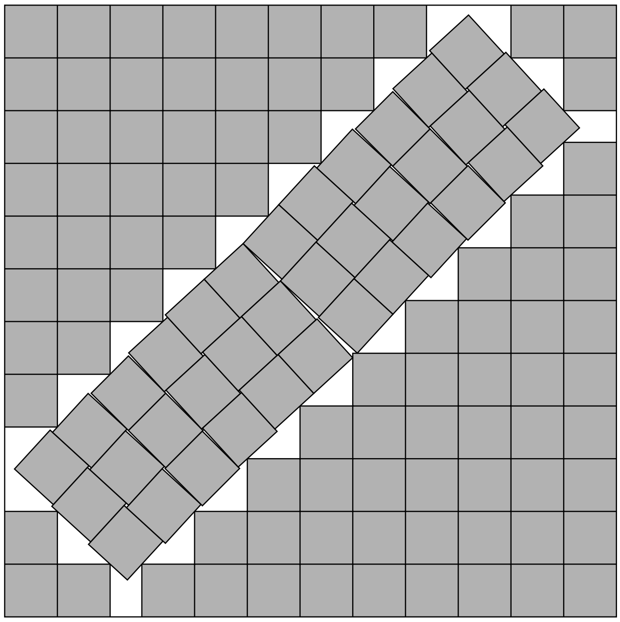
$s = 11$
Proved by Hiroshi Nagamochi
in 2005.

122



$s = 8 + \frac{5}{2}\sqrt{2} = 11.53553390593273\ldots$
Adds three "L"s to the $s(65)$ found
by Frits Göbel in early 1979.

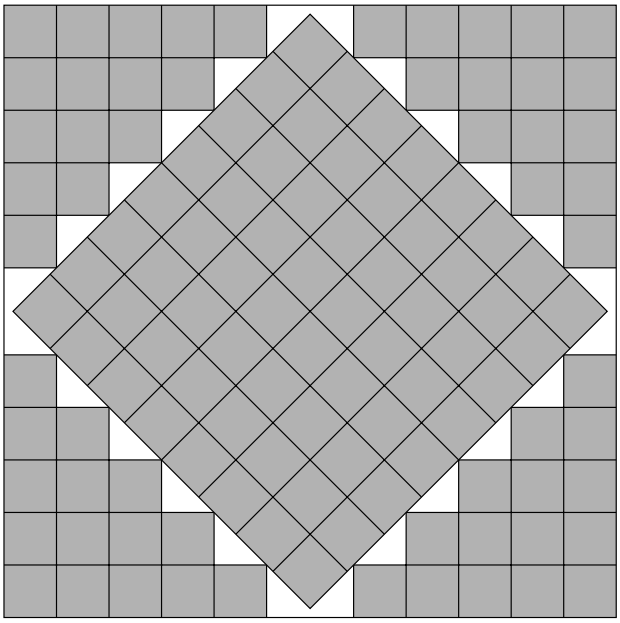
123



$s = {}^{12}$🔒 $= 11.60139979378801\ldots$
Found and improved by David Ellsworth
in December 2024, by extending the
$s(102)$ found by Károly Hajba in
September 2024 and adapting and

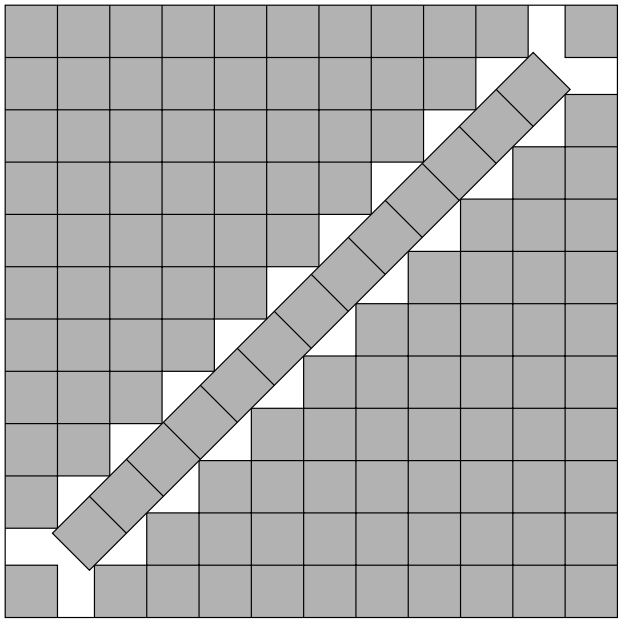
extending the $s(37)$ improvement found
by David W. Cantrell in September 2002.
Improved by David W. Cantrell
in January 2025.
This new technique, which will need to be
applied to about 13 additional packings
previously thought to be finished,
independently found by both
David W. Cantrell and David Ellsworth in
January 2025.

124



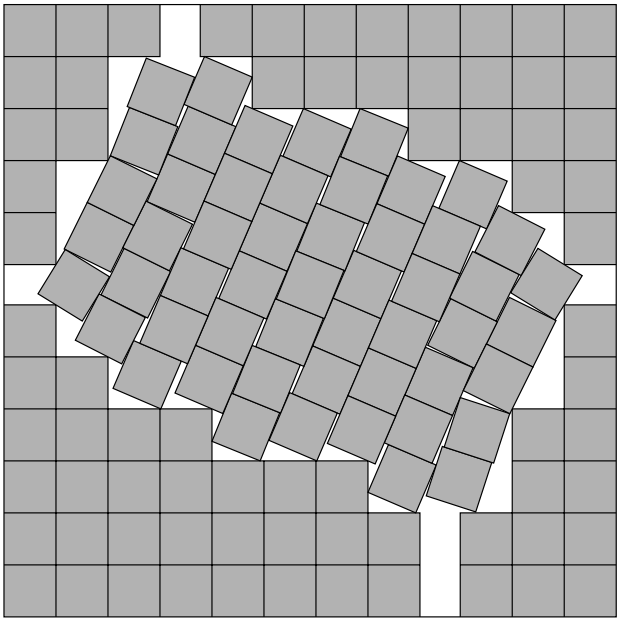
$s = 6 + 4\sqrt{2} = 11.65685424949238\ldots$
Continues a pattern found
by Frits Göbel in early 1979.
Explore group

125

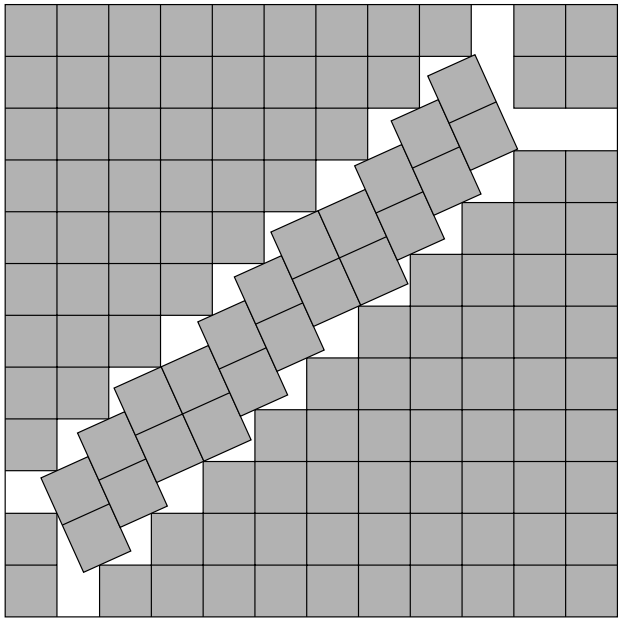


$s = 11 + \frac{1}{2}\sqrt{2} = 11.70710678118654\ldots$
Extends the $s(52)$ found by
Frits Göbel in early 1979.
Explore group

126



127



$s = 11.77473513240654\ldots$

Found by Joost de Winter in August 2026, working with unspecified AI, using an evolutionary beam search with simulated annealing, starting from the latest $s(105)$ as of December 2025.

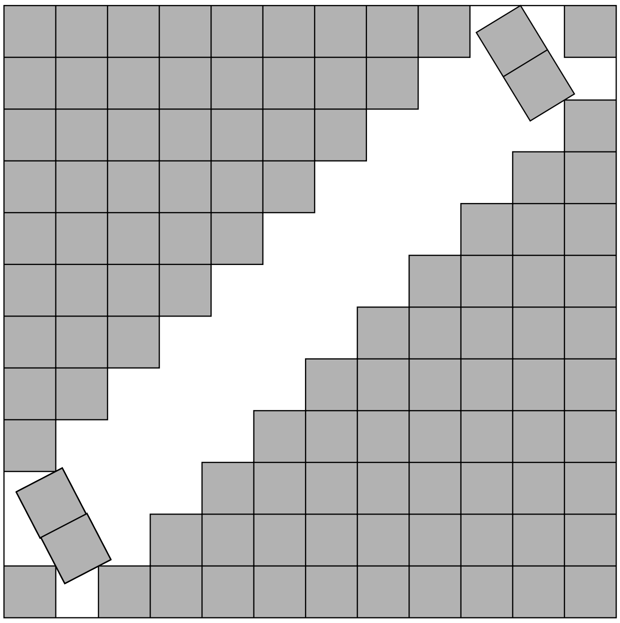
The fourth record-setting packing found that is doubly semi-primitive.

Not yet analytically optimized.

$s = \frac{21}{2} + \frac{1}{2}\sqrt{7} = 11.82287565553229\ldots$

Extends the $s(86)$ found by Erich Friedman in 1997.

128



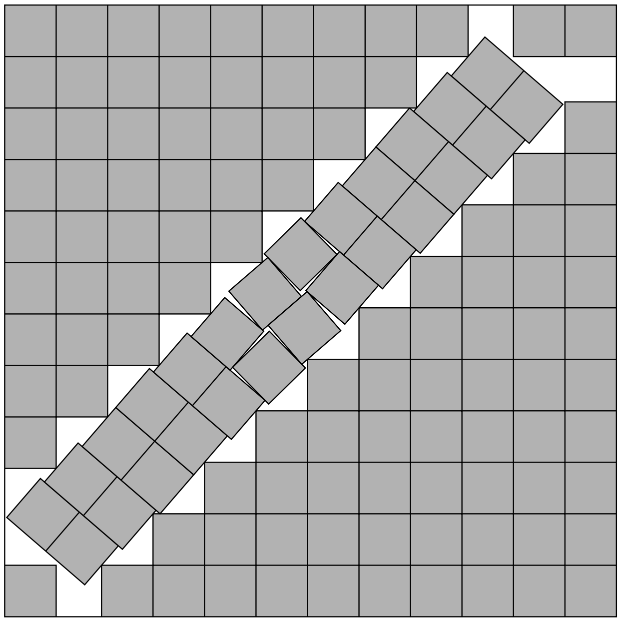
$s =$ 40 🔒 $= 11.82509196821368\ldots$

Found by David Ellsworth in November 2024, based on the $s(69)$ found by Maurizio Morandi in June 2010.

Improved by David Ellsworth in December 2024, including using the technique from the $s(53)$ improved by David W. Cantrell in December 2024.

Improved by David W. Cantrell and David Ellsworth in January 2025.

129



$s =$ 20 🔒 $= 11.88130621809000\ldots$

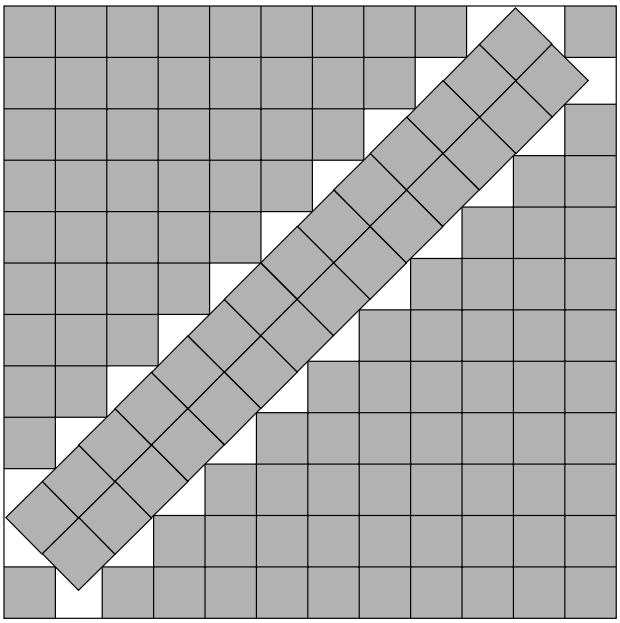
Found and improved by David Ellsworth in December 2024, adapting/extending the $s(37)$ improvement found by David W. Cantrell in September 2002.

Unimproved form similar to the $s(70)$ found by Erich Friedman in 1997.

Improved by David Ellsworth in December 2025, using his modified version of Thomas Schadt's simulated annealing program.

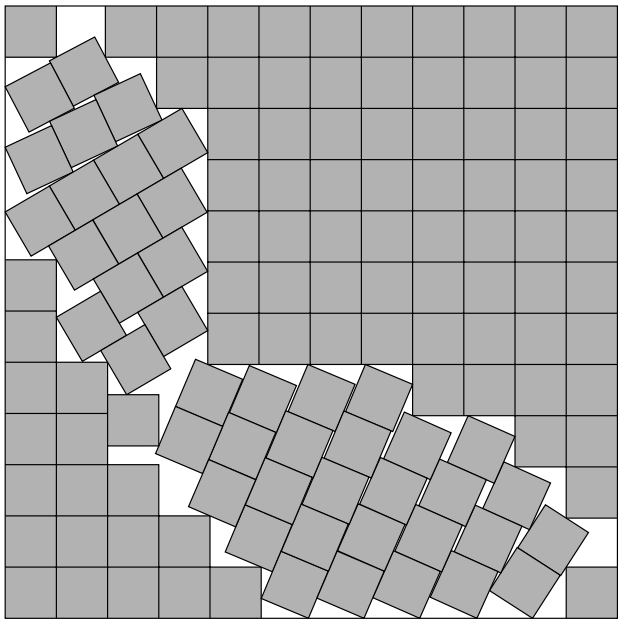
Optimized by David Ellsworth in January 2026.

130



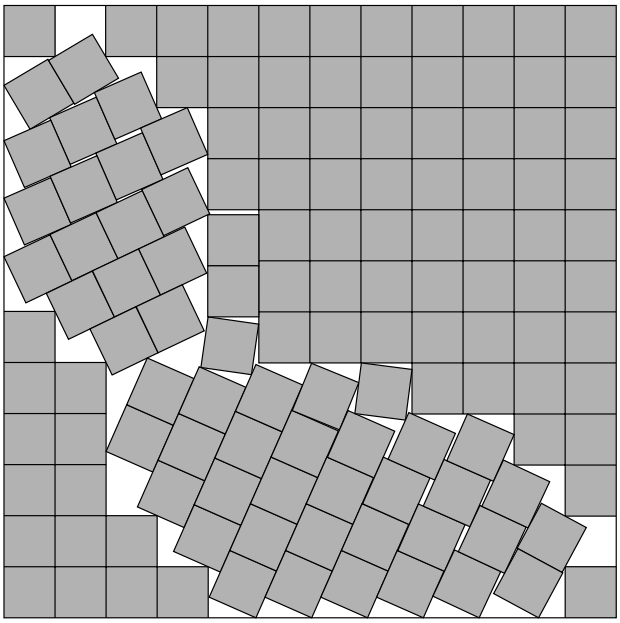
$s = {}^{8}\!🔒 = 11.91119052015898\ldots$
Found by David Ellsworth
in November 2024.
Improved by David W. Cantrell
in November 2024.
Improved by David Ellsworth
in November 2024.
Extends the $s(88)$ found by Erich Friedman in 1997;
adapts and extends the $s(37)$ improvement found by
David W. Cantrell in September 2002.
Further improvement pending.

131



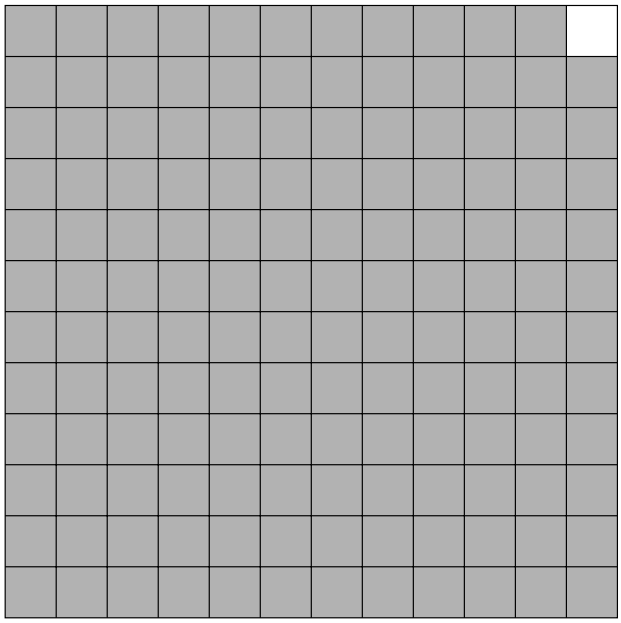
$s = 11.95652543280926\ldots$
Found by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program, starting from the
$s(131)$ found by Károly Hajba
in November 2024.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

132



$s = 11.99137344423646\ldots$
Found by M.Z. Arslanov, S.A. Mustafin,
and Z.K. Shangitbayev in March 2019.
Bounded the $s(n^2 - n) = n$ conjecture to
$n < 12$.
Improved by David W. Cantrell

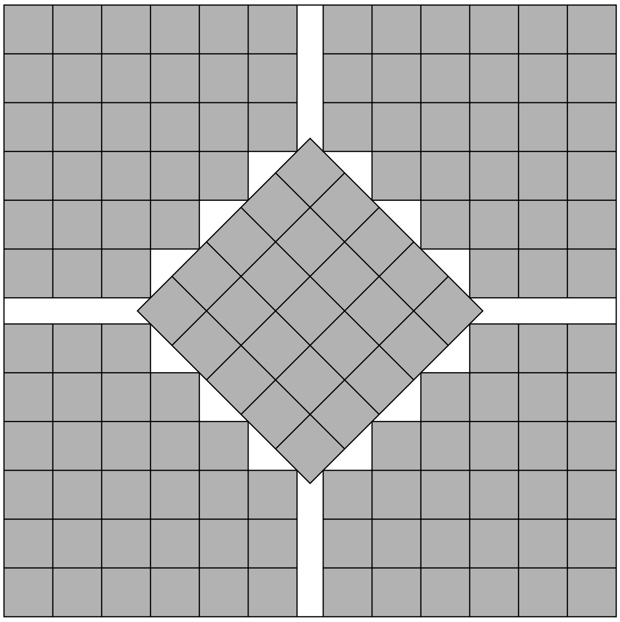
142, 143



$s = 12$
Proved by Hiroshi Nagamochi
in 2005.

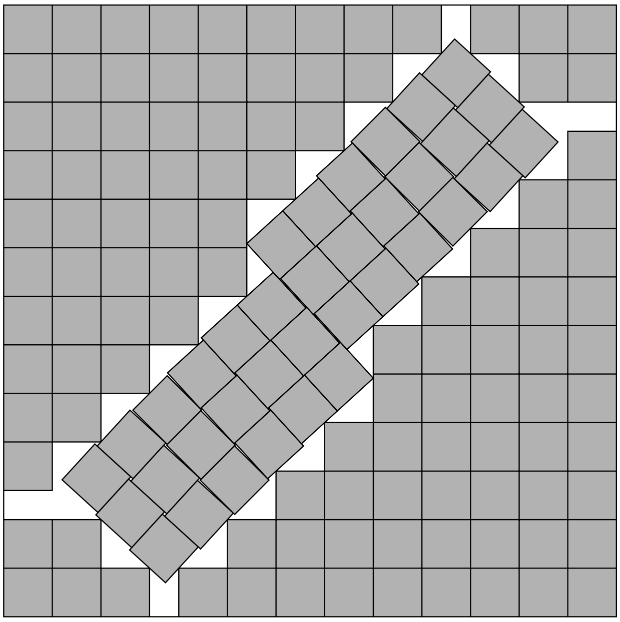
in March 2025.
Improved by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

145



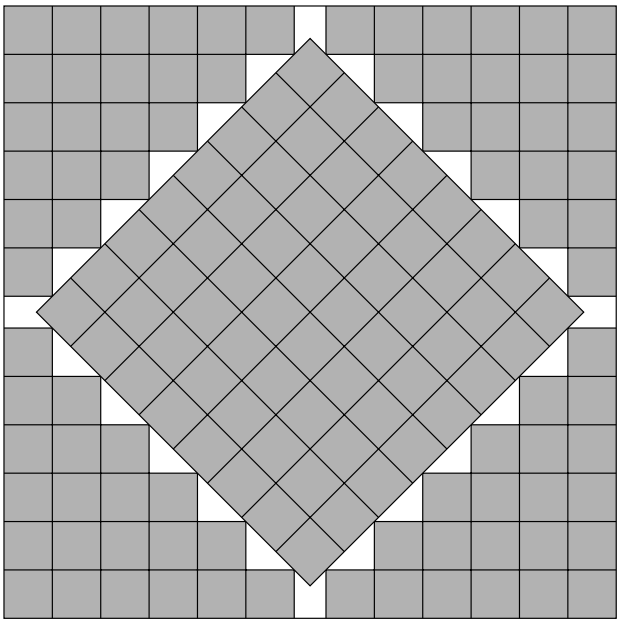
$s = 9 + \frac{5}{2}\sqrt{2} = 12.53553390593273\ldots$
Adds four "L"s to the $s(65)$ found by
Frits Göbel in early 1979.

146



$s =$ [16] 🔒 $= 12.60090777851301\ldots$
Found by David W. Cantrell
in January 2025 by switching to the
rotationally symmetric form of adding an
"L" to the $s(123)$ found and improved by
David Ellsworth in December 2024 and
improved by David W. Cantrell in January
2025 with a technique independently
found by both David W. Cantrell and
David Ellsworth.
This is the first record-setting packing
found that is doubly semi-primitive.

147, 148

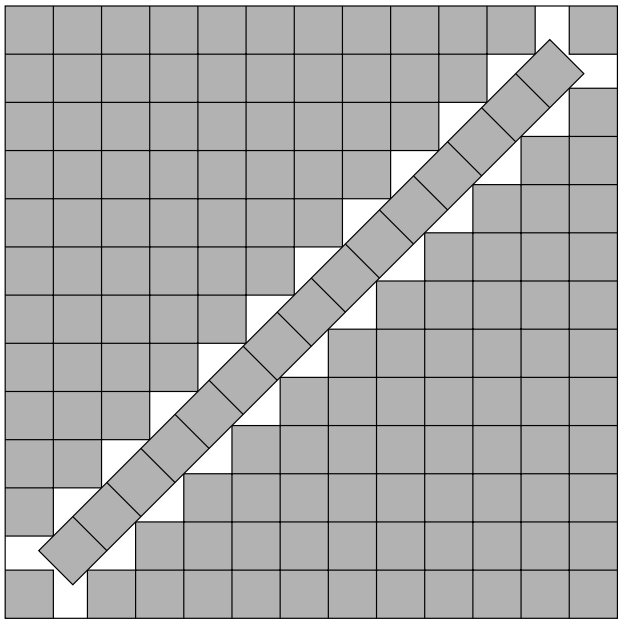


$s = 7 + 4\sqrt{2} = 12.65685424949238\ldots$

Continues a pattern found
by Frits Göbel in early 1979.

Explore group

149

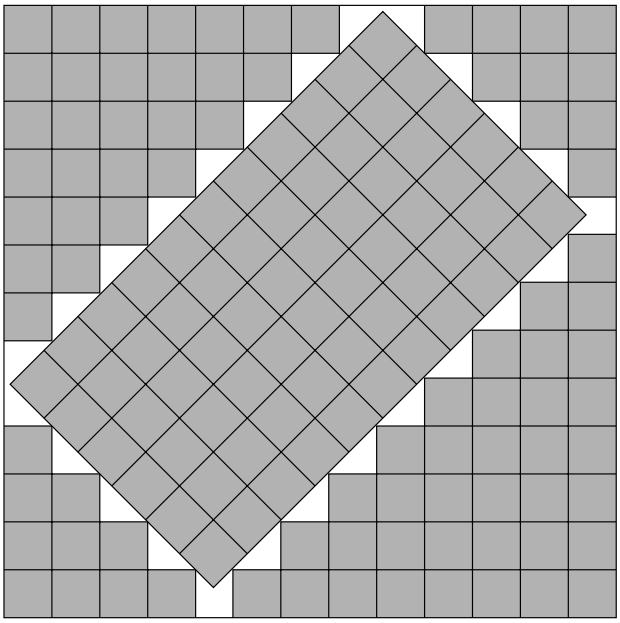


$s = 12 + \frac{1}{2}\sqrt{2} = 12.70710678118654\ldots$

Extends the $s(52)$ found by
Frits Göbel in early 1979.

Explore group

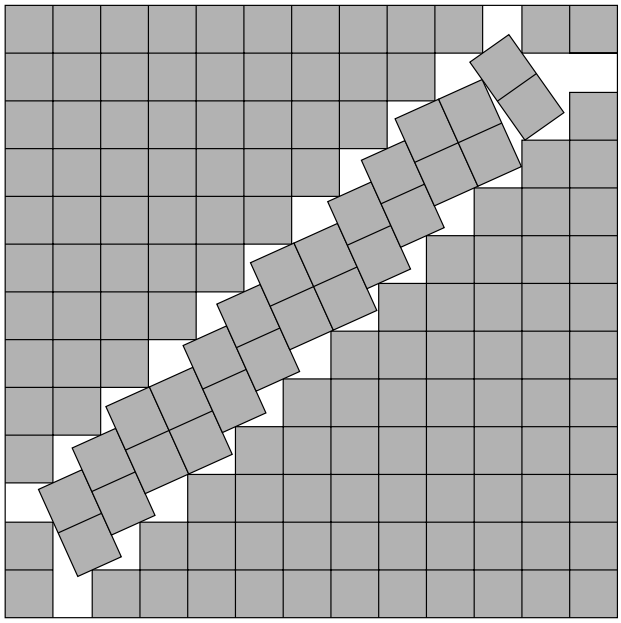
150



$s = 5 + \frac{11}{2}\sqrt{2} = 12.77817459305202\ldots$

Found by David Ellsworth
in January 2025.
Extends the $s(85)$ found by
Erich Friedman in 1997 how the $s(66)$
found by Evert Stenlund in early 1980
extends $s(26)$.

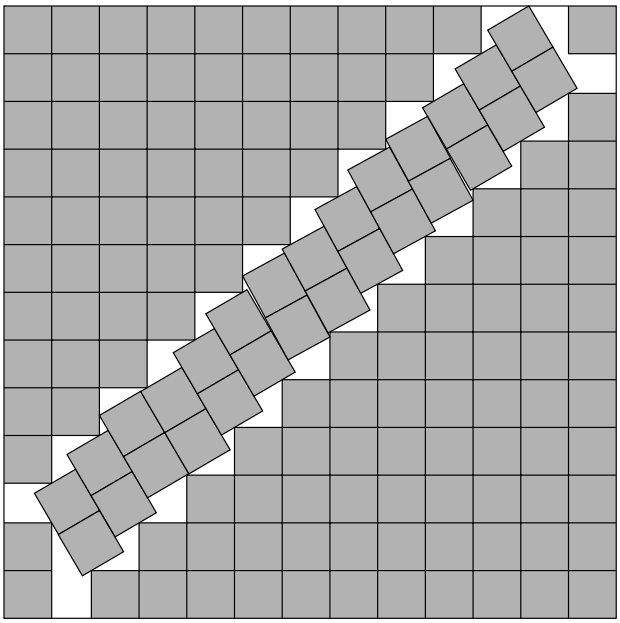
151



$s = \frac{23}{2} + \frac{1}{2}\sqrt{7} = 12.82287565553229\ldots$

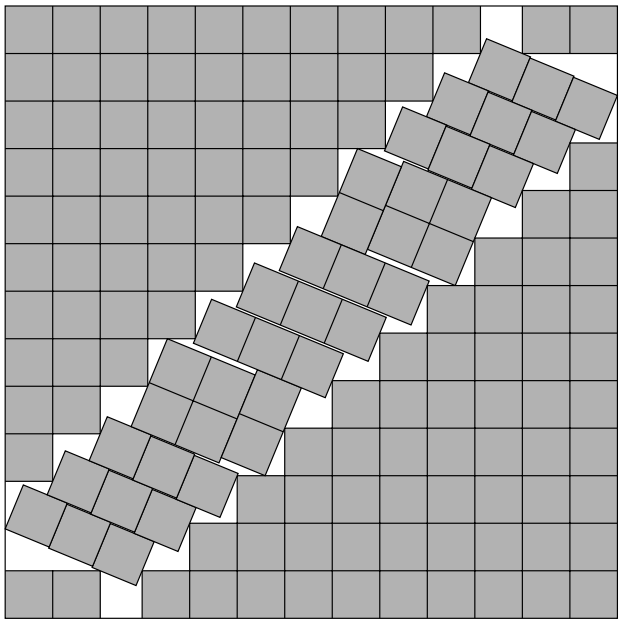
Found by David Ellsworth
in November 2024.
Improved by David W. Cantrell
in December 2024.

152



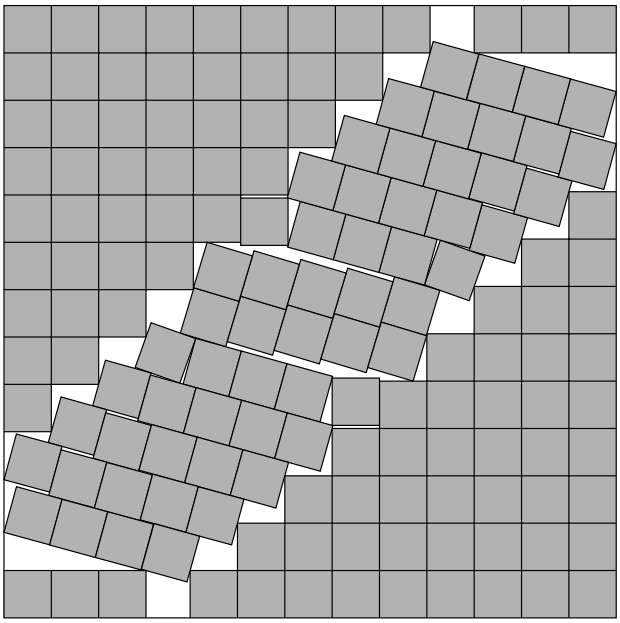
$s = 12.83095954472600\ldots$
Found by David Ellsworth and David W. Cantrell in January 2025, based on the $s(53)$ found by David W. Cantrell in September 2002 and the $s(69)$ found by Maurizio Morandi in June 2010, adapting and extending the $s(69)$ improvement found by David W. Cantrell in August 2023.
Improved by Grigoriy Dyachkov in September 2026, working with Claude Opus 5 and Claude Fable 5, applying a similar improvement as that done to $s(69)$ in July 2026.
Not yet analytically optimized.

153



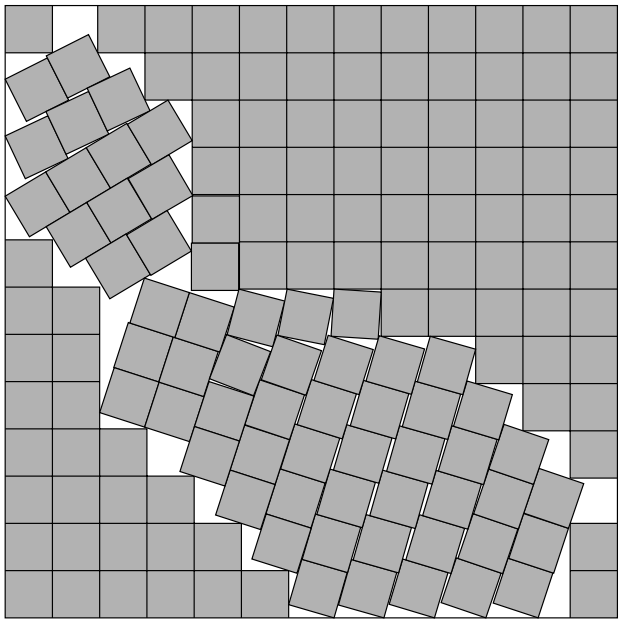
$s = {}^{4}\!\!\text{🔒} = 12.88166675700900\ldots$
Found by David Ellsworth in November 2024, based on the $s(70)$ found by Joe DeVincentis in April 2014.

154



$s = 12.93166712962655\ldots$
Originally found by David Ellsworth in

155

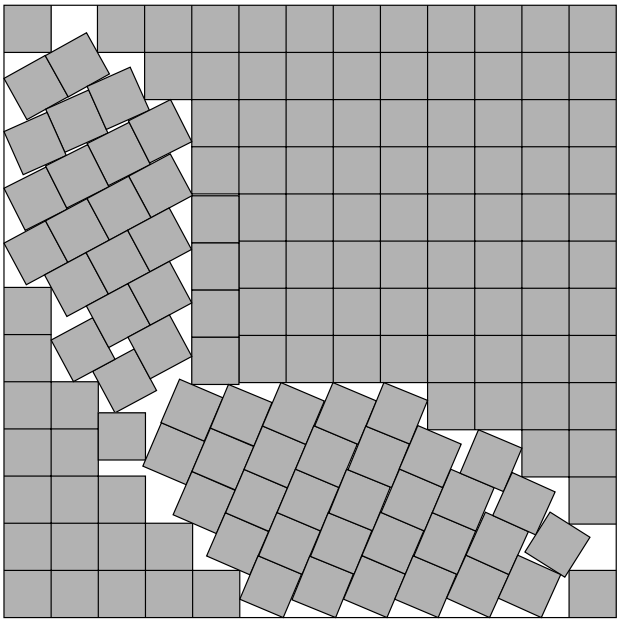


$s = 12.95844711161529\ldots$
Found by David Ellsworth

December 2024, by combining two slightly modified copies of the $s(41)$ found by Joe DeVincentis in April 2014.
Improved by David W. Cantrell in February 2025.
Improved independently by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program. Resembles the $s(179)$ found by David Ellsworth in December 2024.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

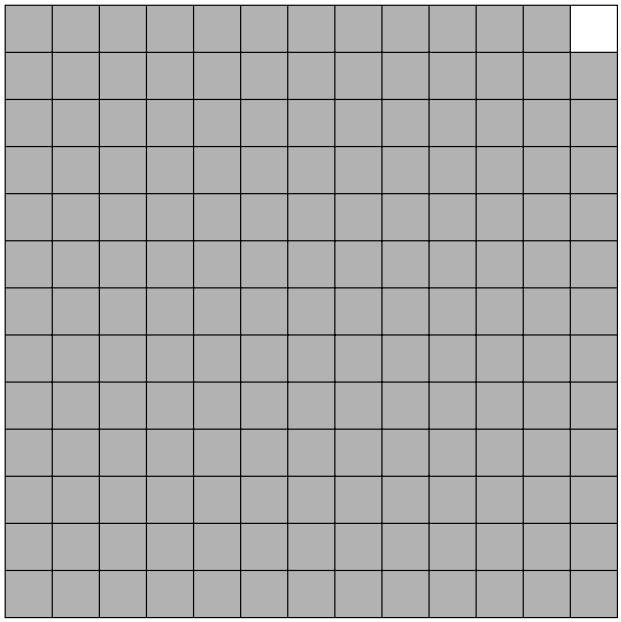
in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(182)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 27 squares removed.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

156



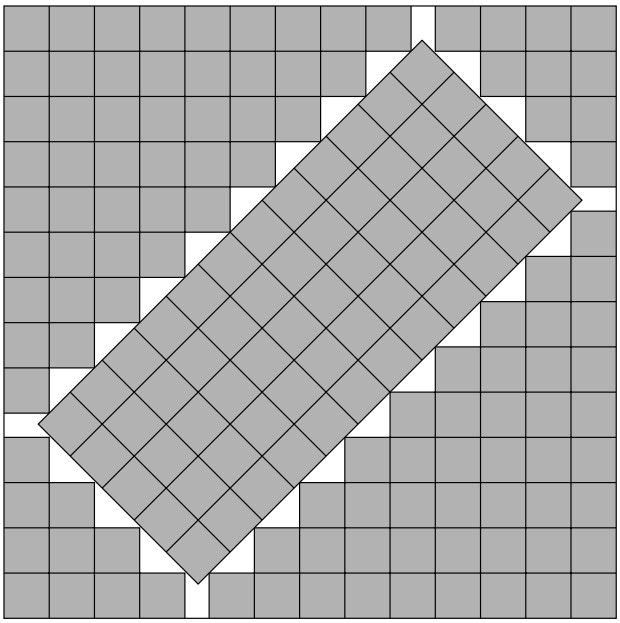
$s = 12.98208376048414\ldots$
Found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019.
Shows $s(n^2 - n) < n$ for $n = 13$.
Improved by David Ellsworth in December 2024.
Improved by David W. Cantrell in March 2025.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

167, 168



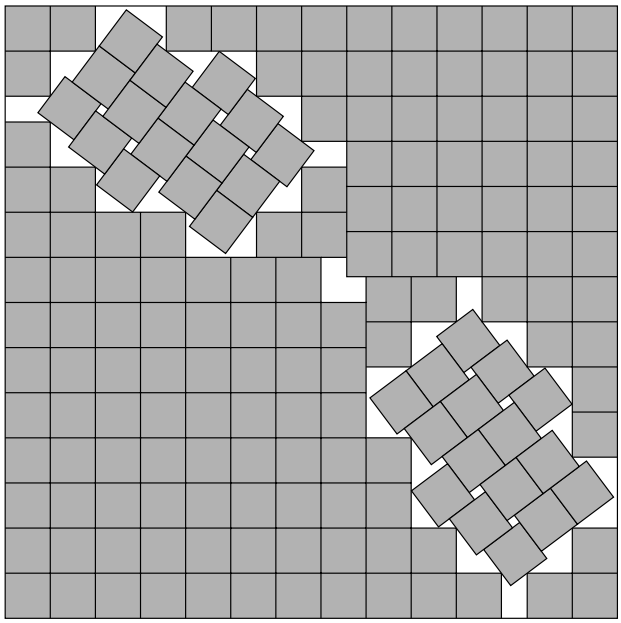
$s = 13$
Proved by Hiroshi Nagamochi in 2005.

170



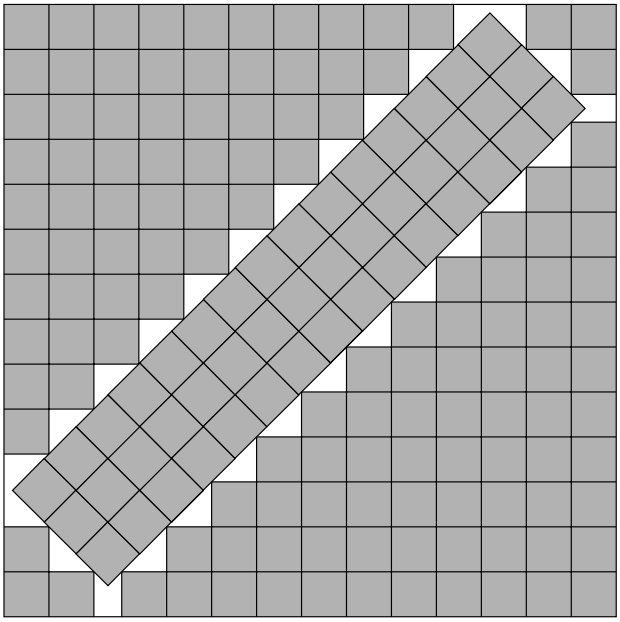
$s = 10 + \frac{5}{2}\sqrt{2} = 13.53553390593273\ldots$
Extends the $s(65)$ found by Frits Göbel in early 1979. This alternative, converting the $s(65)$ augmented by five "L"s into a primitive packing, found by Károly Hajba in November 2024.

171



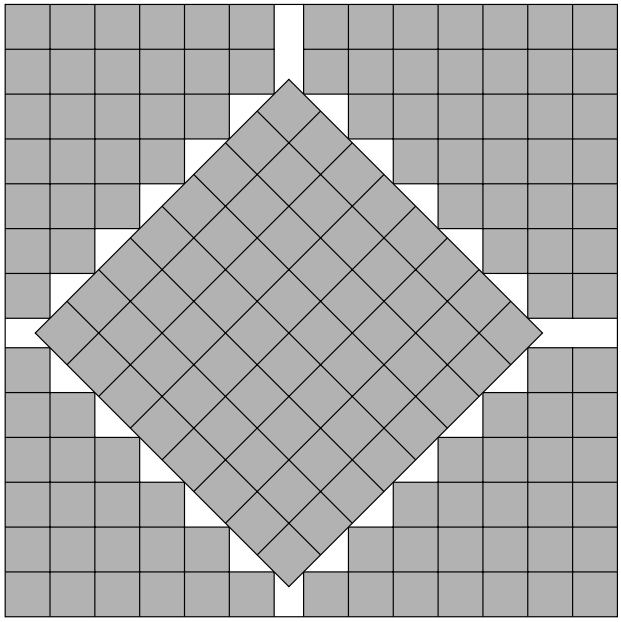
$s = 13 + \frac{4}{7} = 13.57142857142857\ldots$
Found by David Ellsworth in December 2025, by combining two copies of the $s(50)$ found by Thomas Schadt in December 2025 (using a simulated annealing program he wrote, starting from randomness) and optimized by David Ellsworth.

172



$s = {}^{8}\!\!\text{🔒} = 13.618988986601 60\ldots$
Found by Károly Hajba in November 2024, extending the $s(102)$ he found in September 2024. Improved by David Ellsworth in November 2024 by adapting the $s(37)$ improvement found by David W. Cantrell in September 2002. Improved by David Ellsworth

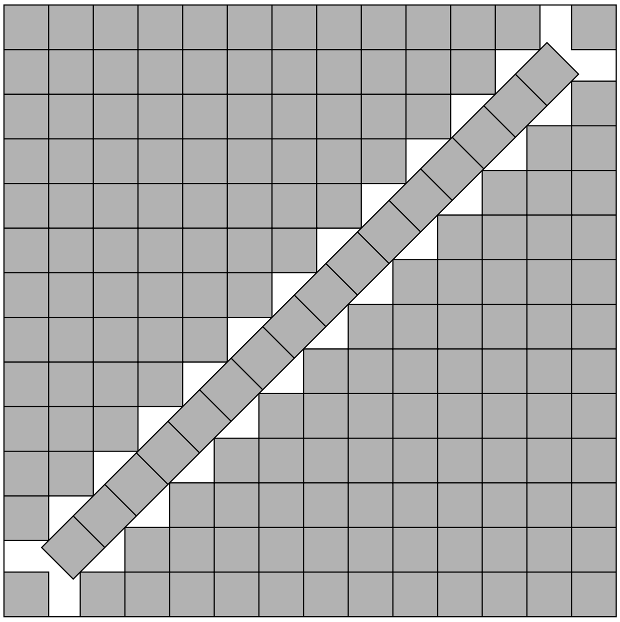
173



$s = 8 + 4\sqrt{2} = 13.65685424949238\ldots$
Adds an "L" to the $s(148)$ that continues a pattern found by Frits Göbel in early 1979.

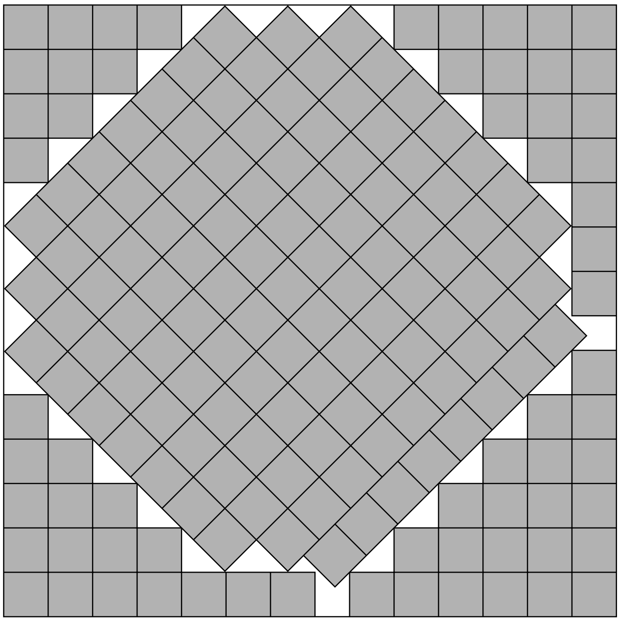
in December 2024.
Further improvement pending.

174



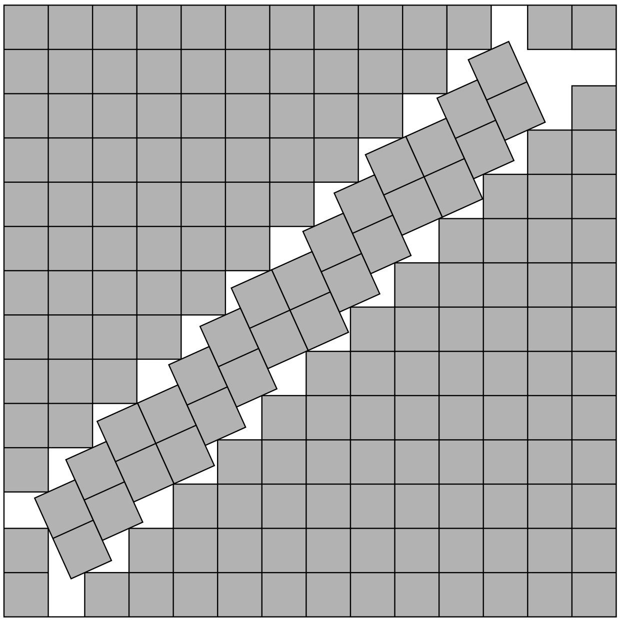
$s = 13 + \frac{1}{2}\sqrt{2} = 13.70710678118654\dots$
Extends the $s(52)$ found by
Frits Göbel in early 1979.
Explore group

175



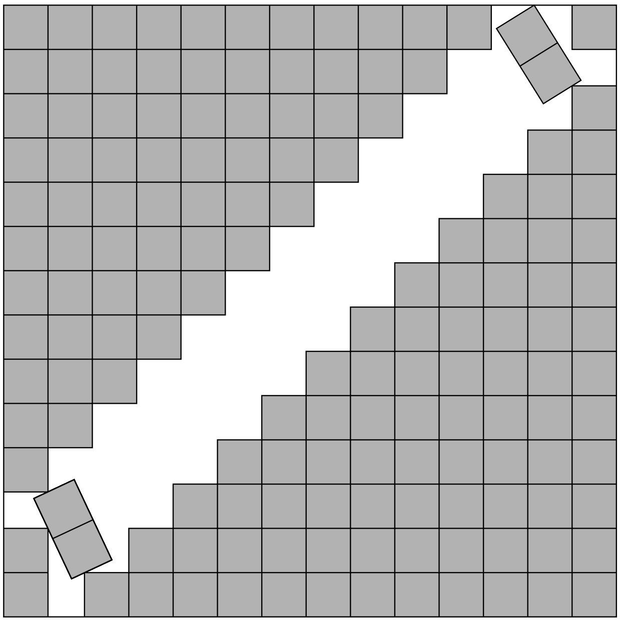
$s = 6 + \frac{11}{2}\sqrt{2} = 13.77817459305202\dots$
Found by David Ellsworth
in December 2024.
Based on the $s(233)$ that continues a
pattern found by Frits Göbel in early 1979.

176



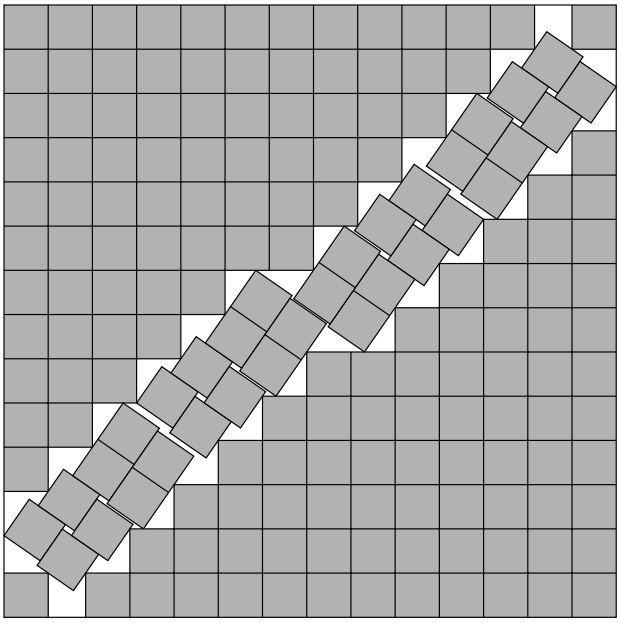
$s = \frac{25}{2} + \frac{1}{2}\sqrt{7} = 13.82287565553229\dots$
Extends the $s(86)$ found by
Erich Friedman in 1997.

177



$s = {}^{32}$ 🔒 $= 13.82302875075647\dots$
Found and improved by David Ellsworth
in November 2024 and December 2024,
based on the $s(53)$ found/improved by
David W. Cantrell in September 2002 and
December 2024, respectively.

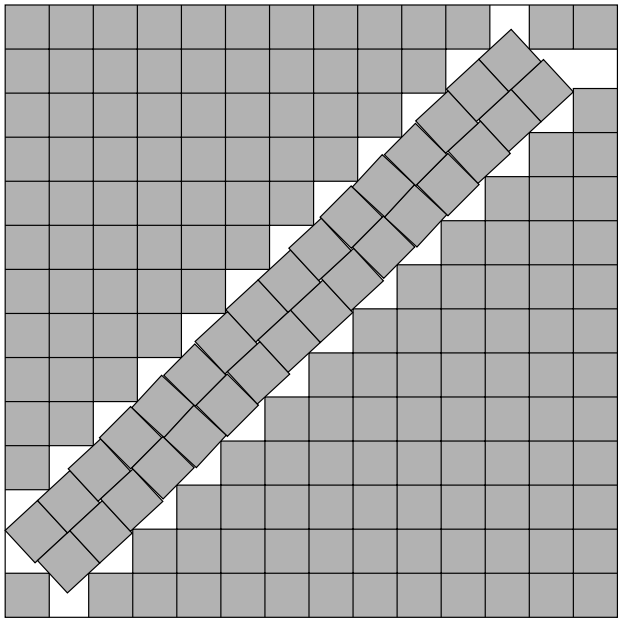
178



$$s = 13 - \frac{1}{2}\sqrt{2} + \sqrt{1 + \sqrt{2}}$$
$$= 13.84666719284348\ldots$$

Extends the $s(54)$ found by Joe DeVincentis in April 2014.

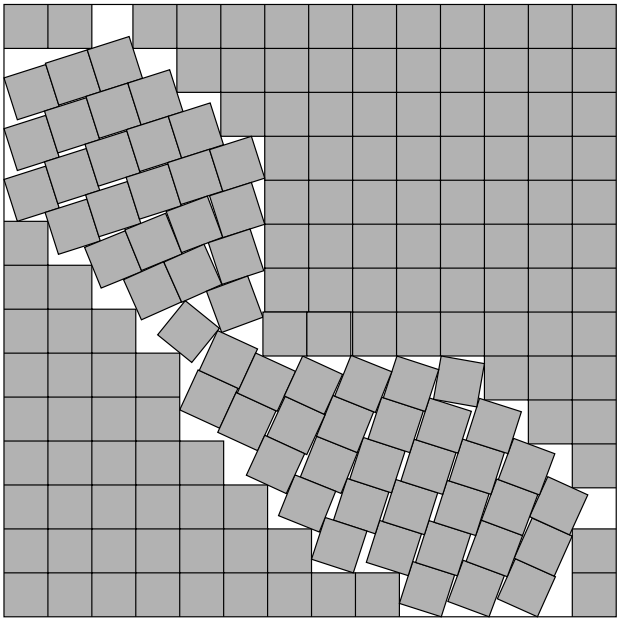
179



$s = {}^{158}\!\!\text{🔒} = 13.89534106997649\ldots$

Found by David Ellsworth in January 2025, using a computer program he wrote.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Optimized by Tej Stead in June 2026, working with Claude Fable 5.

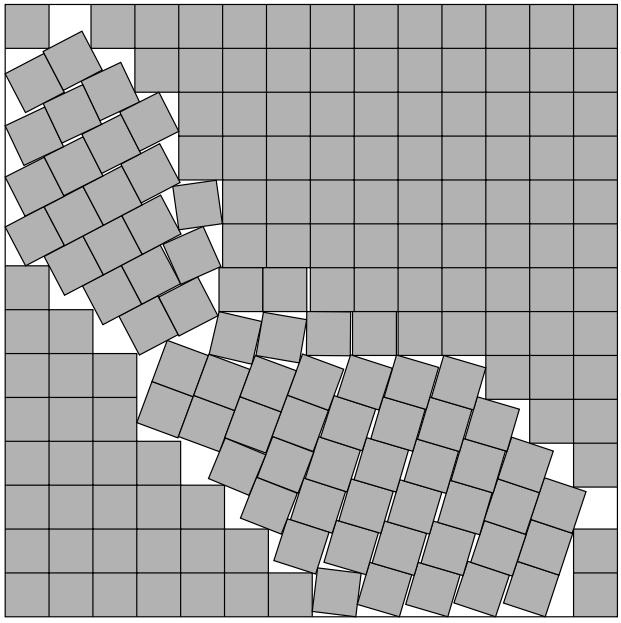
180



$s = 13.93508705291129\ldots$

Found by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(209)$ found by David Ellsworth in January 2026, using his modified

181



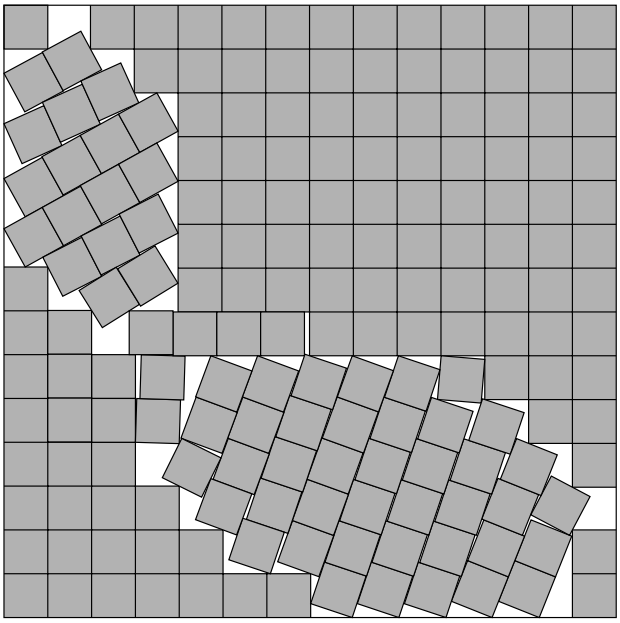
$s = 13.95690672341755\ldots$

Found by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in

version of Thomas Schadt's simulated annealing program, starting from the $s(240)$ converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 31 squares removed, with 29 squares removed.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

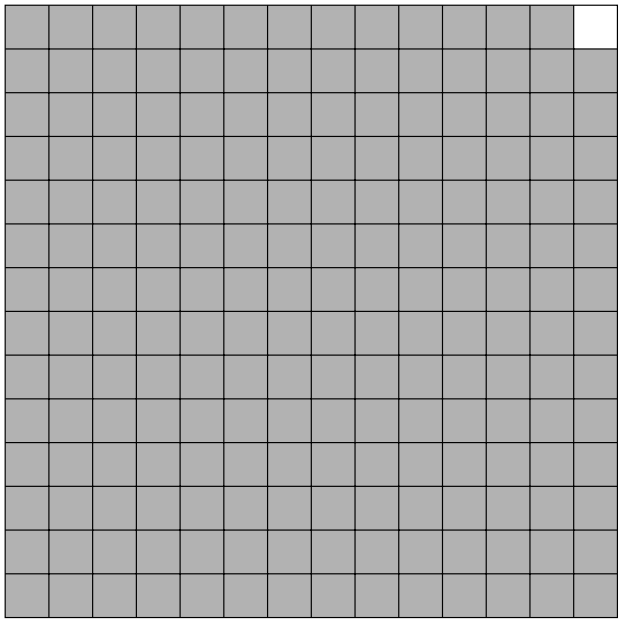
March 2019, improved, fixed, and improved by David Ellsworth in December 2024, December 2025, and January 2026, with 29 squares removed.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

182



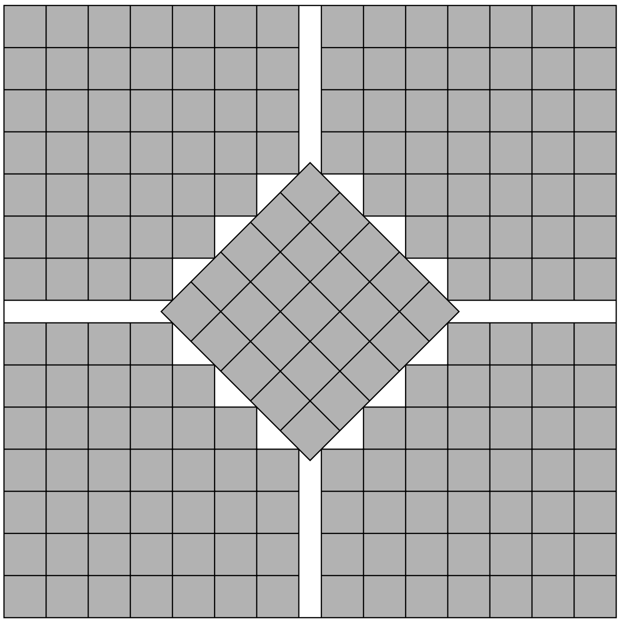
$s = 13.97419105332569\ldots$
Found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019.
Shows $s(n^2 - n) < n$ for $n = 14$.
Improved by David Ellsworth in December 2024.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

194, 195



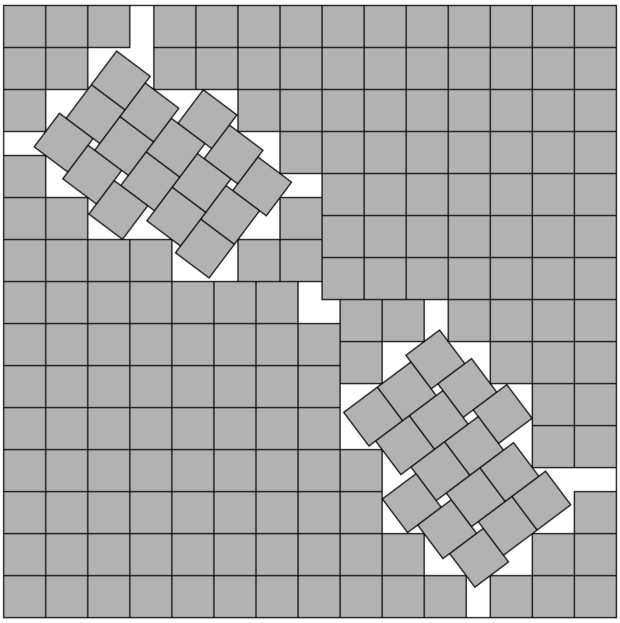
$s = 14$
Proved by Hiroshi Nagamochi in 2005.

197



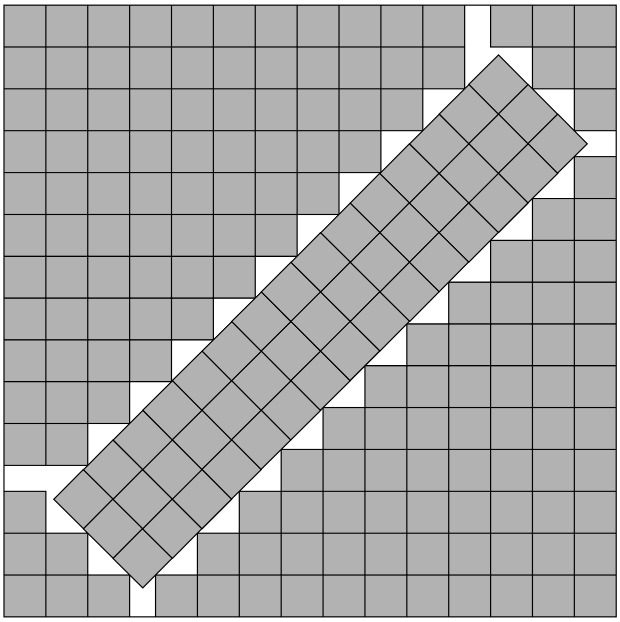
$s = 11 + \frac{5}{2}\sqrt{2} = 14.53553390593273\ldots$
Adds six "L"s to the $s(65)$ found by Frits Göbel in early 1979.

198



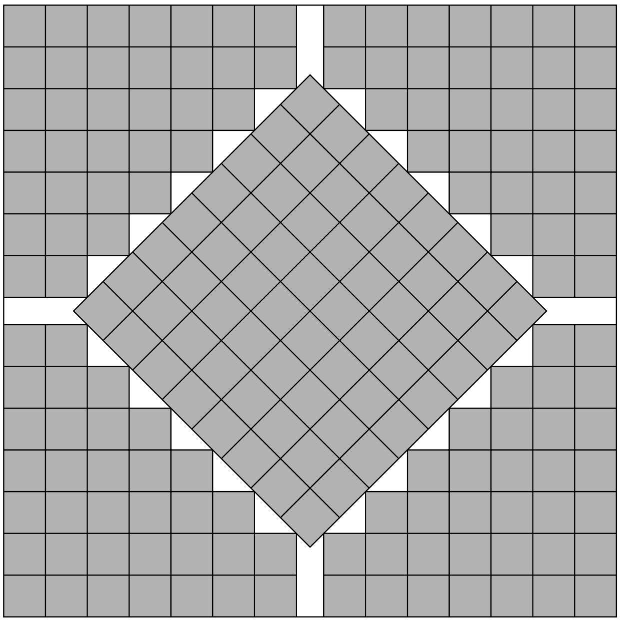
$s = 14 + \frac{4}{7} = 14.57142857142857\ldots$
Adds an "L" to the $s(171)$ found by David Ellsworth in December 2025 by combining two copies of the $s(50)$ found by Thomas Schadt in December 2025 (using a simulated annealing program he wrote, starting from randomness) and optimized by David Ellsworth.

199



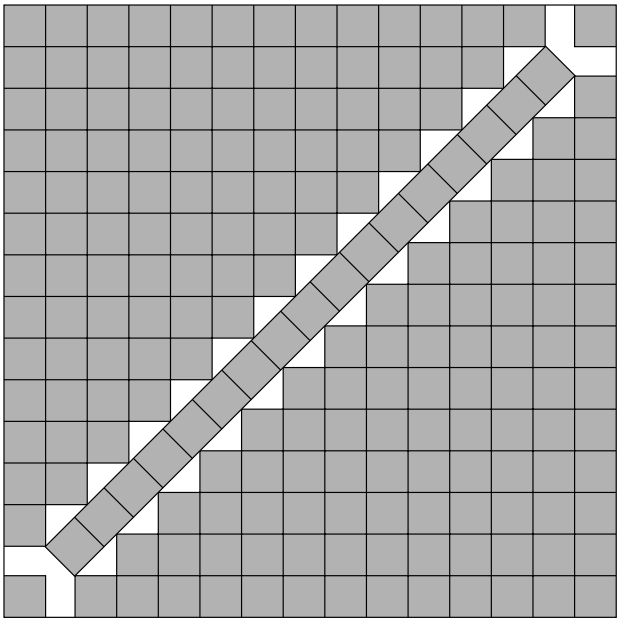
$s = {}^{8}$🔒 $= 14.61898898660160\ldots$
Adds an "L" to the $s(172)$ found by Károly Hajba in November 2024, improved by David Ellsworth in November 2024 by adapting the $s(37)$ improvement found by David W. Cantrell in September 2002, and improved by David Ellsworth in December 2024. Further improvement pending.

200



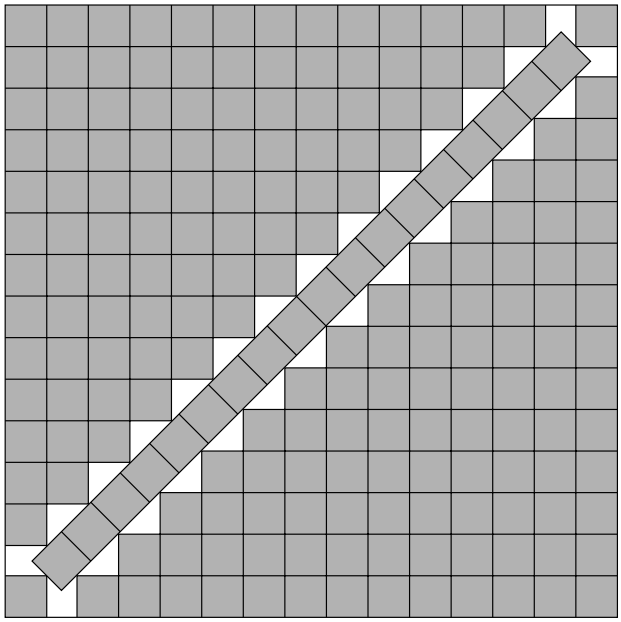
$s = 9 + 4\sqrt{2} = 14.65685424949238\ldots$
Adds two "L"s to the $s(148)$ that continues a pattern found by Frits Göbel in early 1979.

201



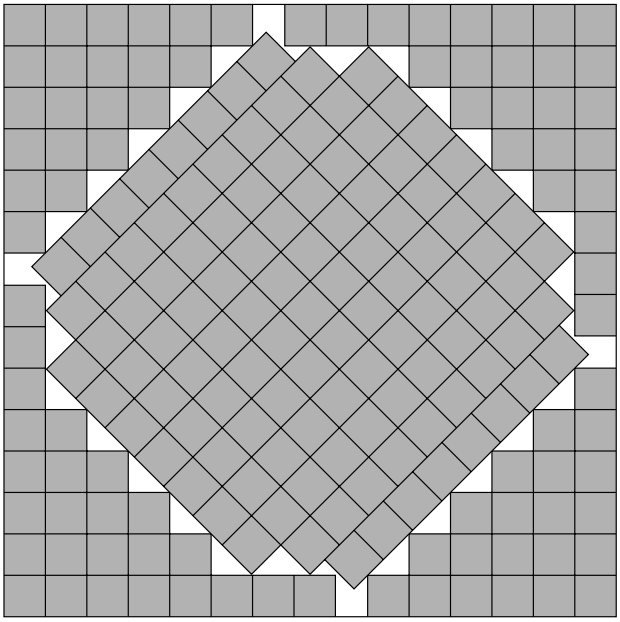
$s = 14 + \frac{1}{2}\sqrt{2} = 14.70710678118654\ldots$
Extends the $s(52)$ found by
Frits Göbel in early 1979.
Explore group

202



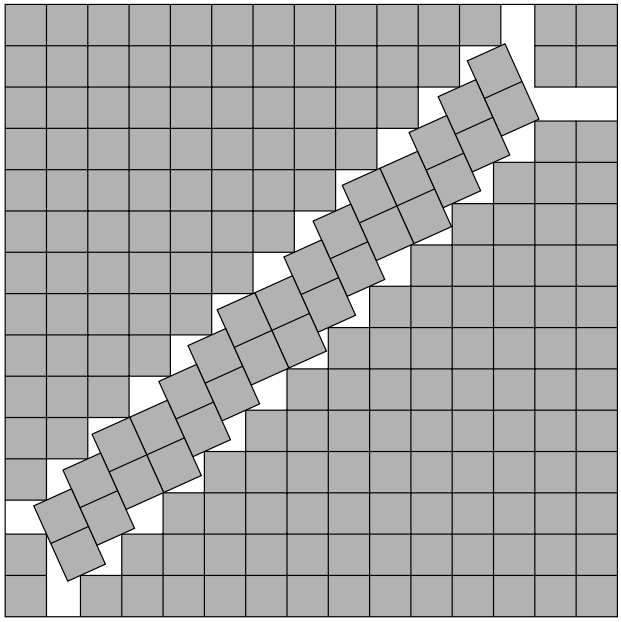
$s = 2 + 9\sqrt{2} = 14.72792206135785\ldots$
Extends the $s(18)$ found by
Frits Göbel in early 1979.
Explore group

203



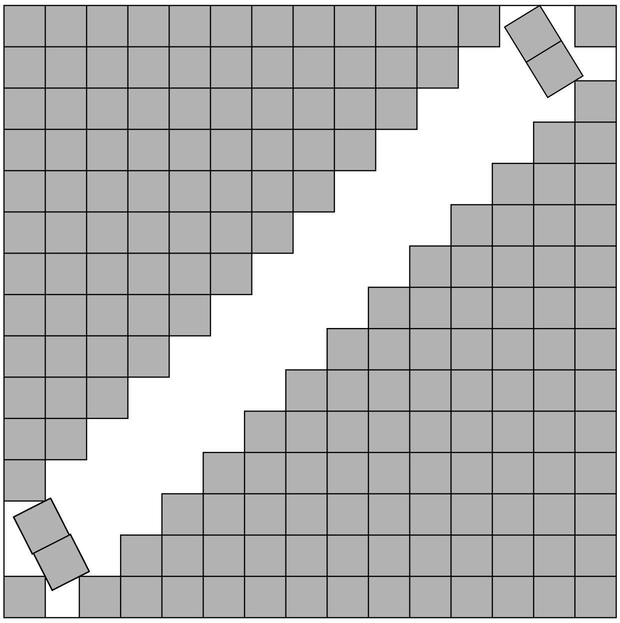
$s = 7 + \frac{11}{2}\sqrt{2} = 14.77817459305202\ldots$
Found by David Ellsworth
in December 2024.
Based on the $s(233)$ that continues a
pattern found by Frits Göbel in early 1979.

204



$s = \frac{27}{2} + \frac{1}{2}\sqrt{7} = 14.82287565553229\ldots$
Extends the $s(86)$ found by
Erich Friedman in 1997.

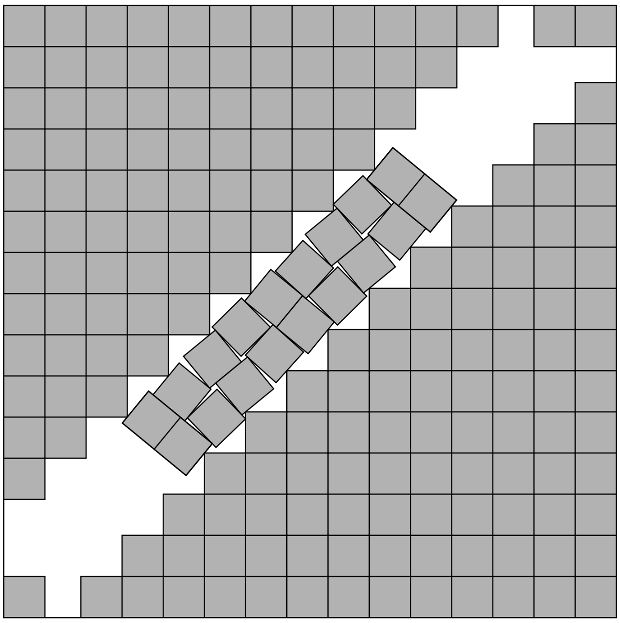
205



$s = {}^{40}$ 🔒 $= 14.82445114612408\ldots$

Found by David Ellsworth
in December 2024, by extending the
$s(128)$ he found/improved in
November/December 2024, based on the
$s(69)$ found by Maurizio Morandi in
June 2010, and using the technique from
the $s(53)$ improved by David W. Cantrell
in December 2024.
Improved by David Ellsworth
in January 2025, using the technique from
the $s(128)$ improved by David W. Cantrell
and David Ellsworth in January 2025.

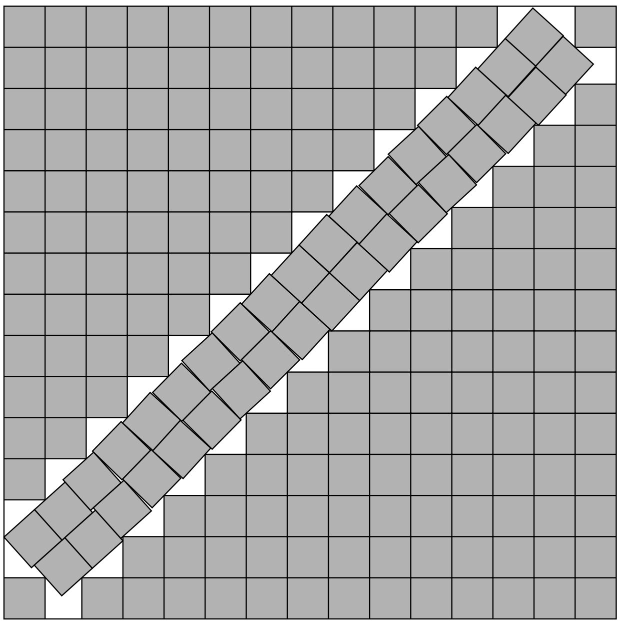
206



$s = {}^{198}$ 🔒 $= 14.87221902560728\ldots$

Found and improved by David Ellsworth
in December 2024, adapting/extending the
$s(37)$ improvement found by
David W. Cantrell in September 2002.
Unimproved form similar to the $s(70)$ found by
Erich Friedman in 1997.
Improved by David Ellsworth
in December 2025, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026 and
Joost de Winter in August 2026.
Optimized by David Ellsworth
in September 2026.

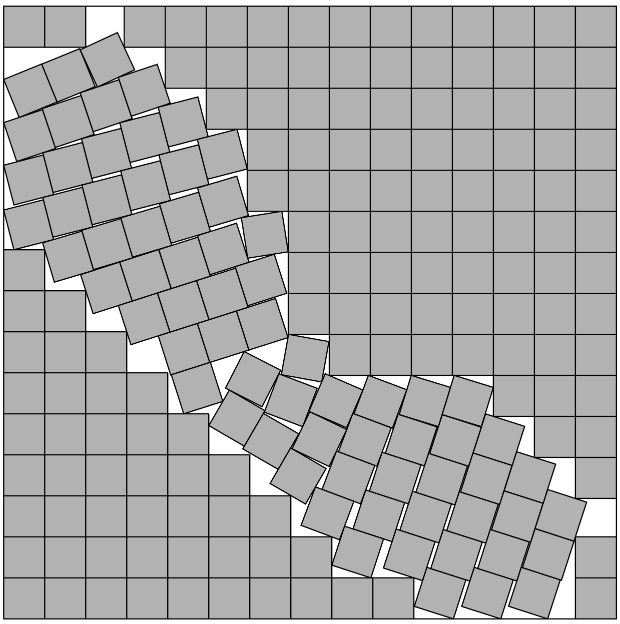
207



$s = 14.89395494255333\ldots$

Found by David Ellsworth

208



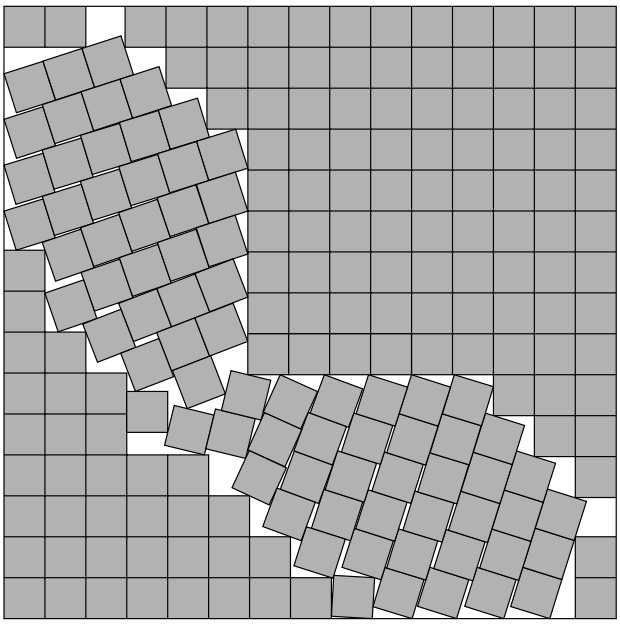
$s = 14.93776656277905\ldots$

Found by David Ellsworth

in November 2024, by extending the $s(88)$ found by Erich Friedman in 1997, and adapting and extending the $s(37)$ improvement found by David W. Cantrell in September 2002 and the $s(130)$ improvement found by David W. Cantrell in November 2024.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

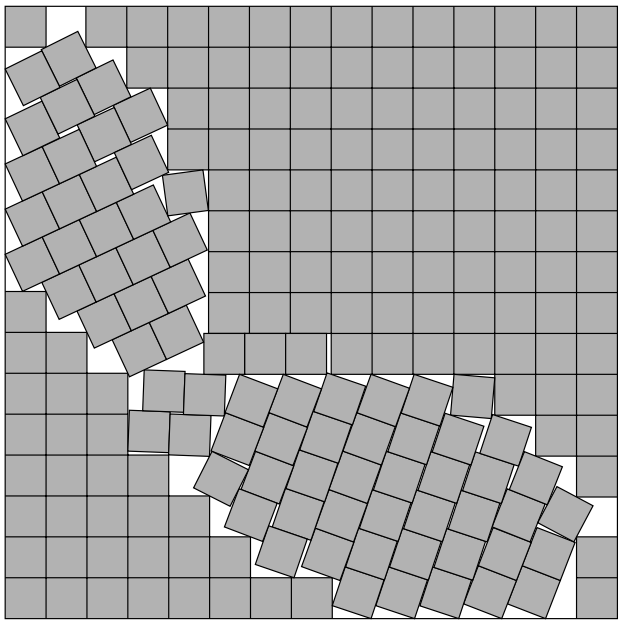
in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(239)$ found by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(272)$ converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 33 squares removed, with 31 squares removed.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

209



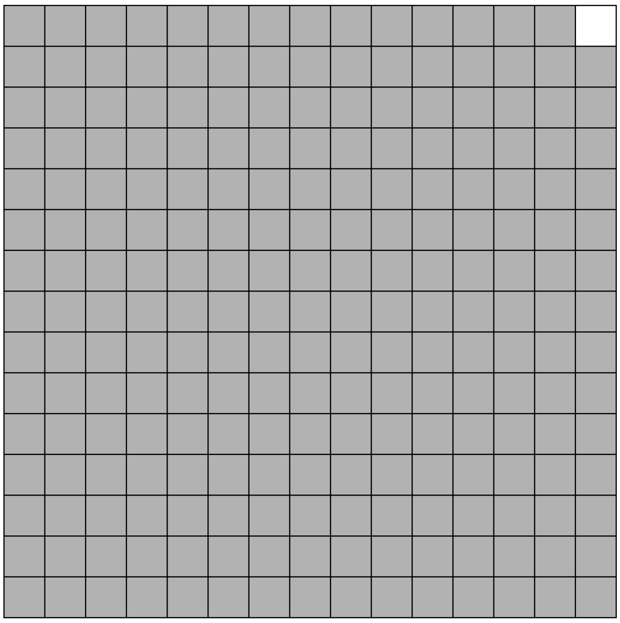
$s = 14.95861500087481\ldots$
Found by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(240)$ converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 31 squares removed.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

210



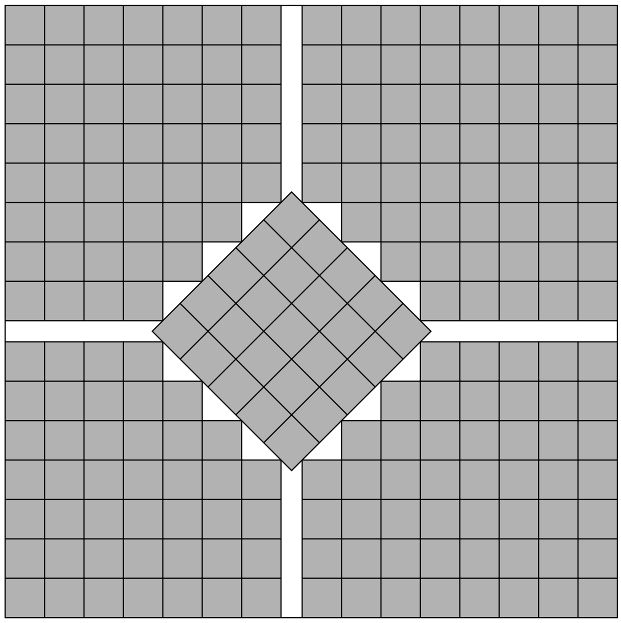
$s = 14.97413341886404\ldots$
Found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019.
Shows $s(n^2 - n) < n$ for $n = 15$.
Improved by David Ellsworth in December 2024.
Fixed by David Ellsworth in December 2025.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

223, 224



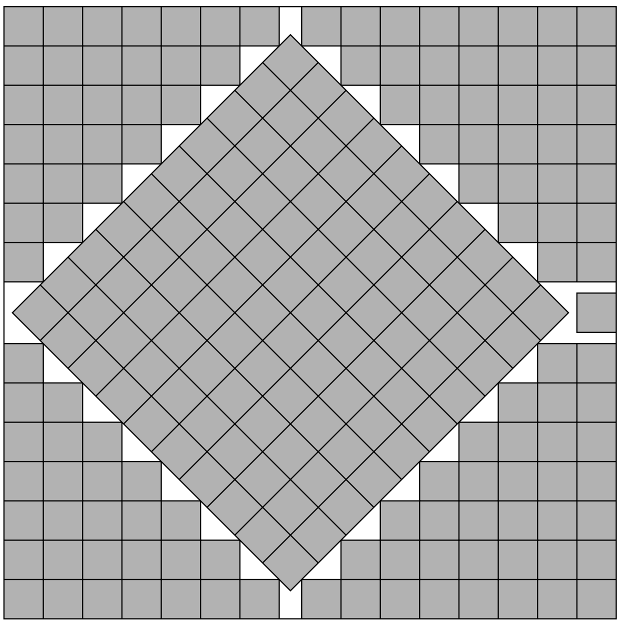
$s = 15$
Proved by Hiroshi Nagamochi
in 2005.

226



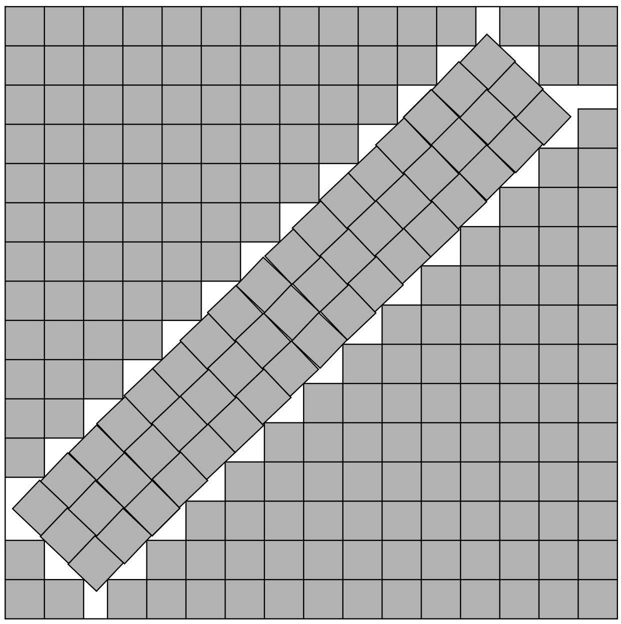
$s = 12 + \frac{5}{2}\sqrt{2} = 15.53553390593273\ldots$
Adds seven "L"s to the $s(65)$ found by
Frits Göbel in early 1979.

227



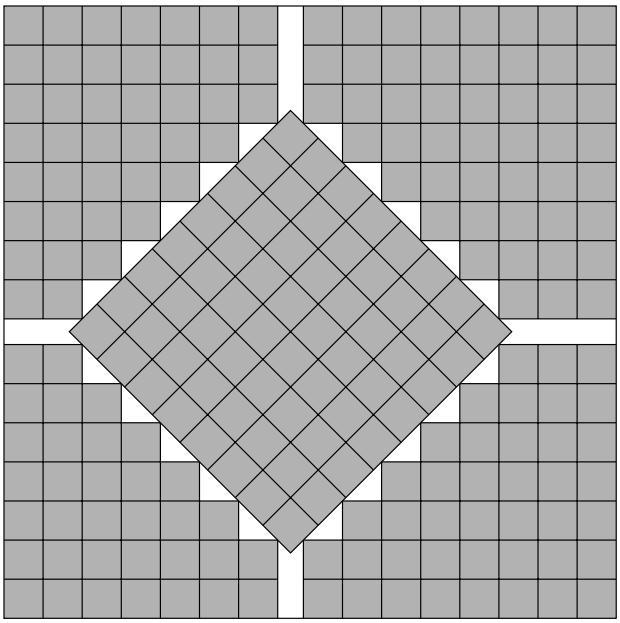
$s = \frac{17}{2} + 5\sqrt{2} = 15.57106781186547\ldots$
Found by David Ellsworth
in January 2025, using a computer
program he wrote.
Continues the $s(26)$, $s(85)$ series found by
Erich Friedman in 1997.

228



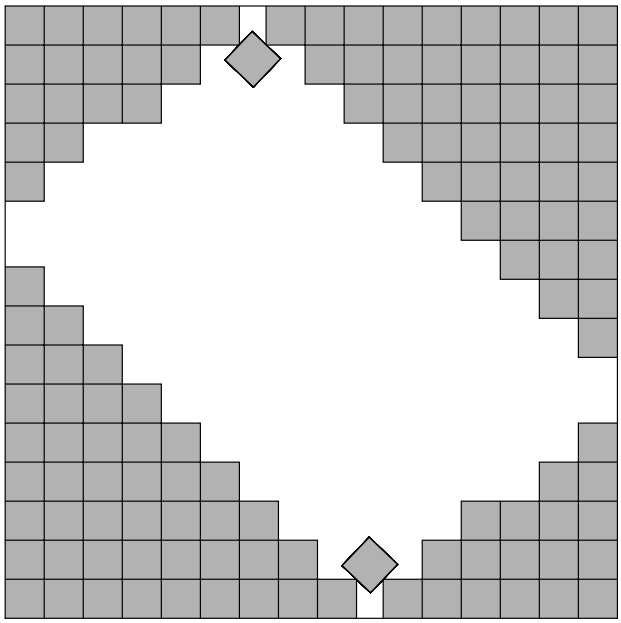
$s = {}^{12}$🔒 $= 15.60902282132495\ldots$
Found by David Ellsworth in November
2024, by extending the $s(102)$ found by
Károly Hajba in September 2024 and
improved by David W. Cantrell and David
Ellsworth in November 2024.
Improved by David Ellsworth in
December 2024.
Further improvement pending.

229



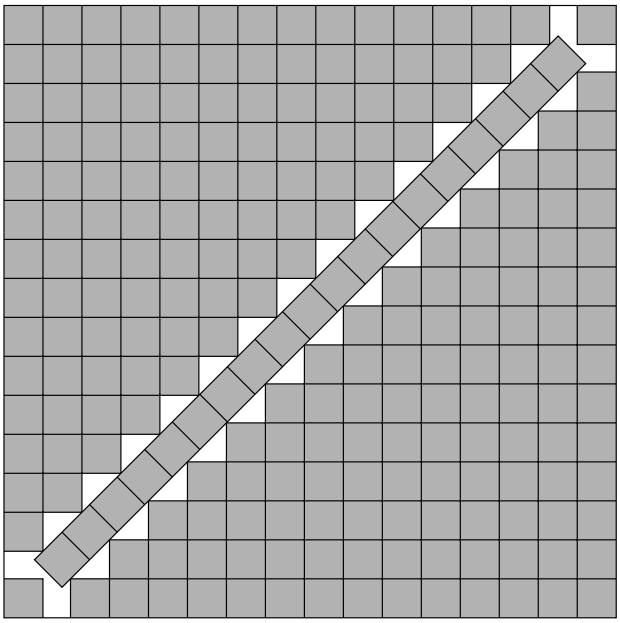
$s = 10 + 4\sqrt{2} = 15.65685424949238\ldots$
Adds three "L"s to the $s(148)$ that continues a pattern found by Frits Göbel in early 1979.

230



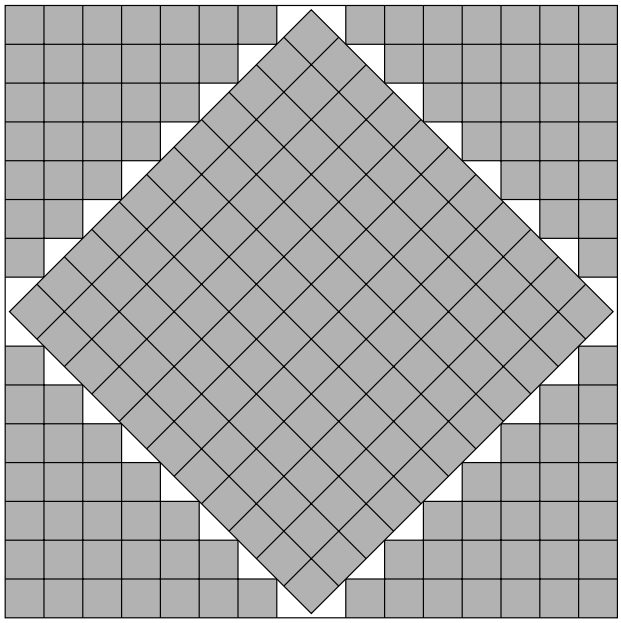
$s = 15 + \frac{28}{41} = 15.68292682926829\ldots$
Found and improved by David Ellsworth in January 2025. Uses a rotational symmetry technique found by David W. Cantrell in January 2025, and the technique from the $s(293)$ improved by David W. Cantrell in January 2025.
This is the second record-setting packing found with a rational side length, thanks to the Pythagorean triple $\{20, 21, 29\}$ determining its tilt angle.

231



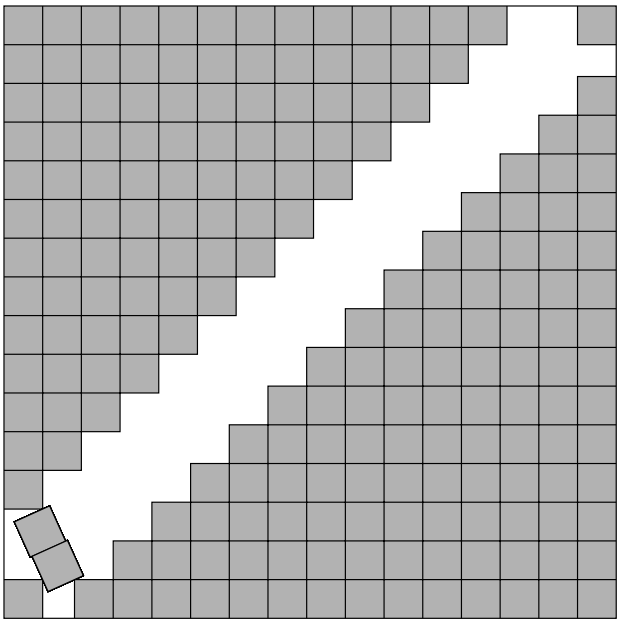
$s = 15 + \frac{1}{2}\sqrt{2} = 15.70710678118654\ldots$
Extends the $s(52)$ found by Frits Göbel in early 1979.
Explore group

232, 233



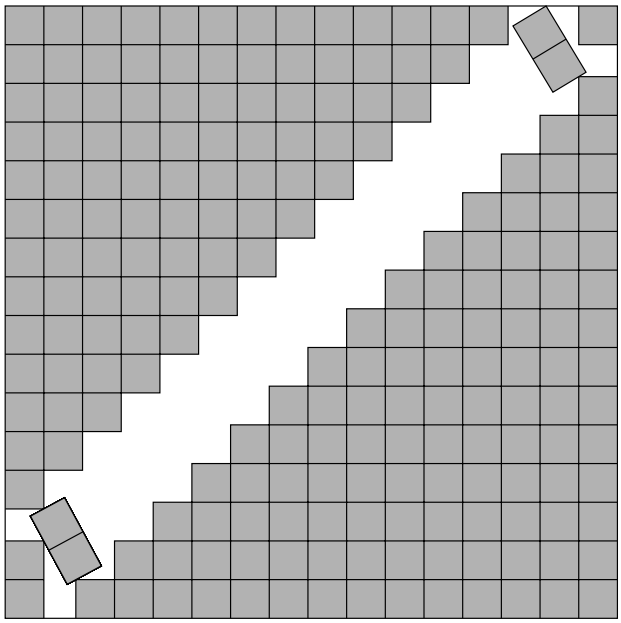
$s = 8 + \frac{11}{2}\sqrt{2} = 15.77817459305202\ldots$
Continues a pattern found by Frits Göbel in early 1979.
Explore group

234



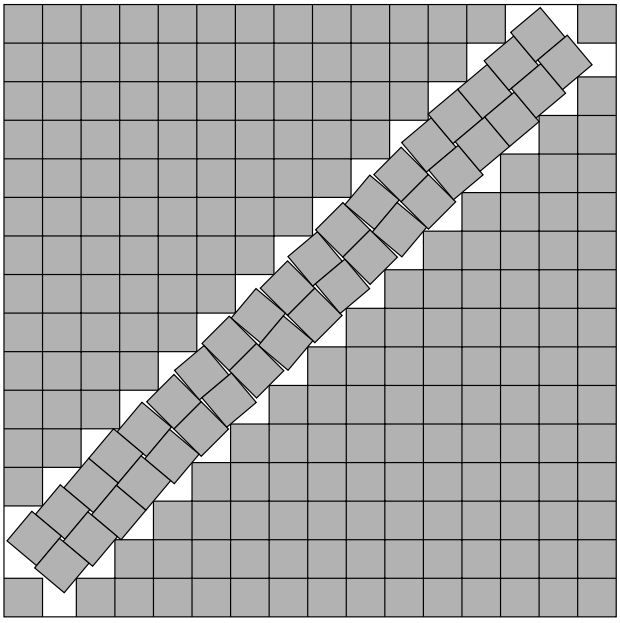
$s = \frac{29}{2} + \frac{1}{2}\sqrt{7} = 15.82287565553229\ldots$
Found by David Ellsworth
in December 2024, by extending the
$s(128)$ and $s(205)$ he found/improved in
November/December 2024, based on the
$s(69)$ found by Maurizio Morandi in
June 2010, and using the technique from
the $s(53)$ improved by David W. Cantrell
in December 2024.
Improved by David Ellsworth
in January 2025, adapting the technique
from the $s(128)$ improved by
David W. Cantrell and David Ellsworth
in January 2025.

235



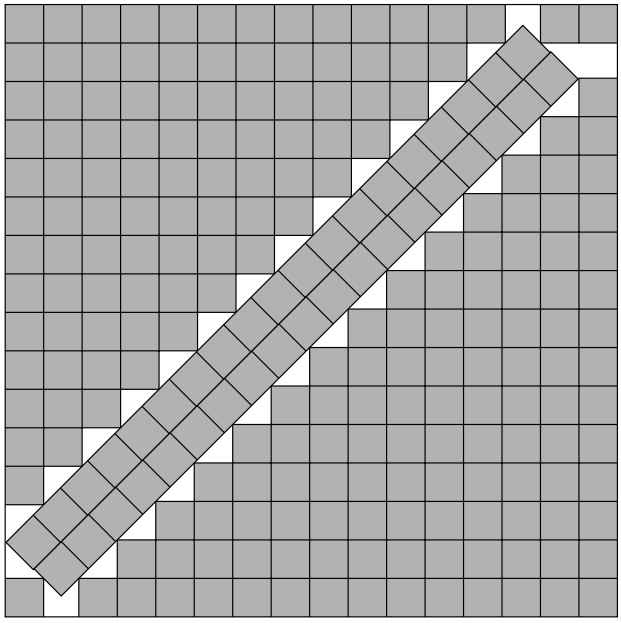
$s = {}^{83}\!🔒 = 15.82660563342856\ldots$
Found by David Ellsworth and
David W. Cantrell in January 2025, based
on the $s(53)$ found by David W. Cantrell
in September 2002 and the $s(69)$ found by
Maurizio Morandi in June 2010, adapting
and extending the $s(69)$ improvement
found by David W. Cantrell
in August 2023.

236



$s = 15.87607539315201\ldots$
Found by David Ellsworth
in November 2024, by extending the $s(88)$

237

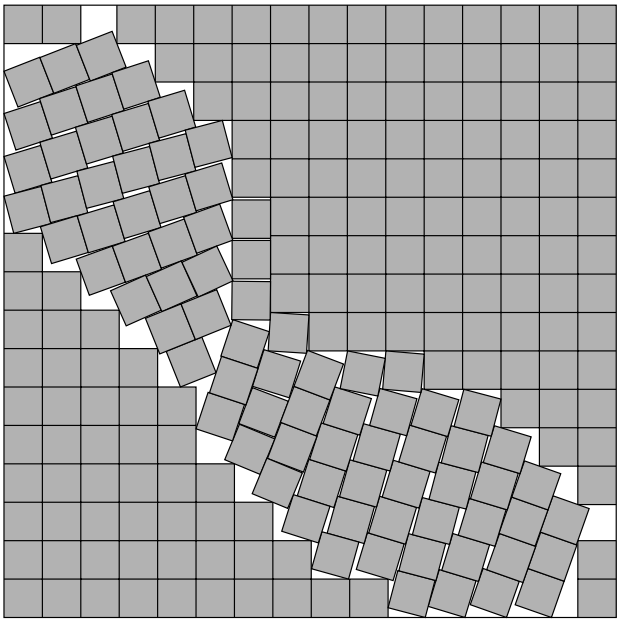


$s = \frac{29}{2} + \sqrt{2} = 15.91421356237309\ldots$
Found by David Ellsworth
in December 2024.

found by Erich Friedman in 1997 and adapting the $s(102)$ improvement found by David W. Cantrell in November 2024. Improved by David Ellsworth in December 2024. Improved by Grigoriy Dyachkov in September 2026, working with Claude Opus 5 and Claude Fable 5. Not yet analytically optimized.

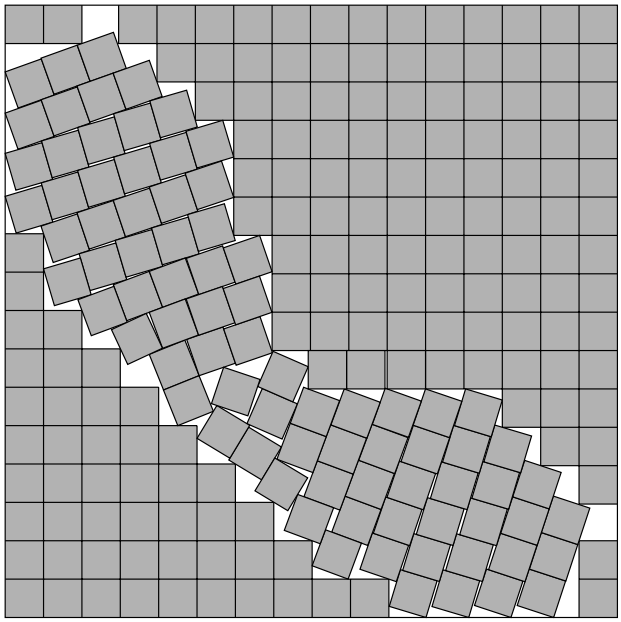
Similar to the $s(70)$ found by Erich Friedman in 1997.

238



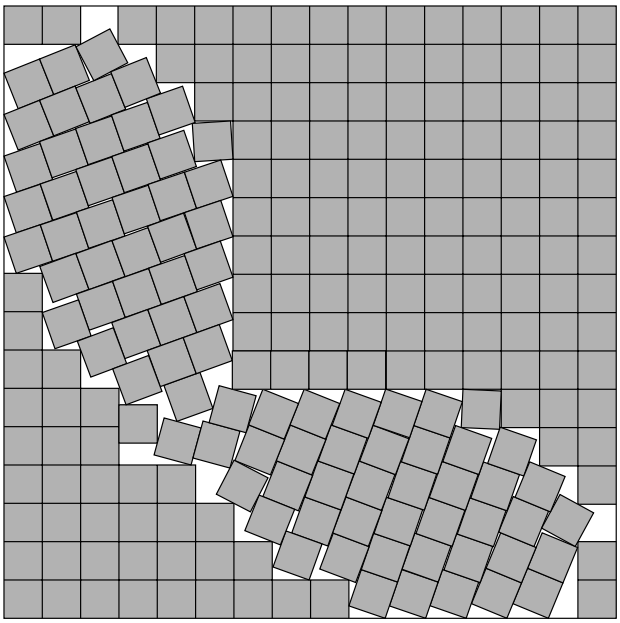
$s = 15.93965520031394\ldots$
Found by David Ellsworth in February 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(239)$ found by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(272)$ converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 33 squares removed, with 1 square removed and 4 straightened. Refined by Grigoriy Dyachkov in August 2026, using unspecified method. Not yet analytically optimized.

239



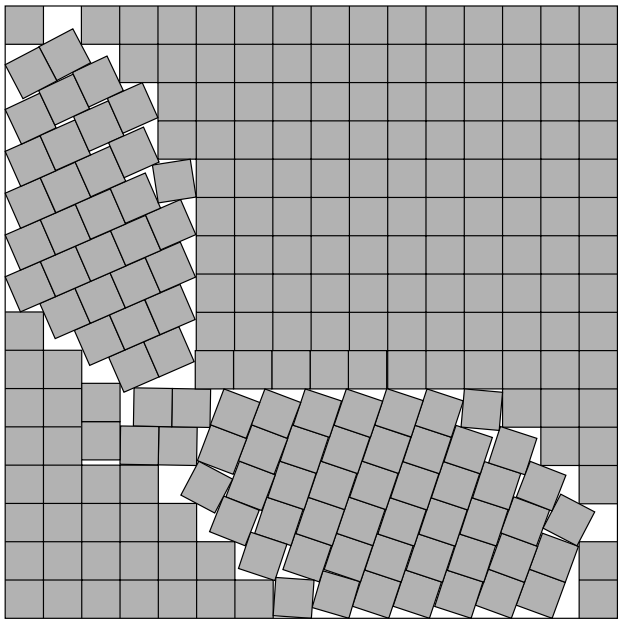
$s = 15.95635358406308\ldots$
Found by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program, starting from the $s(272)$ converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 33 squares removed. Refined by Tej Stead in June 2026, working with Claude Fable 5. Not yet analytically optimized.

240



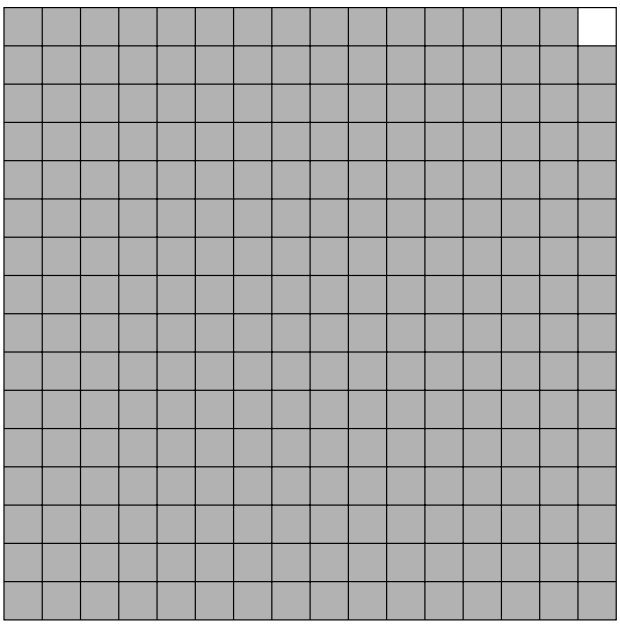
$s = 15.97556282833087\ldots$
Originally found by Károly Hajba
in September 2015.
Bounded the $s(n^2 - n) = n$ conjecture to
$n < 16$.
Converted in December 2024 from the
$s(210)$ found by M.Z. Arslanov, S.A.
Mustafin, and Z.K. Shangitbayev in
March 2019 and improved by
David Ellsworth in December 2024.
Improved by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

241

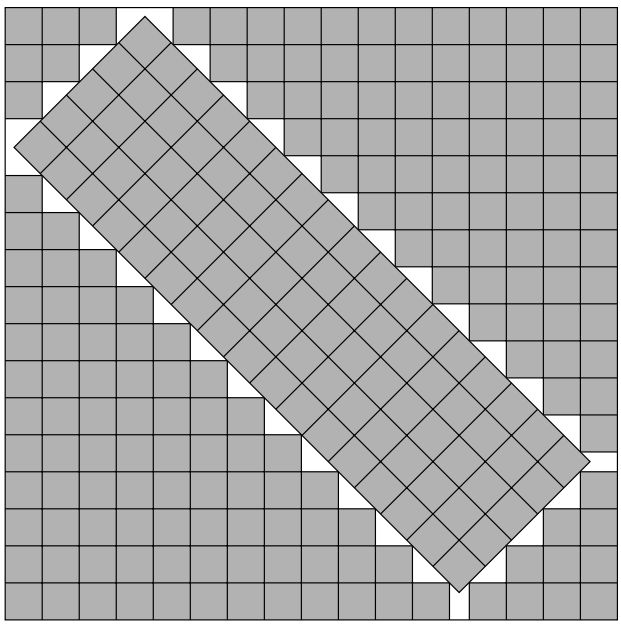


$s = 15.99080517810520\ldots$
Found by M.Z. Arslanov, S.A. Mustafin,
and Z.K. Shangitbayev in March 2019.
Shows $s(n^2 - n + 1) < n$ for $n = 16$.
Improved by David Ellsworth in
November 2024.
Improved again by David Ellsworth in
November 2024, by adapting a technique
from the $s(240)$ found by Károly Hajba in
September 2015.
Improved by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

254, 255



257



$s = 16$

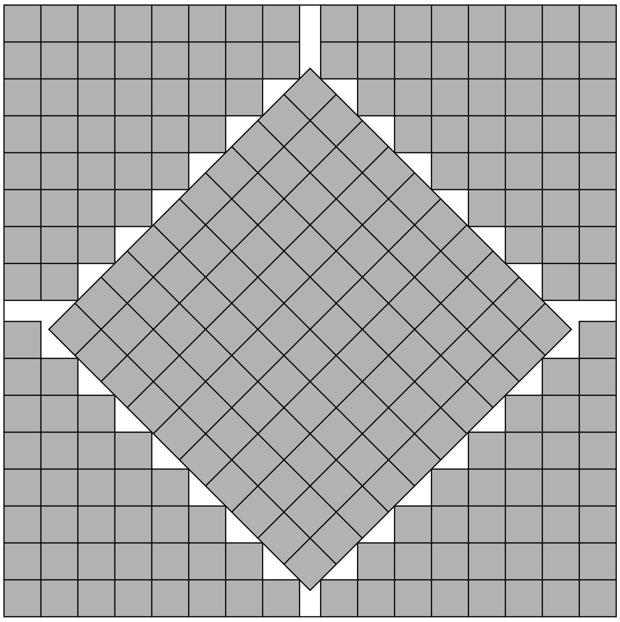
Proved by Hiroshi Nagamochi in 2005.

$s = 13 + \frac{5}{2}\sqrt{2} = 16.53553390593273\ldots$

Extends the $s(65)$ found by Frits Göbel in early 1979.

This alternative, converting the $s(65)$ augmented by eight "L"s into a primitive packing, found by David Ellsworth in January 2025, using a computer program he wrote.

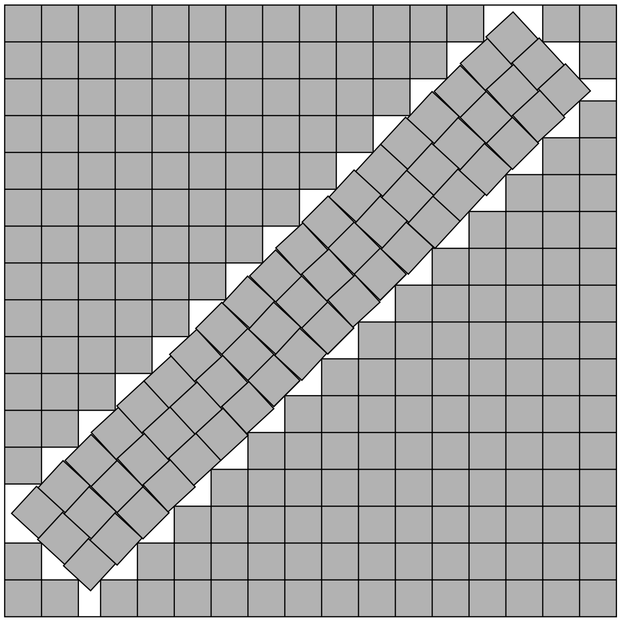
258



$s = \frac{19}{2} + 5\sqrt{2} = 16.57106781186547\ldots$

Adds an "L" to the $s(227)$ found by David Ellsworth in January 2025 (using a computer program he wrote) which continues the $s(26)$, $s(85)$ series found by Erich Friedman in 1997.

259



$s = {}^{8}$🔒 $= 16.60257141234448\ldots$

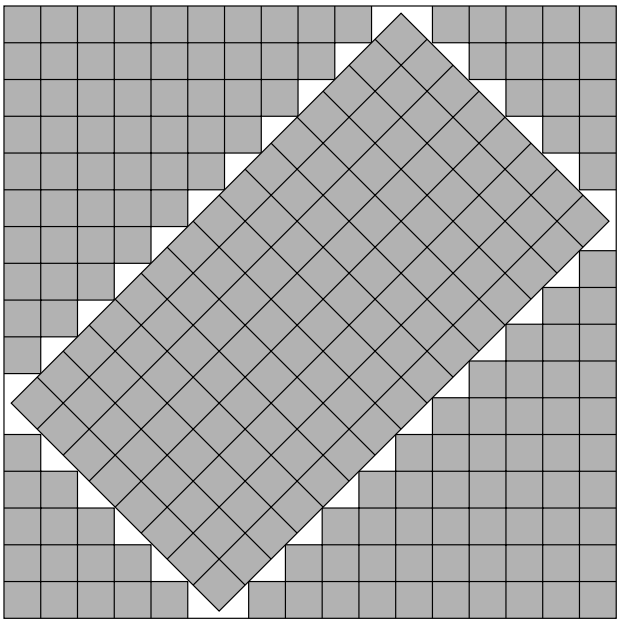
Found by David Ellsworth in December 2024, by extending the $s(102)$ found by Károly Hajba in September 2024 and adapting and extending the $s(37)$ improvement found by David W. Cantrell in September 2002.

Improved by David Ellsworth in December 2024.

Fixed by David Ellsworth in January 2026.

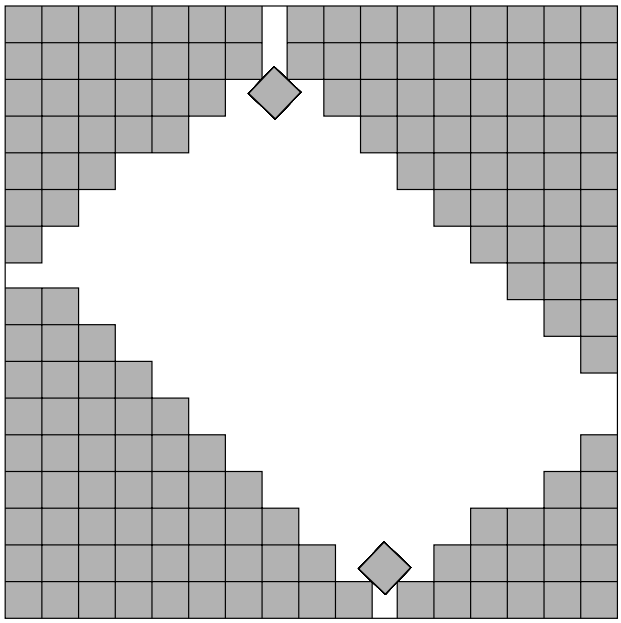
Further improvement pending.

260



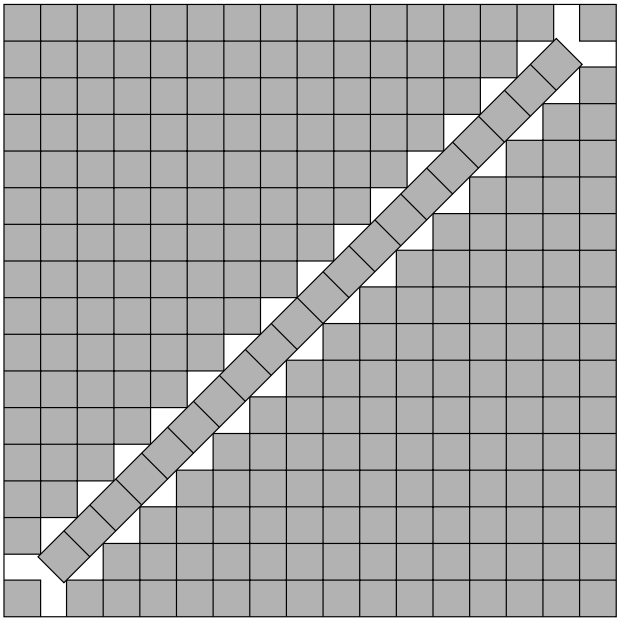
$s = 11 + 4\sqrt{2} = 16.65685424949238\ldots$
Extends the $s(124)$ that continues a pattern found by Frits Göbel in early 1979. This alternative, converting the $s(148)$ augmented by four "L"s into a primitive packing, found by David Ellsworth in December 2024.

261



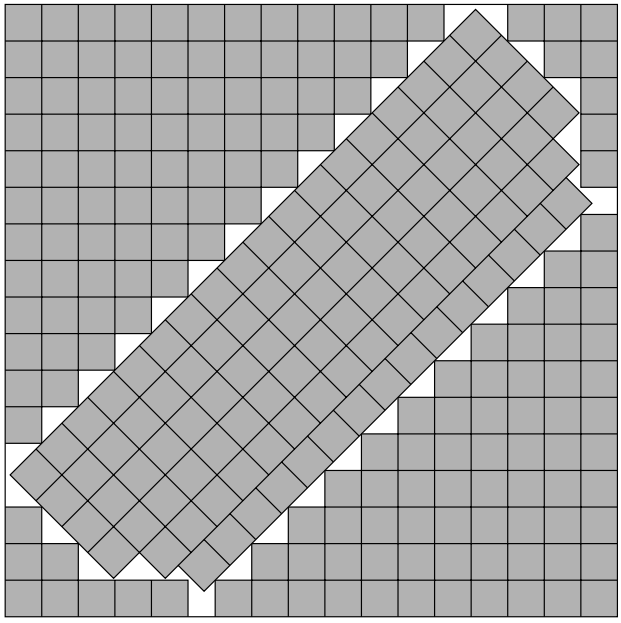
$s = 16 + \frac{28}{41} = 16.68292682926829\ldots$
Adds an "L" to the $s(230)$ found and improved by David Ellsworth in January 2025, which uses a rotational symmetry technique found by David W. Cantrell in January 2025, and the technique from the $s(293)$ improved by David W. Cantrell in January 2025.

262



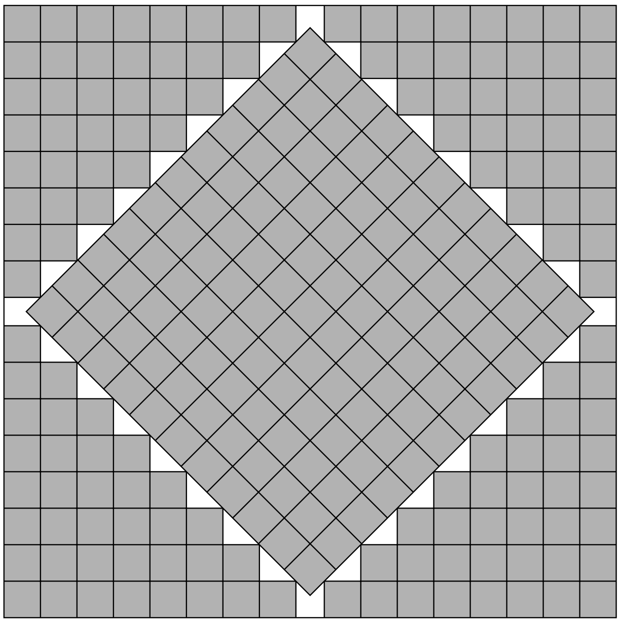
$s = 16 + \frac{1}{2}\sqrt{2} = 16.70710678118654\ldots$
Extends the $s(52)$ found by Frits Göbel in early 1979.
Explore group

263



$s = \frac{25}{2} + 3\sqrt{2} = 16.74264068711928\ldots$
Found by David Ellsworth in January 2025, based on the $s(297)$ he found.

264, 265

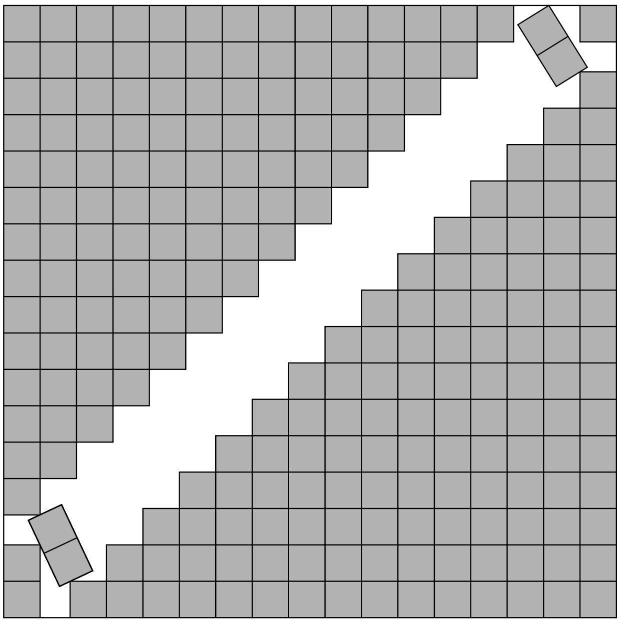


$s = 9 + \frac{11}{2}\sqrt{2} = 16.77817459305202\ldots$

Continues a pattern found by
Frits Göbel in early 1979.

Explore group

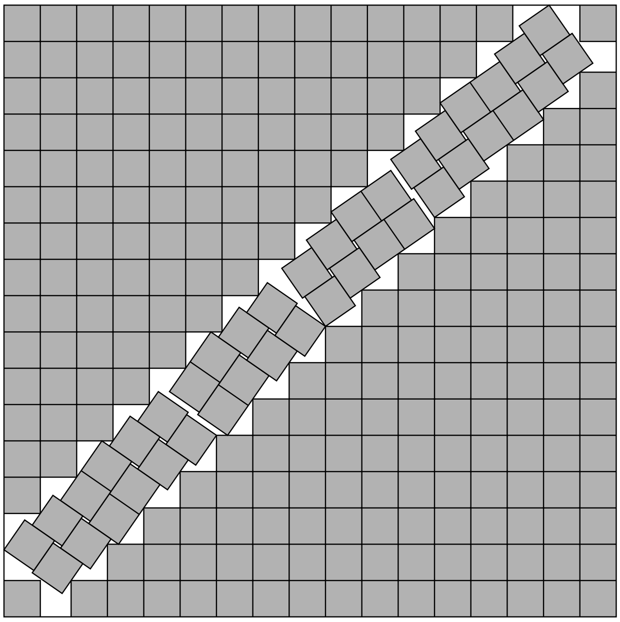
266



$s = {}^{32}$🔒 $= 16.82306208283780\ldots$

Found and improved by David Ellsworth in November 2024 and in December 2024, based on the $s(53)$ found/improved by David W. Cantrell in September 2002 and December 2024, respectively.

267

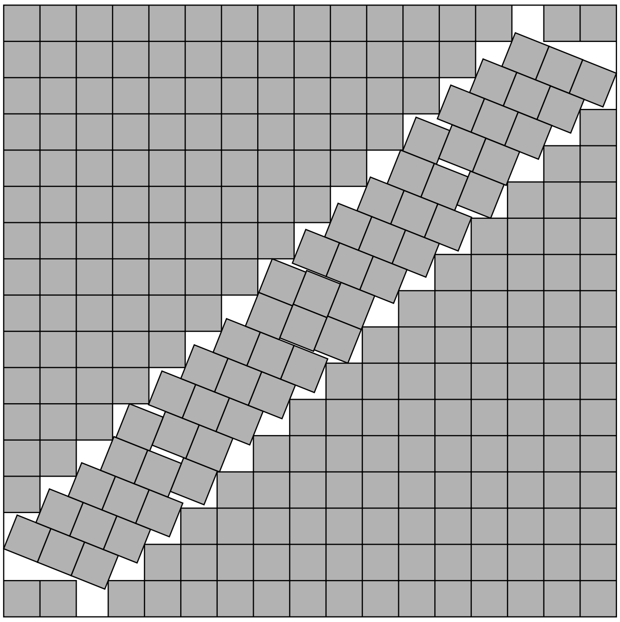


$$s = 16 - \frac{1}{2}\sqrt{2} + \sqrt{1 + \sqrt{2}}$$

$$= 16.84666719284348\ldots$$

Extends the $s(107)$ found by
Károly Hajba in November 2024.

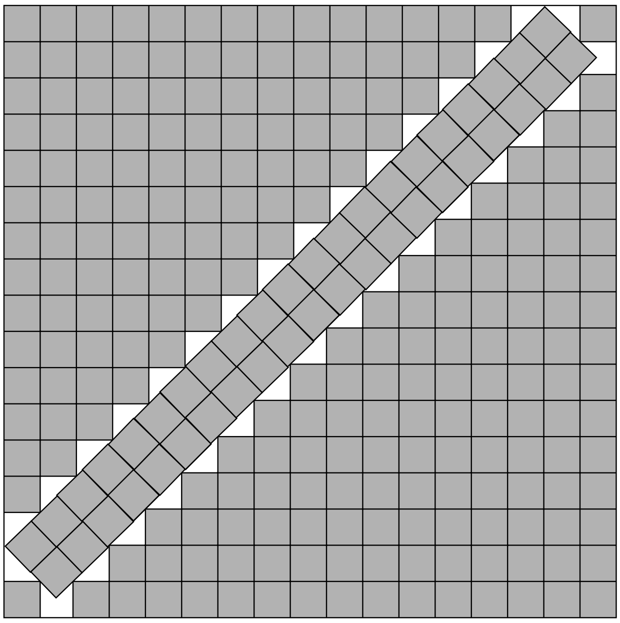
268



$s = 16.87931143465371\ldots$

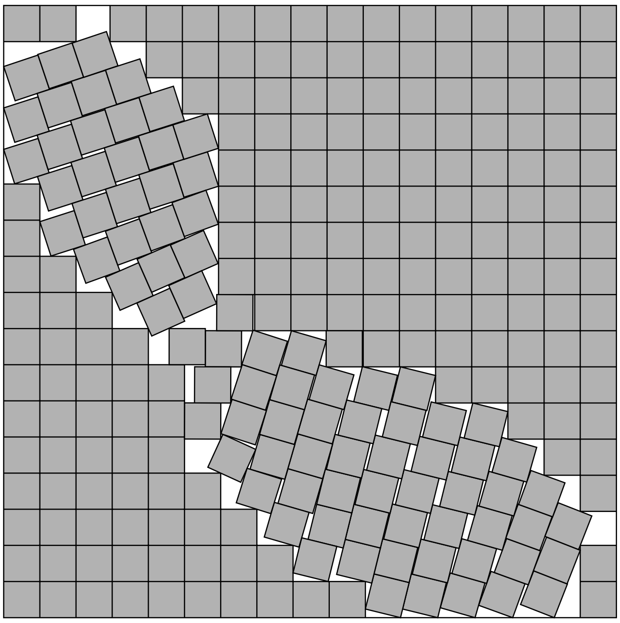
Found by David Ellsworth in November 2024, based on the $s(70)$ found by Joe DeVincentis in April 2014.
Improved by David W. Cantrell in February 2025.
Improved by Grigoriy Dyachkov in September 2026, working with Claude Opus 5 and Claude Fable 5.
Not yet analytically optimized.

269



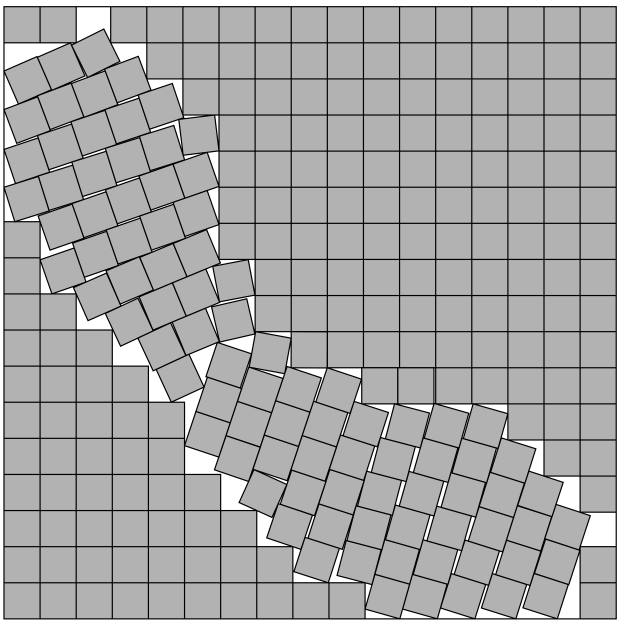
$s = {}^{8}$🔒 $= 16.90596764828402...$
Found by David Ellsworth
in November 2024, by extending the $s(88)$
found by Erich Friedman in 1997, and
adapting and extending the $s(37)$
improvement found by David W. Cantrell
in September 2002.
Improved by David Ellsworth
in January 2025.
Further improvement pending.

270



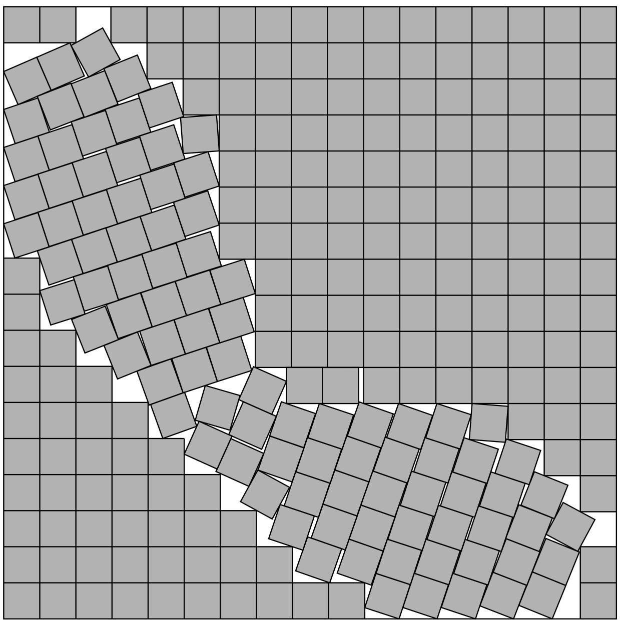
$s = 16.94062059800744...$
Found by David Ellsworth
in February 2026, using his modified
version of Thomas Schadt's simulated
annealing program, starting from the latest
$s(271)$ as of February 2026, with 1 square
removed and 5 straightened.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

271



$s = 16.95499909412532...$
Found by David Ellsworth
in February 2026, using his modified
version of Thomas Schadt's simulated
annealing program, starting from the

272

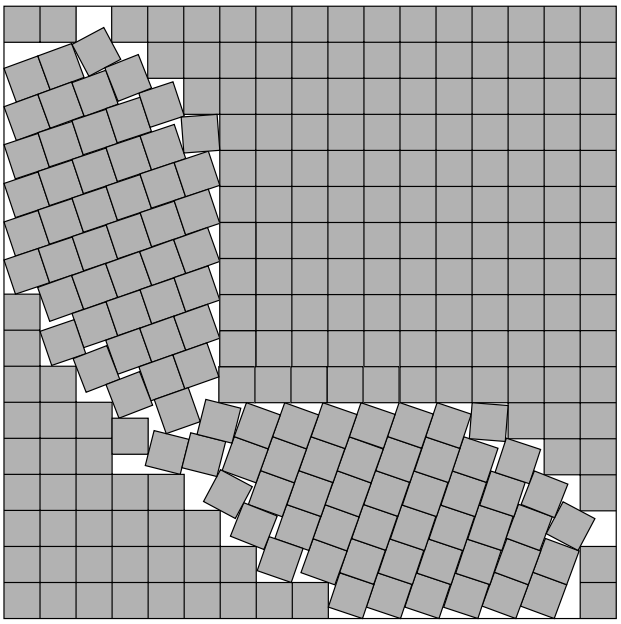


$s = 16.96971602419903...$
Originally found by Lars Cleemann
between 1991 and 1998.
Bounded the $s(n^2 - n) = n$ conjecture to
$n < 17$.

$s(272)$ converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024 and January 2026, with 1 square removed and 4 straightened.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

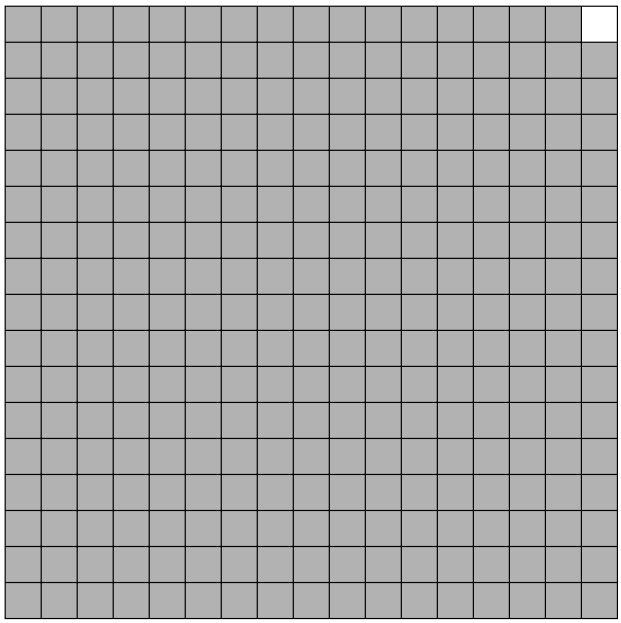
Converted in December 2024 from the $s(210)$ found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019 and improved by David Ellsworth in December 2024.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

273



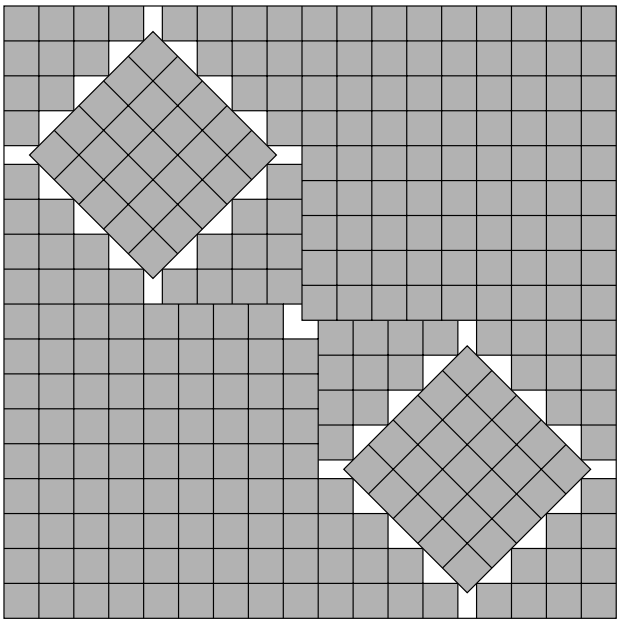
$s = 16.98820725030513\ldots$
Found by M.Z. Arslanov, S.A. Mustafin, and Z.K. Shangitbayev in March 2019.
Shows $s(n^2 - n + 1) < n$ for $n = 17$.
Improved by David Ellsworth in November 2024, by adapting a technique from the $s(240)$ found by Károly Hajba in September 2015.
Improved by David Ellsworth in December 2024.
Improved by David Ellsworth in January 2026, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

287, 288



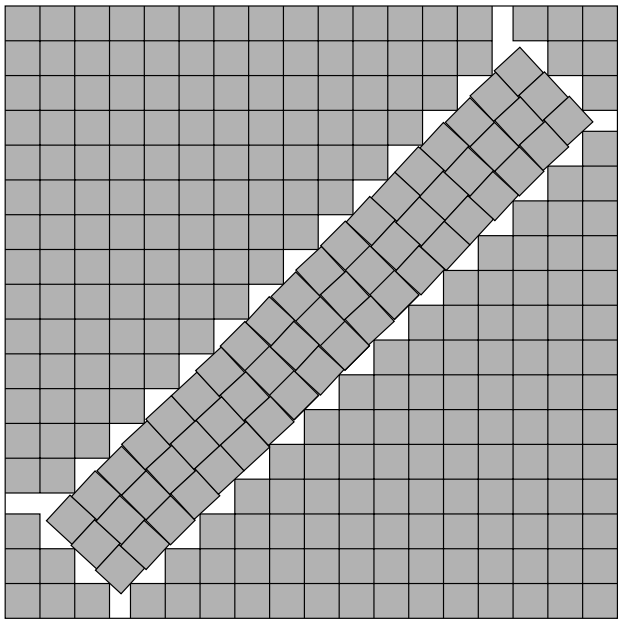
$s = 17$
Proved by Hiroshi Nagamochi in 2005.

290, 291



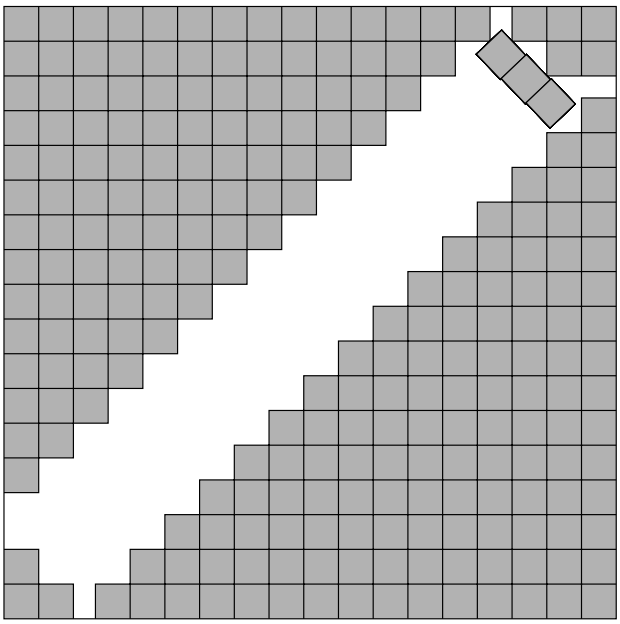
$s = 14 + \frac{5}{2}\sqrt{2} = 17.53553390593273\ldots$
Combines two copies of the $s(65)$ that continues a pattern found by Frits Göbel in early 1979.
This is the first $s(n^2+2)$ that has the same side length as the best known $s(n^2+1)$, and is likely an irreducible primitive packing.

292



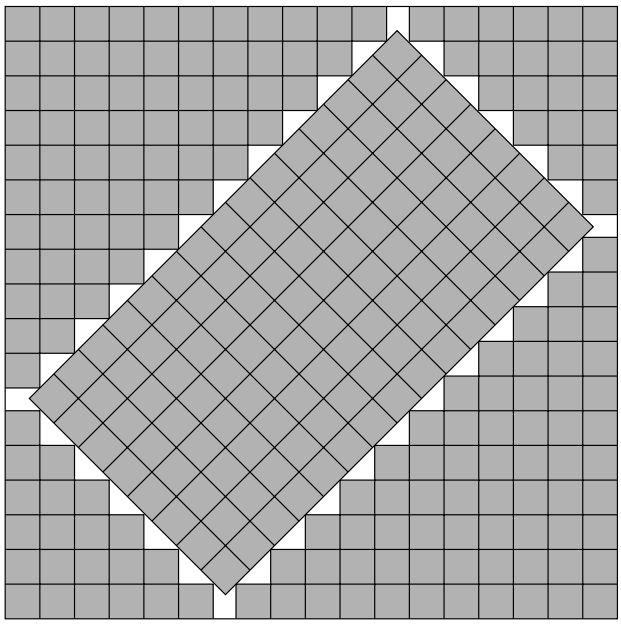
$s = {}^{8}\!\!\text{🔒} = 17.60257141234448\ldots$
Adds an "L" to the $s(259)$ found and improved by David Ellsworth in December 2024 extending the $s(102)$ found by Károly Hajba in September 2024 and adapting and extending the $s(37)$ improvement found by David W. Cantrell in September 2002.
Fixed by David Ellsworth in January 2026.
Further improvement pending.

293



$s = 17 + \frac{26}{41} = 17.63414634146341\ldots$
Found by David Ellsworth in December 2024.
Improved by David W. Cantrell in January 2025.
This is the first record-setting packing found

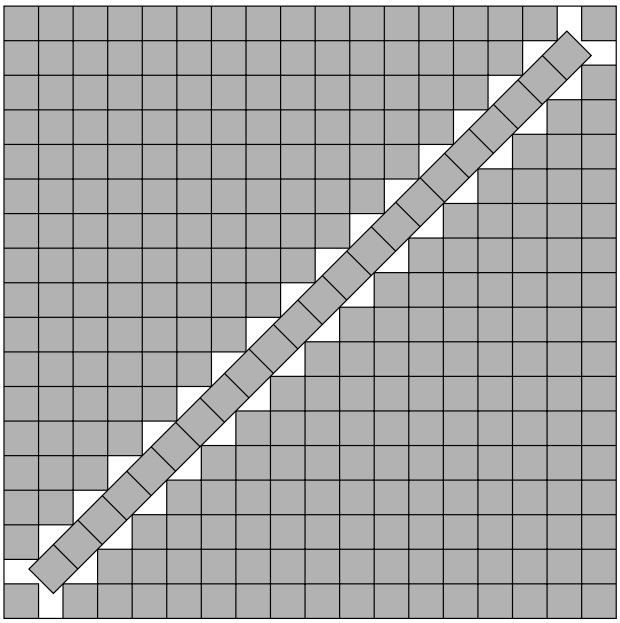
294



$s = 12 + 4\sqrt{2} = 17.65685424949238\ldots$
Extends the $s(148)$ that continues a pattern found by Frits Göbel in early 1979.
This is likely an irreducible primitive packing.

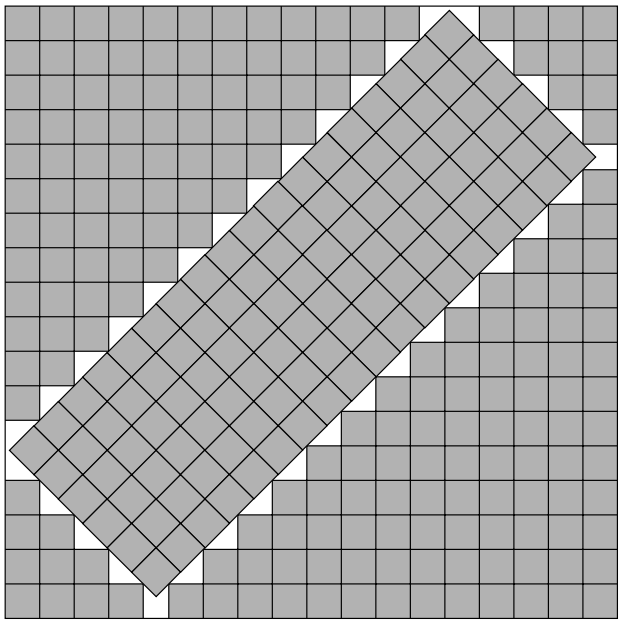
with a rational side length, thanks to the Pythagorean triple $\{20, 21, 29\}$ determining its tilt angle. It was foreshadowed just 3 days earlier by David Ellsworth finding an $s(104)$ with rational side length.

295, 296



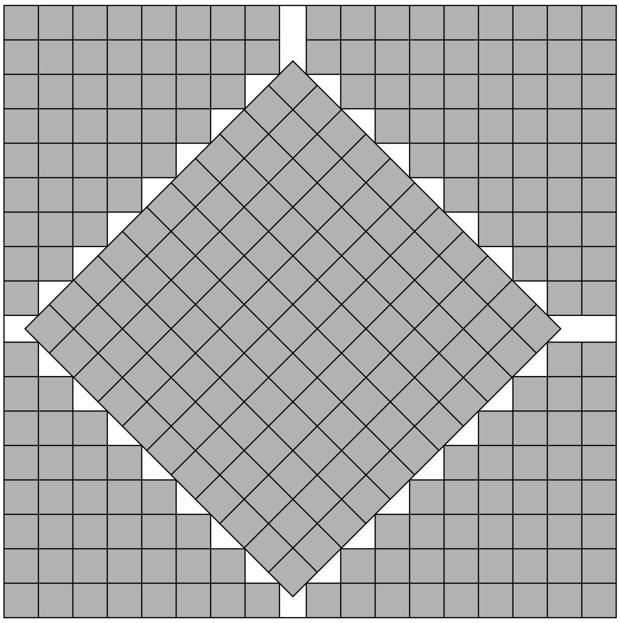
$s = 17 + \frac{1}{2}\sqrt{2} = 17.70710678118654\ldots$
Extends the $s(52)$ found by
Frits Göbel in early 1979.
Explore group

297



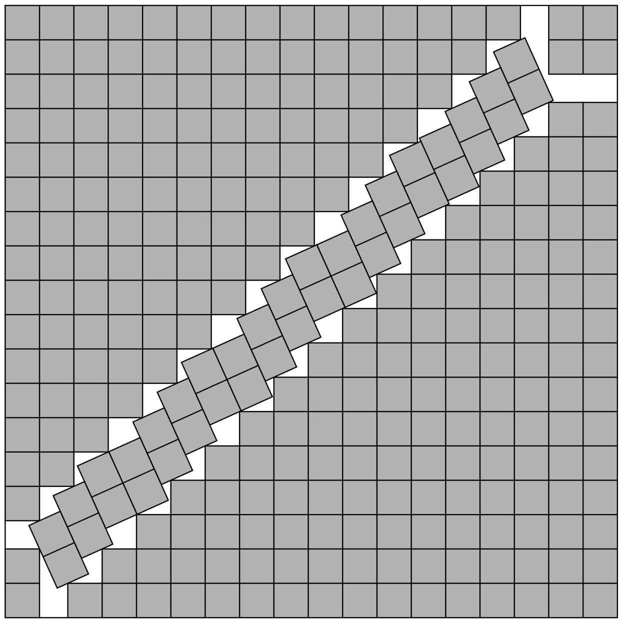
$s = 17.74106074604732\ldots$
Found by David Ellsworth
in January 2025.
Extends the $s(85)$ found by
Erich Friedman in 1997.
Improved by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

298



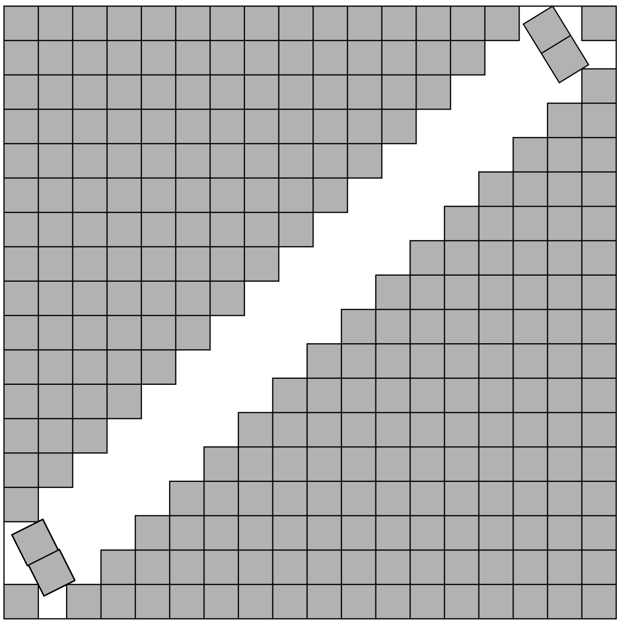
$s = 10 + \frac{11}{2}\sqrt{2} = 17.77817459305202\ldots$
Adds an "L" to the $s(265)$ that continues a pattern found by Frits Göbel in early 1979.

299



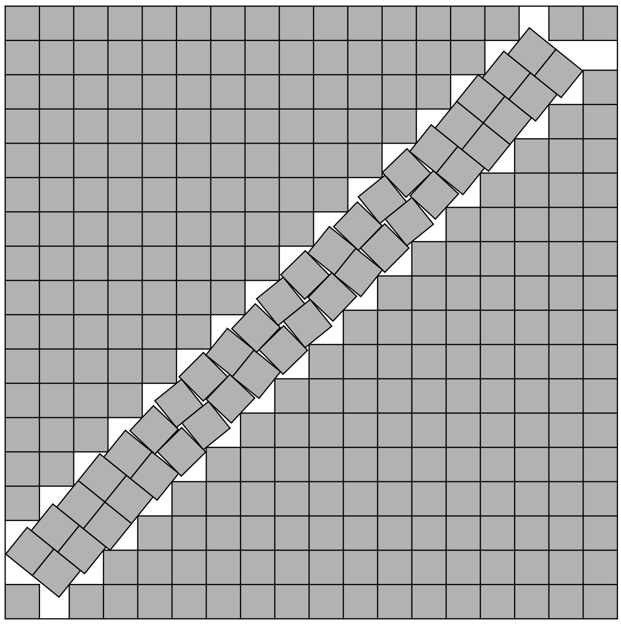
$s = \frac{33}{2} + \frac{1}{2}\sqrt{7} = 17.82287565553229\ldots$
Extends the $s(86)$ found by Erich Friedman in 1997.

300



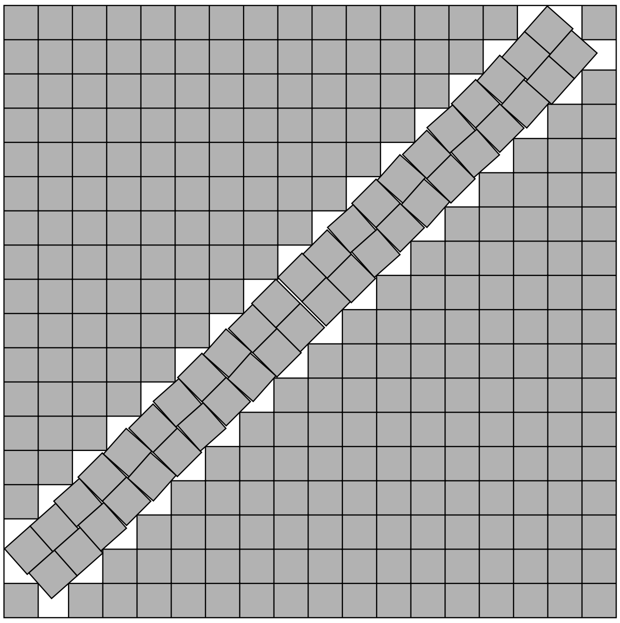
$s =$ 40 🔒 $= 17.82412338847854\ldots$
Found by David Ellsworth in January 2025, extending the $s(128)$ and $s(205)$ he found, based on the $s(69)$ found by Maurizio Morandi in June 2010 and using the technique from the $s(53)$ improved by David W. Cantrell in December 2024.
Improved by David Ellsworth in January 2025, using the technique from the $s(128)$ improved by David W. Cantrell and David Ellsworth in January 2025.

301



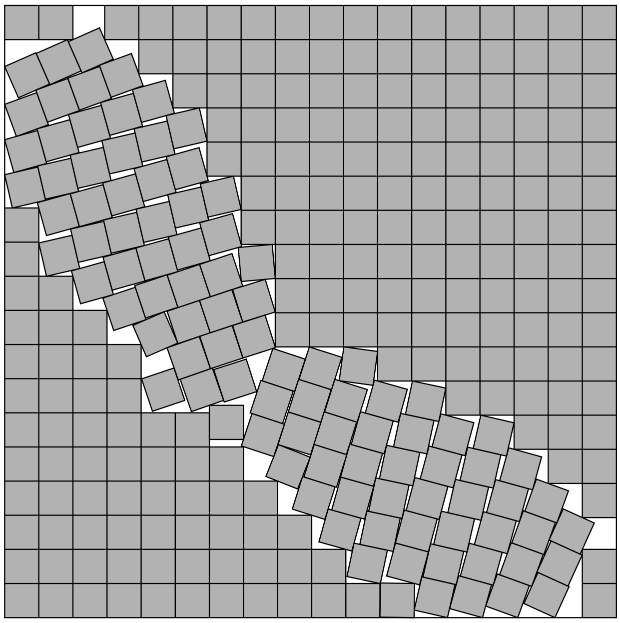
$s = 17.86889155557430\ldots$
Drafted by David Ellsworth in January 2025, including adapting and extending the $s(37)$ improvement found by David W. Cantrell in September 2002.
Refined by David Ellsworth in December 2025, using his modified version of Thomas Schadt's simulated annealing program.
Refined by Tej Stead in June 2026, working with Claude Fable 5.
Not yet analytically optimized.

302



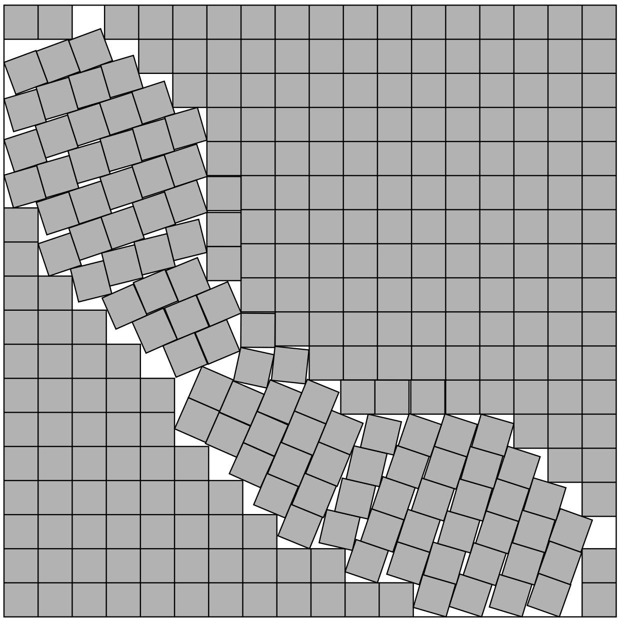
$s = {}^{4}$🔒 $= 17.88674602860566\ldots$
Found by David Ellsworth
in December 2024 (including re-adapting
the techniques from the $s(130)$ and $s(129)$
/$s(206)$ he improved), by extending the
$s(88)$ found by Erich Friedman in 1997,
and adapting and extending the $s(37)$
improvement found by David W. Cantrell
in September 2002.
Further improvement pending.

303



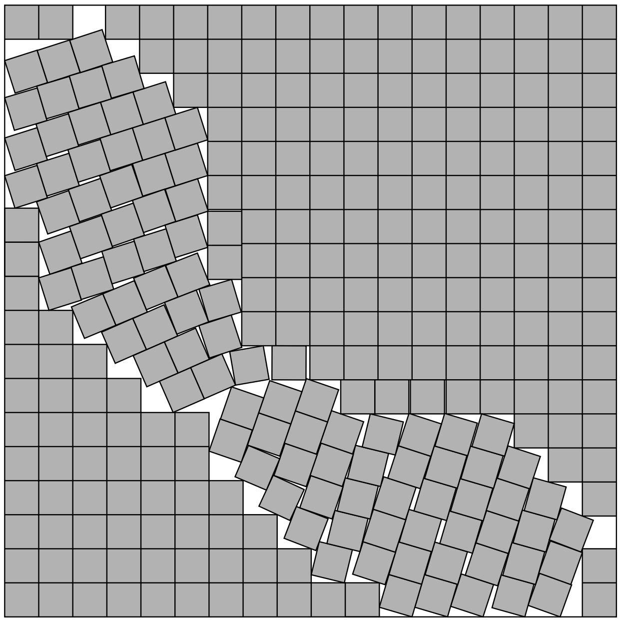
$s = 17.93125509556197\ldots$
Found by David Ellsworth in April 2026,
using his modified version of
Thomas Schadt's simulated annealing
program, starting from the latest $s(305)$ as
of March 2026, with 2 squares removed
and 8 straightened.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

304



$s = 17.94910783564662\ldots$
Found by David Ellsworth in May 2026,
using his modified version of
Thomas Schadt's simulated annealing
program, starting from the latest $s(305)$ as
of March 2026, with 1 square removed

305

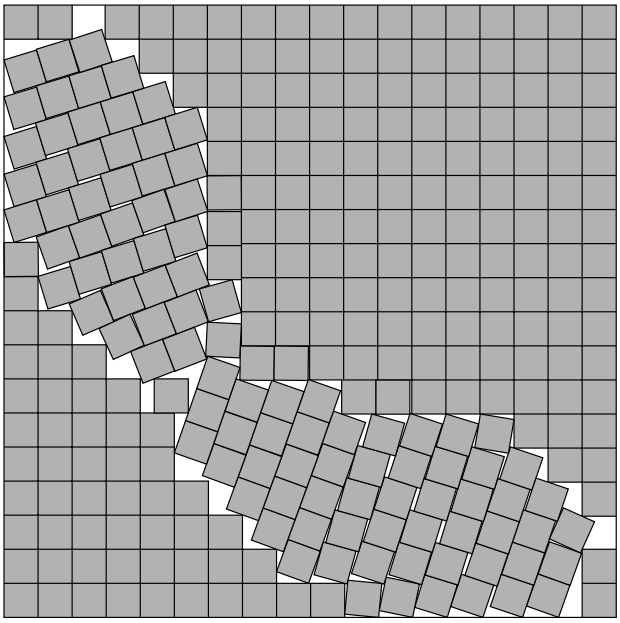


$s = 17.96066201401205\ldots$
Found by David Ellsworth in March 2026,
using his modified version of
Thomas Schadt's simulated annealing
program, starting from the latest $s(306)$ as
of February 2026, with 1 square removed

and 4 straightened.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

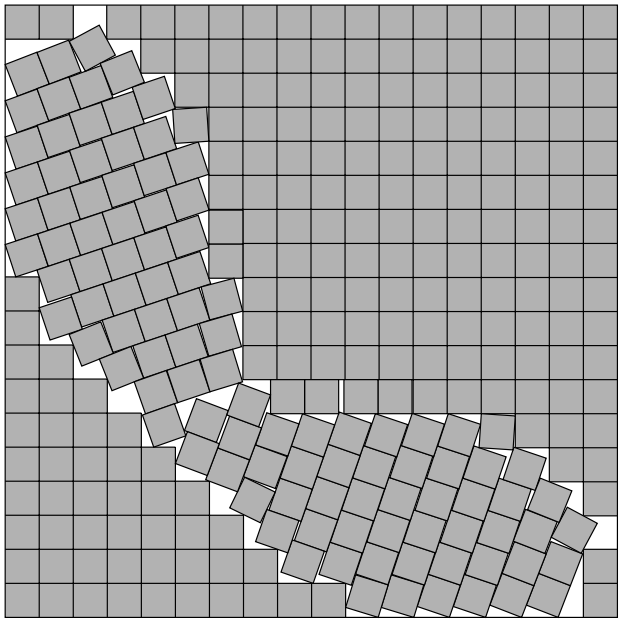
and 4 straightened.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

306



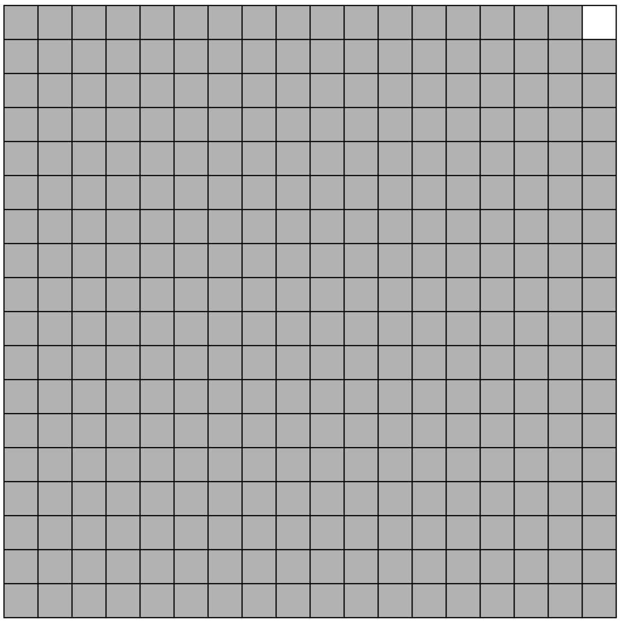
$s = 17.96913960675661\ldots$
Found by David Ellsworth
in February 2026, using his modified
version of Thomas Schadt's simulated
annealing program, starting from the latest
$s(307)$ as of February 2026, with 1 square
removed and 4 straightened.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

307



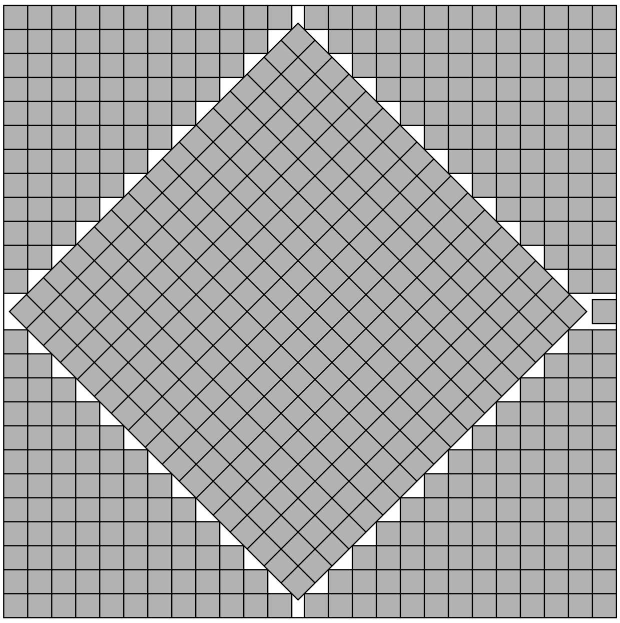
$s = 17.98272201579610\ldots$
Found by M.Z. Arslanov, S.A. Mustafin,
and Z.K. Shangitbayev in March 2019.
Shows $s(n^2 - n + 1) < n$ for $n = 18$.
Improved by David Ellsworth in
November 2024, by adapting a technique
from the $s(240)$ found by Károly Hajba in
September 2015.
Improved by David Ellsworth and
Károly Hajba in December 2024.
Improved by David Ellsworth
in January 2026, using his modified
version of Thomas Schadt's simulated
annealing program.
Refined by Tej Stead in June 2026,
working with Claude Fable 5.
Not yet analytically optimized.

322, 323



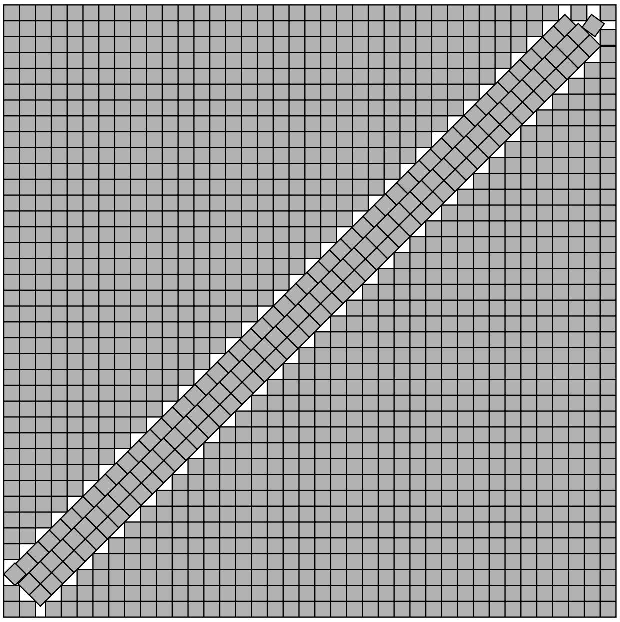
$s = 18$
Proved by Hiroshi Nagamochi
in 2005.

626



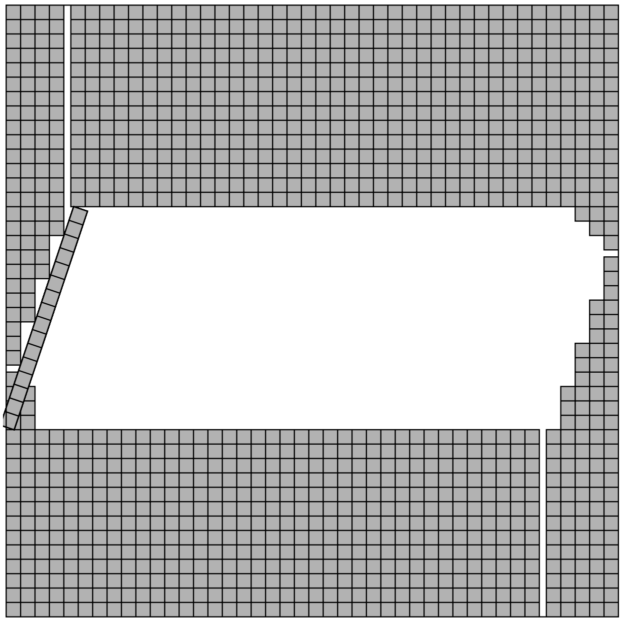
$s = \frac{27}{2} + \frac{17}{2}\sqrt{2} = 25.52081528017130\ldots$
Found by David Ellsworth
in January 2025, using a computer
program he wrote.
Continues the $s(26)$, $s(85)$ series found by
Erich Friedman in 1997.

1453



$s = 38.62799971540531\ldots$
Found by David Ellsworth
in December 2024.
Extends the $s(17)$ found by John Bidwell
in 1998, and the $s(83)$ found by
Károly Hajba in September 2024 and
improved by David W. Cantrell in
November 2024.
See also $s(260)$, $s(446)$, $s(791)$, and
$s(1097)$, none of which are optimal.
Improved by Grigoriy Dyachkov

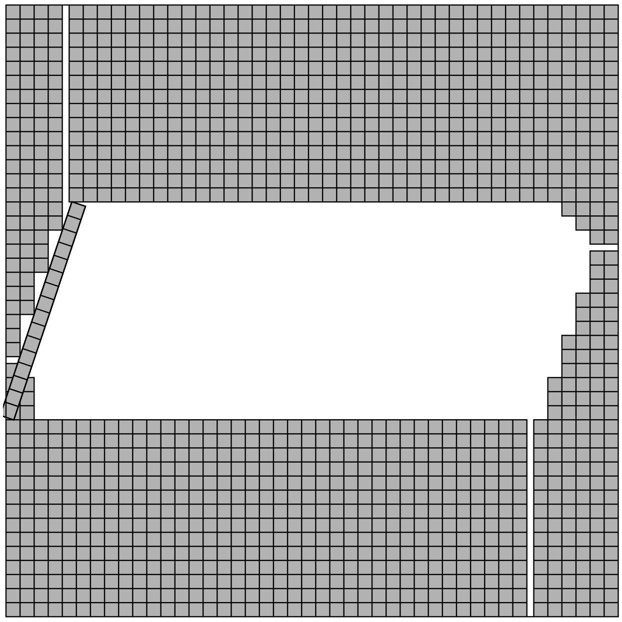
1765



$s = {}^4$🔒 $= 42.48797851186022\ldots$
Found by Károly Hajba
in November 2024.
Bounds $\{s(n^2+1)\} \geq \frac{1}{2}$ to $n < 42$.
Beats the $s(1765)$ Göbel square.
Improved by David Ellsworth
in November 2024.

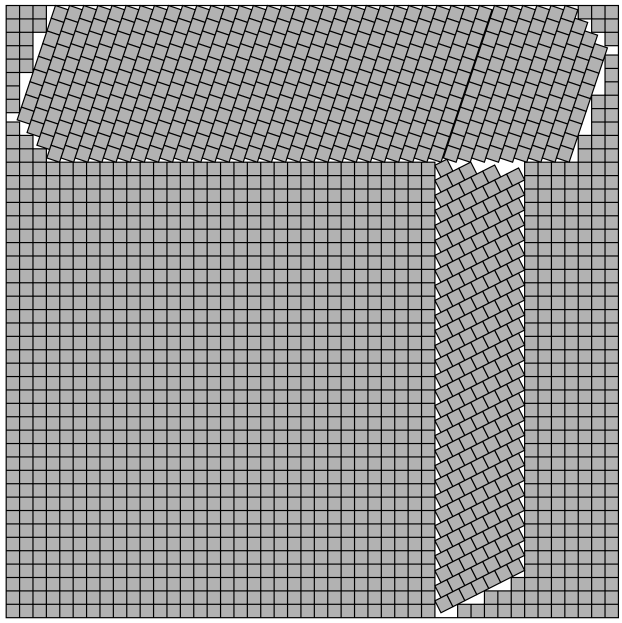
in September 2026, working with Claude Opus 5 and Claude Fable 5.
Not yet analytically optimized.

1850



$s = {}^{4}$🔒 $= 43.48797851186022\ldots$
Originally found by Michael J. Kearney and Peter Shiu in June 2001.
Bounded $\{s(n^2+1)\} \geq \frac{1}{2}$ to $n < 43$.
Now adds an "L" to the $s(1765)$ found by Károly Hajba and improved by David Ellsworth.

2043



$s = 45.69642783940993\ldots$
Found by David Ellsworth in December 2024.
Beats the $s(2043)$ Göbel strip.
Improved by Grigoriy Dyachkov in September 2026, working with Claude Opus 5 and Claude Fable 5.
Not yet analytically optimized.

For more details, see Erich Friedman's paper on the subject: Packing Unit Squares in Squares: A Survey and New Results (or David Ellsworth's edit).

Other articles:

- Analytic Minimization of Underdetermined Nonlinear Systems
- Göbel squares
- Göbel strips
- $s(n^2 - n - 1)$ pattern
- Rigid packings